Chapter 1: Introduction to Mobile App Development

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Hey, this is a full course on how to build a real mobile app with cloud code.

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So, from nothing to a working app on your phone, whether it's Android or iOS.

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I run a business that does around 300K a month, and it's almost entirely built on Cloud Code now. I've also shipped several web apps and SAS products. And the number one most requested video I

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get is, "How do I take all this awesome knowledge and use it to build mobile apps?" That's exactly what I'm going to show you guys how to do today. By the end of this video, you'll have a working

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app running on your actual phone. It's not going to be a mockup. It's not going to be uh some sort of emulator. It's going to be a real app that you can show people as well as the App Store and Play

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Store links. And you don't need any prior mobile development experience. If you guys can type an English sentence into a terminal, you'll be able to follow along just fine. In terms of what to expect, first I'll show you guys a

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finished app that's running on my iOS phone in under 2 minutes. Then we'll set up everything from scratch. So, Claude Code, uh the mobile development environment, and then your phone as a testing device. Then I'll walk you

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through how mobile apps like actually work. And there are a few different frameworks you can use to build these things, but I'll focus on a couple in particular. I'll talk about React Native. I'll talk about Expo and so on

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and so forth. Then we'll actually build an entire app together. I'll show you how to test them on your phone in real time as we build. Then we'll just repeat that exercise another three or four times so you guys get really used to

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building apps with different levels of functionality. And at the end, I'll show you how to take these apps and submit them to the App Store and then the Google Play Store. Okay, so hopefully

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you guys are as excited as I am. Uh use the bookmarks and chapter headings down below to save your spot and then jump around as needed. And I'll catch all y'all in the video.

Chapter 2: Showcasing the Habit Tracker App

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So I have an app right over here that I want to show you guys. It's just a simple habit tracker app. Obviously, doing it in this form is kind of annoying. So, what I've instead done is

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I've loaded this up locally on my computer. And I want you guys to know from an interface perspective, it's the exact same thing. It's just you are going to need to run these on your computer in order to do the design and

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so on and so forth. That's the whole idea. Um, and as you guys can see here, it's got a simple onboarding page. I've set the resolution here to mobile. You

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know, you can create your first habit. I don't know, maybe I want to do like a a training habit of three times a day with a specific color. And then as you guys

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could see, you now sort of have that laid out. You know, if I tap this, I can sort of fill that up logically. Um, this is more or less the exact same thing as

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any, you know, mobile app that you might see in the app store that does the same thing. And I did this in legitimately less than 5 minutes using cloud code and then a couple of cute little prompt

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templates that uh allow you to build these sorts of things and scaffold them really quickly. Now, the development system I'm going to show you guys in today's video is going to involve

Chapter 3: Understanding the Development System

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developing it as a web application first. So, that is something that is accessible in a web browser. The benefit there is obviously you can design something and then push it out for the

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web. So you can have it on a website. Uh and then you can also port it over to your phone. Then we can optimize it for phones as well as things like tablets and so on and so forth. And it's

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probably the simplest and and most straightforward one. But I also want to be clear that it's not the only way to build an app. I'm also starting off with a cute little habit tracker example here

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just because this usually involves most of the functionality that people want in an app. Usually you want some sort of database of some sort. You want a way to

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read the database. You want a way to, you know, send information to the database, modify it. Then you want some sort of front-end feature that, uh, you

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know, interacts with the database in a particular way. You want, you know, some sort of authentication setting, which I'll show you guys how to build. Uh, and then, you know, you want some sort of like cute little functionality. But by

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no means is this the only sort of app you can build. You can build literally anything. As we progress through the the course, I'm going to show you guys how to build apps that use AI to do really

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cool things. For instance, this app, Cal AI, is a good example of, you know, weaving AI into traditional app functionality. you can take a photo of something and then, you know, send it

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off to an API. They just sold for $50 to $100 million, I think, to My Fitness Pal. So, you'll get to the point where you can build apps just like this. We'll use similar sorts of designs and stuff

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like that, um, by the by the end of this course. And so, the progression I want you guys to think about is we're going to start over on the left by building a local habit tracker app. That's going to

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include no API. So, we're not going to send or receive requests to any third party services. We're just going to store it all locally. Um, we're not going to have any sort of database.

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We're just going to do it sort of directly on our phone or in the local storage. From there, we're going to add an API on. And so, you know, we're going to be using functionality that somebody

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else has created, but we're going to wrap that in our app. Then, we're going to add a database on. Then, we're going to push this over to a cloud solution.

Chapter 4: Building the App from Scratch

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And then, finally, we're going to have cloud. We're going to have an API. We're going to have authentication. So, we're going to have some sort of login screen and then a database more similar to that app that sold for, you know, between $50

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to $100 million. So, whatever your goals are, you'll be able to meet them in this course. I just want to make sure that everybody's on the same page here. So you don't see me developing a habit tracker app and be like, "Oh my god,

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this is such a crappy thing. Why the hell would anybody you design the 100th habit tracker app? The the habit tracker app is the means to an end. It's not the actual end itself." Okay, without

Chapter 5: Setting Up Cloud Code

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further ado, let's actually talk about how we're going to build this, including cloud code, setting up our development environment, and then also setting up our phone as a testing device before doing some some real basic dev, just to

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show you guys what this looks like in practice.

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So step one of building mobile apps with cloud code is having Cloud Code. And so in order to do that, just go to claw.ai/lo.

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You'll see a page that looks something like this. Obviously, their design may have changed by the time that you guys are taking a look at it. And then you can either continue with Google or enter

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your email. So I'm just going to enter my email address. After that, you'll get an email sent that allows you to sign in. And then once you're signed in, you'll have a variety of different plans

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shown to you. Now, I'm actually already on the highest plan over here called the Max, but you guys are fine signing up with the Pro, at least for this demo.

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Anybody scawking at the price, 20 bucks a month is probably the highest return on investment you will ever get. I use Cloud for more or less everything. Now,

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uh, as mentioned, this several hundred a month plan legitimately runs a $300,000 a month business. So, talk about ROI.

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Personally, I would just get at least one month, learn as much as you can, and then if you want to experiment with local models and other approaches and stuff like that to save on costs, feel

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free to do that. It's just best to learn at least initially from the source. So, I'm not going to click downgrade because I already have my account. But assuming that you do, it'll take you to a quick payment page and then you can customize

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your clot. And then once you're done, you'll have a page that looks something like this. So, I'm just going to say, "Hey, what's going on?" And uh Claude will get back to me essentially giving me a response. What's on the docket?

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People even say that anymore. I don't know. Now, the thing to know about Claude, if it's the first time you're using it, is this right here is your Claude chat. Okay? But in order to do

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things like develop and design mobile apps, we got to move away from the vanilla cla, we got to move towards a

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little more complex of a feature called claude code. Now claude code has a bunch of different ways to visualize and use.

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Okay? And this little space invader down here is going to be omnipresent if you're doing coding, so get used to it.

Chapter 6: Using the Cloud Code Interface

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Um, you can do so directly in the claude code web app here. You can also download the desktop app, which little button at

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the bottom lefthand corner. So you can download for Mac OS or Windows or you can use a third-party service like anti-gravity or Visual Studio Code to

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manage all of your files and then also communicate with Claude. And in my case, I much prefer that last option. And that's what this looks like here. You

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basically have a little cloud code window open. You have all your files on the left hand side. And then what's really cool is because you have this

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much granularity and access to all of the files, uh, you know, you can usually build much more sophisticated sorts of apps. So that's what I'm going to be

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using throughout this course. But I do want you guys to know you can use whatever interface you want. The reality is these are all just wrappers around like the brain of the model, the the

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actual intelligence. Whether you communicate with, you know, an uh integrated development environment style app like me, or whether you communicate with claude code, you're just using the

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claude.ai/code website, all of this is is the same thing at the end of the day. Now, if you want to follow along with my tutorial,

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how to set it up in this sort of interface. Well, you've already signed up to Claude. So, you only need one more thing, and that's the actual app itself.

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So, in my case, I'm using this app developed by Google called Anti-Gravity.

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Anti-gravity is pretty great. It's just really simple. It shows all your files and stuff like that. It's technically a Google product, so they're going to try and use their own models, or at least

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get you to, but I'll show you guys how to avoid that. just head over to that download page, check your whatever operating system you're in, and then

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download the correct uh executable for you. So, in my case, that's Apple Silicon. Then, all I have to do is click on this little anti-gravity 1dmg.

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Obviously, this will be a little different if you guys are in uh you know, Windows or something else. And then in my case, all I have to do is just drag this little anti-gravity thing over to the applications folder. Now,

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after I do that, I'll get a page that looks something like this. So, there'll be a little anti-gravity logo. There'll be a little agent panel on the right hand side. may seem really intimidating.

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Don't worry too much about this. Just going to close the agent view and then head over here to where it says extensions. And the extension that I'm looking for is called the Claude Code for VS Code extension.

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So, I'm just going to type Claude Code for VS. And then you can see that we have the Cloud Code for VS Code extension right over here. Now, in my case, I've already installed it, but all

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you have to do is just give it a quick install. I'm just going to click auto update here and then maybe go install specific version and then update to the most recent just so you guys can kind of

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see what it would look like when it's installing. So, boom. I just installed mine. What's really cool is after you're done. Okay. Uh, you'll see this little

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clawed icon pop up everywhere. So, I'm just going to exit out of this. And then I'm just going to double click anywhere on the page to open a new file. And then you'll see that little claw icons

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anywhere. So, now if we just click this and then exit out of all this other confusing stuff, you'll see that we basically now have a chat with Claude and we have that little space invader.

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So, now I can say, "Hey, what's going on?" And then it can get back to me saying, "Hey, how can I help you today?" If that doesn't happen, you may need to

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log in, in which case you /lo, type claude account with subscription. That's then going to open up a page, which will basically do the same thing that you did

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a moment ago where you where you logged into your anthropic uh account. In any case, the whole idea is to get a screen that looks something like this. As long

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as you have a cute little Claude Code guy somewhere on the page, you're good to go. From here on out, I'm just going to send a couple of brief prompts over

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to Claude Code and have it build us something really quick just so you can convince yourself that it's indeed working. So on the lefth hand side, there's a little open folder button. I'm

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just going to click open folder and then I'm going to go new folder and I'll say, I don't know, example project. The reason why I'm doing this is because, you know, we want to be able to see the

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files that Claude outputs. Now, every time I do this, just because of the fact that Google's trying to push their own products, it's going to change the interface and so on and so forth, right?

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So, I'm just going to have to reopen up a little Cloud Code terminal and exit out of the the native Google one. Once we do that, we'll have Claude saying, "Welcome back." And I'll say, "Hey, make

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me a really simple onepage site in the current directory." Then open it. This really simple onepage site isn't a lot

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of context. Might ask you some questions, might not. The whole idea is I'm just going to show you guys how easy it is to build a web property. Okay, so now we have a web property set up here.

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Claude actually just built this. I could say adjust this So, it greets me, Nick.

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And now, if I go back to that page here, um, you know, after it makes the edit, you can see that it just swapped the name from world to Nick. Okay, so

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hopefully you guys see it's not very complicated or difficult to build things. Usually the more difficult part is just building the right things and then getting those things onto the

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internet somehow, but don't worry about it. I'm going to show you guys how to do all of that as well. So, now that we've figured out how Cloud Code works and done a quick little demo, let's move

Chapter 7: Transitioning to Mobile App Development

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into our mobile app development environment. It's clearly one thing to build a website like I just did here, right? And this is currently existing on

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my computer. But if I want to get this over to my phone and then not only do I want to get it over to my phone, but if I want to share it with, you know, potentially hundreds of thousands of

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people and make something viral, we got to go one step further. The way to build an app nowadays tends to revolve in one

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of these three uh methods. Now, don't worry too much about the terminology or the platform names. You don't need any experience or any fun familiarity with

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this stuff. I'm going to walk you guys through it all regardless. But basically, there are three current methods. The first is using something called Expo and React Native. And this

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is uh actually the preferred approach that I'm going to show you guys how to do. Essentially, what we're doing is we're making use of a particular set of programming languages. So, JavaScript

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and TypeScript. We then have one codebase. And this codebase allows us to push out to our iOS apps. It also allows

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us to push to uh Android and then as mentioned you can even open up like a web app as well. So you can have it like you know on a website somewhere. Um you

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do cloud builds via the expo I think like app building system. Uh that's more technical terminology but essentially that just offloads a lot of the work

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onto a standardized tool that just knows how to do the sorts of builds that we're talking about get things ready for deployment. It's got a really huge

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ecosystem of supported apps and other files. Then it also allows you to do hot reload on device and that's valuable especially for testing purposes because

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you typically have to reload the app over and over and over again. And so this is what we're going to be doing um for our app. We're going to be using Expo and React Native. And I'm going to

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walk you guys through, you know, how all of that works in a moment. But I also wanted to run through a couple of the other common ways because hey, if you clicked on a course that's multiple

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hours and then you decided to learn all about app development, you might as well learn some of the lingo so that when you're talking to your app dev friends and they say they're using capacitor or

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flutter and firebase, you know, you can actually reason with uh with them reasonably well. Okay. So everybody else

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basically uses tools like flutter and firebase. And so this takes advantage of a different language called Dart. Okay,

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which is I think customdeveloped uh by Google specifically for the purposes of designing apps for their Android ecosystem. It also involves their own UI

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toolkit. The value there is basically Google has their own really opinionated like design system. If you've ever been on like a an Android app, for instance, you'll know the buttons tend to look

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pretty similar. You know, the modals tend to look pretty similar. All that stuff is very like opinionated. It's very set. And the benefit there is when you have an opinionated design system,

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it's pretty easy to put something together. You just kind of borrow their design. And you don't have to build that stuff from scratch. But obviously the downside is you get limited in what you

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can do with the design. You know, if you think back to my little habit tracker app, the streaks, which I'll walk you guys through in a moment, you know, this uh design is a really particular style

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that you probably wouldn't see in an Android app right out of the box. We're using slightly different fonts like serif fonts. The um little containers and stuff like that have different sort

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of like border and uh you know, corner radius. That's like the curve. You know, you have different types of capitalization and stuff like that. and

Chapter 8: Exploring App Development Frameworks

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some of the interactivity, the little bounces and stuff like that just look different. So that's that that's that opinionated design. This is me making one up myself and and ultimately I think that's what most people want. They want

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the ability to design their own app the way that they want to. What's cool is they also allow you to use their own backend. So Firebase and all of the uh

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apps are very like well optimized to use Firebase. Okay, which typically means you know for enterprise level stuff you get faster reads, faster writes and so

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on and so forth. Um these are those material design widgets that I was talking about. it's part of their their UI toolkit. Then it's also compiled to native ARM processors which again is

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like a highly optimized thing. It allows the app to run better on you know their their devices. Um so there's nothing wrong with using this approach. I just am choosing not to in this course

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because you know I don't really want anybody have to worry about languages that aren't really well understood. I don't really want you to feel locked into Google's in my opinion not as nice

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UI. And then I don't want you to have to worry about you know choosing the Firebase database or uh you know high level performance optimization. And I want you guys just to like worry about

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getting an app out. Finally, there's Capacitor, which I believe some might think is even easier to use than Expo and React Native. Um, it it's it is

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really easy to use. You basically just like build a website. Okay, so literally this website right over here, and then you just wrap it with features that allow you to export that to, you know,

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the app store and whatnot. Um, it uses languages that most people are really familiar with like HTML, CSS, JavaScript, and stuff like that. Uh, you can access your device APIs via plugins.

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So, it adds like an additional layer between the device and then, you know, the functionality that you want your app to have and then your website. And then it's also really, really easy to

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migrate. But, you know, you're here to learn how to build apps first, not necessarily websites. I have a lot of great tutorials that show you guys how to build the best websites ever. Um, I

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find if you are optimizing for everything right off the the get- go, you're probably not as good at anything in particular. And so this to me strikes a really good balance between like hyper

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optimized, you know, appdev in a specific language, in a specific database type, and then really general sort of website stuff. I know the app

Chapter 9: Getting Started with Expo and React Native

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developers in the comments are probably already about to skewer me, but that's okay. Let's just get you up and running with Expo and React Native. So, how do you actually get set up here in Expo and

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React Native? Well, the first thing I'm going to do is I'm just going to use a built-in slash command called slashclear. And that's just going to get rid of all of the conversation history

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that I had with um you know Claude previously. The benefit there is it's also going to keep my token count a little bit lower and uh ensure that whatever I'm working on is just as as

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efficient as humanly possible. Now inside of my little folder example project with this index.html that I put together here, what I'm going to do is just rightclick this. I'm just going to

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delete it because I don't really want, you know, anything in my folder for what I'm about to say next. And then all you have to do is say, I want to build a mobile app with Expo and React Native.

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Set up my workspace for me. Now, if it's the first time you're using Claude, let me just zoom out of this and then click on this button right down here, then

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click settings. And there's actually a uh little toggle that allows you to allow dangerously skip permissions. So, that's personally what I click by doing

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that. Uh Claude just, you know, can do whatever the heck it wants. I don't really have to wait for it to to work and so on and so forth.

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Okay, great. So, if we head back over here to folders, which I actually see everything that's laid out, you'll see that Claude now has like created a whole workspace. And um it really is as easy

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as that. Rather than try and, you know, download a template from somewhere or whatever, what it's doing is it just it already knows about Expo's setup. It already understands the different files

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and stuff like that that you need in order to create like a working app. Um and you know, now now we basically have something. So, what I'm going to say is

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like create a simple demo application, some habit tracker for demo purposes, and then let me launch it on my phone.

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Okay. So, what it's going to do now is it's going to plan out a habit tracker demo app, probably pretty similar to the one that I showed you guys, although I'd imagine with much simpler features. And

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then it's going to go ahead and do the building. And this is not rocket science. As you guys could see, I've just made a couple of simple requests.

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Um, you know, it's not as easy as it seems necessarily. There are a couple of technical things and issues that you'll run into if you don't know what you're doing. So, I'll walk you through that

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when it comes up. All right. Now, it's taken about a minute and a half to 2 minutes here to plan out the habit tracker demo app. You may be, you know,

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really chomping at the bits to just skip to the end and so on and so forth, but I encourage you to to watch alongside me.

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you know, I'm trying to strike a good balance between submitting a message and then immediately jumping to the finished output. Um, I don't want to do that

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because I think it's important for you guys to realistically see how the tokens are generated and what the model is saying at every step. But, uh,

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understand that any sort of development with agents involves offloading your own brain a little, right? Your own intelligence. And so, you do have to

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wait a fair amount of time for it to work. Now, it'll give you a bunch of fancy pants code. code and this code might be kind of intimidating. What this is doing is it's just it it is currently

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performing the writing action of like putting this in a file somewhere so that it can work with it. If you guys see

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errors pop up and stuff like that, don't worry too much about this. What's really cool about claude is it's capable now of

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solving its own problems, which uh I like quite a bit. And as we progress through the course and I show you guys better ways of doing things, you'll also

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find that you can instruct Claude very high level using what's called a system prompt to get even more autonomy where

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you want it to and then less autonomy in places that you don't. Okay, so what is it telling us? It's saying the app is now ready to run. Here's what I built.

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The today tab, the manage tab, and some data persistence. So this is currently storing locally. I just asked it to build demo, which I usually does in a

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local way. There's no database. There's no nothing. Now, it's actually giving us some instructions here. It's saying, "Hey, could you install Expo Go from the

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App Store or the Play Store, then run this function, and then scan the QR code with your phone's camera or the Expo Go

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app?" You're probably like, "What the hell does this mean?" Well, we actually want to get this thing live on our phones, right? So, that's what I'm going to show you guys how to do. This demo app that they just built. In order to

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see it, what we have to do is we have to go down to the App Store. Okay? And in order to show you guys that, I'm just going to use this iPhone mirroring app so you guys could see everything that

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I'm doing on my phone. Um, I don't think there's any way to make this any bigger, unfortunately. So, just zoom in as much as you guys can and uh take a look at

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what I'm doing. First things first, we're going to have to go uh app store right over here. Then once I'm in the app store, I'll go search. And what I

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want to do here is I want to just type Expo Go. So now I'm on my phone literally doing this. Um, and I've already downloaded and installed Expo

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Go. So, in my case, it'll say open, but this is the one that you want. So, you're going to install it, and then that'll move us over to a page that looks something like this. If you have

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any recently opened apps, then uh they'll pop up down there. That's my little streaks app that I was showing you. Okay, great. Once we have this, you do need to log in or sign up. And so,

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I've actually already created an account here, just uh Nikki Jer. So, I'm going to give that button a quick click. And now, you'll see it's actually giving you some instructions. It'll say, "Start a

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local development server with this command, npx expo start." Now, one thing that you'll find is AI from time to time

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will give you instructions suggesting you have to do stuff yourself. Well, one of the highest star things you can start doing right now is just asking to do it

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for you. It's trying to hand this off to you because of the way that it was trained. And basically, the way that it was trained is sort of like following

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instructions. You know, it was it's read a bunch of guides on how to do this sort of thing and it sort of compiled all that. And so, it thinks that it needs you to launch this sort of thing. It

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needs you to like submit this command and stuff like that. but you don't actually have to um do this for me in a

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new terminal window to me does like 80 to 90% of the work. So what it's going to do now is it's basically going to open up a new terminal window. Okay,

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it's right over here. And if I zoom way in this new terminal window here now is being managed by AI. Okay. And one thing that you're going to see immediately is

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it's going to pop up with this big sexy QR code. And all we have to do in order to actually run our app now is I'm just going to open it up on my phone. I'm

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going to scan the QR code, open up the EXP link. My case, I have to enter my passcode.

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And now it'll jump directly over to the app. I'm just going to run this uh locally here for a sec before showing you guys on the iPhone mirroring app.

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And then we actually have the little habit tracker app that is running on our on our phone. Okay. So, let me just show you guys what this actually looks like by running the iPhone mirroring app.

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Again, moving that over and then connecting. Uh there are equivalents for Android, of course. So, I'm just using iPhone here for simplicity. And you see we now we now have our app right now.

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This isn't perfect. There are a couple of issues already. You can see down at the bottom it says uncaught for promise ID 1, whatever. Don't worry too much about that. Just focus on like the core

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functionality. You know, we actually have something that's working, right? We have a couple of basic apps here. And you know, it looks like when it's done, it says all done for today. And so,

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what's really cool is this is just running on my phone. And it's running um basically using an emulator to show you what this would look like if it were

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actually like fully uh built. and and live and stuff like that. Okay, so that is the mobile development environment that we're going to be using. Um, we're

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going to be using this Expo Go sort of like QR code scan thing and then we're just going to be showcasing it locally right over here. Um, I don't just do it

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using an mirroring method like this. I'd highly recommend if you guys are actually testing this to like do it on the phone itself. Um, just understand that I have to do this in order for you

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guys to see what this would look like um, for tutorial purposes. But what's really cool is when you do it on your phone, um, you know, you you you get to see if there are any like navigation or

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UX issues. Sometimes there are issues with like you, you know, I don't know, this sort of like pull and and drag functionality. Sometimes there's hidden

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functionality that you won't know, uh, where you drag to the left and the right and so on and so forth. And, uh, yeah.

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Anyway, I guess what I'm saying is I'd recommend just doing it on your phone.

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And so, one thing that I'm realizing with my phone is it actually created an entirely new page called um, create a new habit. And so here I'm capable of sort of creating my new habits and and

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closing them out and stuff like that. I wouldn't have known that if I didn't go on my phone. Okay. So now what we've done is we've set up both cloud code

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then a mobile development environment and we've also shown you guys how to build the testing loop onto a mobile device. From here on out we can actually

Chapter 10: Designing a Successful App

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get started with the best and most exciting and enjoyable parts the actual app development itself. Let's do it. All right. So we've clearly seen that it's

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possible to build an app. Now that we know that you can do more or less anything under the sun with technology, how do you actually design like a good

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app? And what you'll find is in our current lovely society, uh we can more or less do whatever the heck we want.

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It's actually just about taking all of that power and potential and then orienting in the right direction. In our case for mobile app design, what good

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apps look like are apps that satisfy and sort of go through all five of these steps. So, in general, any good app, and

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I'm just going to use our habit tracker as an example, needs what's called a core function. And that is the one thing essentially that if you removed

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everything else from this app, the app would still be the app. Okay? Like for instance, in our habit tracker app, if you removed all of the accessory

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additional features away from our, you know, habit tracker app, all the cute little colors in that prior example, uh, all the different types and so on and so

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forth, maybe the settings page. If you stripped all of that away, the one thing that would remain is there's probably going to be a way to like create a habit

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and then tap on it and every time you tap on it, it it logs it to some database. That's sort of like the core functionality, the fact that it tracks a habit. Right? Now, after we're done

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defining that one thing, okay, what we're doing next is we're turning that one thing into a feedback loop. And so

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in any sort of like user experience design, any sort of uh game design or anything like that, what you need is you need a cycle that maps an action to a

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reward. And ideally that action and reward loop is under about 30 seconds or so. Now obviously this whole idea of a

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loop is sort of uh you know it depends on whether you're working in with a game let's say you know if you're doing like a mobile game for instance the loop is usually going to be a lot tighter.

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there's going to be a lot more rewards and it's going to be like a lot more dop dopamineergically stimulating than something like a habit tracker app or something like a calorie tracker app.

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But what's cool is you can still take all of those same ideas and features right the fact that you perform an action and then receive some sort of reward um whatever it is that you are

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building. And so for instance in our case right what we want to do is every time somebody you know does a habit okay

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aka creates one and or fulfills one we need some way of rewarding them. And usually the way that we're going to reward them is we're going to reward them using some sort of visual stimulus.

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Okay. So just because I want to keep track of everything here, I'm just going to move this down a bit. And then I'm going to just sort of um annotate here.

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And I'll say the whole idea with our core function is we want to track a habit. And the whole idea with our core loop is we just need some sort of like

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stimulating reward. Okay? And variety of different ways to do this. We want like a nice sound to occur every time I tap the button for instance. We want some

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sort of like nice animation to occur every time I tap the button. If we do uh our habit goal for a certain amount of time, maybe we want cute little confetti

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to rain down from the the top of the screen. We just need some sort of like visual reward stimulus. Some apps go really far with this. Um I'm using one mobile app right now called Opel, which

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is like a it's like an app blocker, which basically eliminates my ability to use YouTube and, you know, X and all these different social media platforms

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between certain hours of the day. And what I really like that they do is they will like create a little gem for you.

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And the more days that you block an app, the cooler and more exciting the gem gets. You know, at the very beginning of the day, they they're like, "Hey, here you just unlocked this normal basic gem." And you're like, "All right, cool.

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Whatever. It's a normal basic gem." But a week or two in, you know, after you've been blocking it consistently for a long period of time, the gem is now super shiny. It's amethyst. It's, you know,

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uh, kind of in the name Opal. And I'm not sponsored by them or anything. I just really like their app. um you know it just gets cooler and cooler and cooler and so human beings were very

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like tactile creatures right in general anytime you're designing an app you need to take advantage of that tactile nature um you know your phone has the ability

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to uh perform haptics right like vibrate for instance when you fulfill one of these things like there there are multiple ways to stack that on but

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basically what we want is you know user taps habit and then we want the uh uh habit

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to reward reward, okay, the user. And this mechanism right over here, this can get about as complex as you want. You

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could add on like financial, monetary rewards, their apps, and uh and have it sort of accountability partners now where like you give a certain amount of money to a friend if you miss a whatever

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that is. That core loop in our case is going to be really simple, but you can make that as complex as you want. Okay. After that, we need accessory features.

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So, you know, we started with the core function, then we defined the core loop of actions that the user is going to take with the core function. Now we need some form of like accessory. So what

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wraps around that core loop and core function and I don't know just adds to it. So for instance um every one of the things I'm about to say support that

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main loop of you know creating a habit um you know tapping a habit essentially to fulfill it and then sort of receiving some sort of immediate reward. Uh well

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you're probably going to want a list of all the habits and uh uh uh times that you've tracked all these individual habits in the past. So, for instance, when we set up our database, you're

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going to need some way that, you know, a user can log the fulfillment of a habit and then, I don't know, a week later can still go back and see, oh, you know, on May the 4th, I actually logged X habit.

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On May the 3rd, I actually logged Y habit. Why? I don't know. I mean, that that that's valuable in and of itself.

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Maybe you could tie that to some sort of like visualization chart or something like that, right? So, I'm going to say chart plus logging. That's a really cool

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accessory feature. Um, you know, you should be able to customize your habits.

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If you think about it, not all habits are the same. Habits that occur, you know, on a daily basis, just once might be sort of like a zero or one sort of habit. It's like you either do it or you

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don't. But habits that uh I don't know, you need to do multiple times a day.

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Let's say you have a habit of like I want to drink water. Maybe you know you count the number of cups that you drink or something like that. So we need like multi-

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tier habits for instance. And then maybe you also want and I don't think I'm going to add this to our specific app at least initially, but maybe you also want

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like a social feature, right? What good is tracking a habit if you can show it off a little bit? And so I'm seeing a lot of these sort of like habit tracker apps implement social functionality

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where you can like add your friends. You can get little notifications when the other person submits a habit or something like that. Um, so these are all just ideas and examples of accessory

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features that don't actually fundamentally change the core function or the core loop, but you know, they allow you to do slightly different things. They allow you to cover a little bit more ground. And it's important that

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your app has, you know, all all three of these. Okay. After that, I'd recommend a surface area check. So typically when

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new people start developing mobile apps and really just any apps in general, they get really really excited. And so they'll they'll instead of just do one core function, they'll do 10. And then

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instead of uh one core loop, they'll do a 100. And then instead of just a couple of accessory features, they'll do a thousand. And then before you know it, your app has like 5 million pages and

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it's just very very complicated. Well, in a world where you can do everything, typically it's not about doing everything. It's about doing one specific thing fairly, really well. And

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the best way to facilitate that is by just making sure that your app isn't super complex. And so, you know, as part of implementing these features, we're probably going to need a couple of

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different screens, right? We're going to need like, I don't know, a habit creation screen, a habit tracking screen, and so on and so forth. Um, that can be thought of as digital surface

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area. If that digital surface area gets too large or there are too many paths and whatnot, the user is going to get very confused. And so, we we don't want to do any of that. We want to make sure that there's just a very simple core

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loop facilitated by a handful of screens. And uh you know it's simple enough that that the user just uses the app once. We don't need to explain it to them. We don't need to like basically

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get them a PhD in our habit tracking app in order to to to learn to use it. Uh one run through is basically everything they need in order to onboard and know

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how to use our app properly. Okay. And then finally we need some sort of retention hook. Uh and so what I mean by retention hook is we need some form of

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way to create an unfinished state that the user has to come back to after a certain period of time in order to finish it. So, in our case, you know, if

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we're doing a habit tracker app or something like that, uh there variety of different ways to do this, but essentially, you know, you could have some form of challenge or something

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where it's like, uh, oh, hey, you know, you you join the app and then you have a 3-day challenge where you log a habit 3 days in a row, and it's sort of inherently retentive, right? Because in

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order to achieve the 3 days, you need to go on the app three separate times over the course of a 72-hour period. Uh likewise, you could maybe have it so

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that you know you you start the app, you click a button, and then it actually checks in with you, let's say a few hours later to see how your progress towards the habit is, whether whether or

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not you've done the habit. Maybe uh the very first thing we do in an onboarding in our onboarding for a habit tracker app is it's like, okay, create the a

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habit and then we'll check in with you before you go to bed at the end of the day to make sure that you've done the habit. Then all you have to do is just tap one button and then you know you can

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you can log it. Obviously a variety of different ways to take it there, but the whole idea is like you don't just build an app so that people use it once and then uninstall it, right? Any app that's

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ever trying to tell you that that's their goal is lying to you. The whole job in app development is you want people coming back to the app and so you have to take advantage of some sort of

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dark patterns in user behavior like retention in order to do it. Uh but retention hooks are very very important.

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And so what I'm going to do at least to start is I'll just do like a 3-day challenge. And it's just going to be, hey, can you log whatever the habit is that you want to do? You know, drink

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water. uh I don't know do 20 push-ups uh I don't know read for an hour or whatever can you log that every day for three days and assuming you can you have

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your app okay so what we just did is at a very high level we just designed the app you know it's one thing to uh build it in code and whatnot but it's another

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thing entirely to understand like why you're doing what you're doing and so I just wanted to take a couple minutes that we were all on the same page that like that is how you design an app you

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go through the core function then you uh ideate the core loop then you figure out some accessory features you do a surface area check make sure your surface area

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isn't super bloated. Then finally, you make sure you have some form of retention hook. And if you have all five of these things, you now basically have an MVP. Um, well, at least you have the

Chapter 11: Crafting the Minimum Viable Product

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MVP that you can give to Claude to have it turn it into the actual thing. You have like a scope. Okay. So, what I'm going to do now is I'm going to take everything that I just talked about and

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I'm just going to plug it into Claude code obviously in text. And then we're going to take the little fledgling app that I developed and then turn it into

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something that actually abides by all all five of these core features. The very first thing I'm going to do is I'm going to head back to our little claw code window inside of anti-gravity. And

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then I'm just going to go slash and then nit.

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Now, the reason why you go slash in it after you build out like a brief little uh demo of the app is basically this is

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a set of instructions that teaches claude every time it's initialized what the app is without it having to go through every single file on its own.

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And so for instance, what this just wrote, if I actually go down to the claude.md, which is the system file, okay, this is where it's actually storing all that information, is it's

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saying this file provides guidance to cloud code when working with code in this repository. Here is the architecture of our app. Here's the theme. Here's the state. Here's the

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path. Here's everything that you need in order to basically run it. And so the value there is you basically just eliminate um a couple of tokens so you're not actually spending as much

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money because most of the time you're build by token or your token usage consumes your plan. And then it also just knows where all the files are.

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Okay. So after you create your cloudmd, then you can just go back/clear. You'll see this little token meter on the right hand side will go back to basically

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nothing. And then we can continue with um you know actually having it design this app and make it better. And so what I'm going to do is I'm going to build the app according to the specifications

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that I'm talking about. And then just to make my life way easier, instead of me writing all this stuff out just in the chat with my fingers, I'm just going to use my voice. And I'm using a voice

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transcription tool called Aqua right at the moment, which is where I can basically just talk into my mic and then it'll uh, you know, record a bunch of text. So, that's what just happened. And

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you can see that it just pumped it right in there. What I'm going to do is I'm going to go back here and then I'm just going to discuss my app with Claude.

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Hey, so I really like the app as it stands, but I'd like to level this up with an app design framework that uh you know, high-quality app devs and

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designers use where essentially every app that I build requires a core function, then a core loop, some accessory features, uh surface area

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check to minimize the number of screens, and then some sort of retention hook. So the core function for this app is it needs to be able to create and then

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track habits. The core loop for this app is basically every time a person uh creates a habit and then tracks it, they

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need to be rewarded in some way. It needs to be visually stimulating. There needs to be some form of haptic feedback and then ideally there's some sort of sound like a like a chime or something.

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Also, we need some form of challenge.

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So, uh you know, if they're embarking on a 3-day habit challenge, let's say, which might occur immediately after onboarding, at the end of that 3-day

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challenge, we also need to reward them for the fulfillment of their efforts.

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The accessory features for this app are going to be something like uh logging.

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So the user should be able to see all of their prior habits tracked, some sort of, you know, accountability thing so that they can look back and then maybe

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see a graph or a chart of just how consistent they've been. Um, and then ideally we need a way to create multiple types of habits, not just one. Uh, being

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aware that, you know, a habit where you log it once per day is different from a volume based habit where you need to maybe do it three or four times a day.

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uh for surface area check just make sure that we don't have more than somewhere between five to seven screens in our app. We want it to be as simple as possible. And then in terms of retention

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hook, the thing that brings people back to the app, we want to create challenges for the user and basically have some sort of ongoing thing that checks in

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with them via push notifications probably once a day or maybe a couple times a day. um just consistently knocking on their door, seeing whether

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or not they've done the habit, whether they're ready to start the habit or or hey, you know, don't forget about XYZ habit uh whose intention you said

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earlier today. Stuff like this just gets people coming back to our app and is ultimately responsible for a fair amount of our usage.

Chapter 12: Implementing App Features and Functionality

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Okay, so I just said a giant wall of text and because I did, it's not actually going to paste all of that text directly in like verbatim. You're not going to be able to see it. Instead,

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it's just going to say pasted text number one plus eight lines. So, what I'm going to do is I have this whole thing here. If I press enter, it'll now

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populate. Um, and now that I've given it both the framework and also every little step in that graph, um, I'm going to have it actually implement and and add things to the app that we built earlier.

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Just that simple little tracker. One more thing I'm going to do is instead of doing all of the testing and demoing on my phone like I was doing earlier

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because the iPhone mirroring app just can't get any bigger. We're going to be doing this inside of Chrome directly.

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The functionality is going to be the same, but this will just allow me to zoom way more in and then have you guys actually be able to see the thing. Um, but as a final check, I'm going to be

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going back through my phone using the Expo setup that we talked about earlier.

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And that's important just because if you can't actually physically, you know, play with the haptics, we can't hear the chime come out of the phone and stuff like that, you're not really getting the

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whole experience. Okay. And this just thought for a minute and 29 seconds here. And then it got back to me saying, "Here's how I'd map your framework onto the current app." So, six screens in total. There'll be an onboarding.

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There'll be uh first launch flow, which will pick three habits for you and then start a 3-day challenge. Today, which is a daily checklist, that's that core loop

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that we talked about. history, which will be a calendargraph view of all of our past completions. And then some sort of consistency chart. A manage button,

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which is where you can do all of the stuff like adding, removing a habit, configuring the type, and so on and so forth. A challenge, which is where you can see the challenges that you're

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currently enrolled in. Then finally, uh some little settings, which is where you can set your notification preferences, your theme, and so on and so forth. We'll do push notice uh on that. Okay.

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And then as things get better and better, you'll actually enhance the core loop. So, uh, you'll go haptics, then you'll do sound, and then you'll also go

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visual as well with a little confetti particle burst on completion using this library called reanimated. You know, if I was a crazy crackedout appdev and I

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was looking at this stuff, maybe I'd say, "No, I don't want you to use the Expo AV library. I want you to use this other library." Or, "I don't like reanimated. I like this other thing."

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Or, you know, confetti particle bursts are very CPU heavy. They're not like ideal, so let's not use them. But, we're just getting up and running at this point. So, I'm not going to be very opinionated in all the things that I

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want us to do. I just want to show you guys proof of concept. You can get pretty far without uh you know without actually having to worry too much about every individual step. And so what it's

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done here is it said, "Hey Nick, this is a substantial build. I want to tackle it in a bunch of phases." But just like I talked to you about earlier, uh you know, Claude is going to try and

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conserve both tokens and then also just like it's not going to try really hard or at least as hard as it realistically can in every scenario. So what I'll always do when it presents me a

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situation like this is I say, "No, I just want you to do the whole thing." So that's what I'm going to do here. I'd like us to do all of it, including phase

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40 minutes, 34 seconds

1 through 4. Um, once done, open in Chrome, not Expo, so I can test locally.

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40 minutes, 43 seconds

Okay. So, I'm just going to send that in and then we'll see where it goes. All right. And it spent something like 5 minutes thinking and then opened a couple of Google Chrome tabs. You'll

40:51

40 minutes, 51 seconds

find that it'll do stuff like this just as part of its own internal testing loop from time to time. Um, we can expedite that as well by installing a specific

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40 minutes, 58 seconds

library which I'll run you guys through later as we get a little bit more complex with our app. But if I go to Google Chrome here, you'll see that we now have this tab open, localhost 8081.

41:08

41 minutes, 8 seconds

And so anytime you're designing or developing any sort of app, um, you will always usually open up a local server of some sort. Don't worry too much about it

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41 minutes, 15 seconds

on like a technical POV. What we're doing is we're just like starting up a server to host our app on the back end.

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41 minutes, 20 seconds

And so you can see if I just zoom in here to make it really easy that we we're actually already getting started with the app. Um it hasn't actually gone through and built the functionality yet.

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41 minutes, 27 seconds

I think it's just like scaffolded a couple of pages. And obviously there are a couple of issues like the text here is invisible. But you know if I give this a click you can see that we we actually

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41 minutes, 35 seconds

have like the apps right here. So initially it's saying hey choose your habits uh exercise meditate and then drink water eight times a day. Then it's

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41 minutes, 44 seconds

going to invite us into a 3-day challenge with that little button. Uh that's still screwed up. And then, okay, I'm just going to make this a little smaller.

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41 minutes, 52 seconds

And then let me just make it look like an actual mobile app would realistically.

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41 minutes, 57 seconds

You can see that we now have sort of uh these three bottom tabs as well. Okay, so here's sort of the 3-day kickstart.

42:06

42 minutes, 6 seconds

Okay, we're 0 to 3 days right now. In order to drink water, every time I click this button, I don't think you guys can hear this yet, so let me move this. Let

42:13

42 minutes, 13 seconds

me just um uncheck the audio. You guys can now hear this little like chime/beeping noise. Ideally, I don't

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42 minutes, 20 seconds

know, like hopefully you can. Um, and then once you're done, once you make it to all eight, we now have one of three complete. Same thing with this meditate.

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42 minutes, 27 seconds

Same thing with this exercise. But then you obviously have like that cute little chime with something saying all done for today. Okay. And then we can also uncheck, which is an important part of

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42 minutes, 35 seconds

the core loop as well. I'm sure you guys could see, but there are probably some situations in which like you can't, you know, you don't actually want to uh uh remain in a habit or maybe you locked it

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42 minutes, 43 seconds

by mistake if that thumbmed or something, whatever. So, that's the today page. It looks pretty good. The history page over here is where you can see the long running streaks, which is

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42 minutes, 50 seconds

nice. So, here's an activity graph. I don't actually have any activity cuz, you know, I just started with this stuff. Kind of curious if I go back here and I log my habits. Okay. And you can

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43 minutes

see the activity graph has actually popped up there, which is quite nice.

43:03

43 minutes, 3 seconds

Looks like there's also a consistency chart which logs the consistency on a 7-day basis of how far you've gone with exercising, meditating, and let me just

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43 minutes, 11 seconds

tap through all of this so it gets to eight. And then also drinking water. And it looks like it's at 3%. And there's also a manage a habit button. And I'm

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43 minutes, 20 seconds

just uh highlighting this so I could see the text underneath. This is obviously a bug that we're going to have to fix, but you can see you can select a daily habit or a count style habit. I don't know. I want uh sleep habit, let's say. Okay.

43:32

43 minutes, 32 seconds

And this is going to be a daily habit.

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43 minutes, 34 seconds

Sleep before, I don't know, let's say 9:00 p.m. And then we'll do a daily reminder. Reminder at 9:00 a.m., it

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43 minutes, 42 seconds

looks like. Um, and then I'll press enter. And then now it looks like we have a sleep before 9:00 p.m. happen. If I go back here, we can we can also tap that, which is kind of cool. Awesome.

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43 minutes, 52 seconds

So, I mean, like this is more or less everything that I'm going for. In order to test the full 3-day kickstart, we're probably going to have to like accelerate that somewhat and build in uh

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44 minutes

our own little like testing feature loop so we could see, you know, like this is one day, this is another day, this is another day. When all those 3 days hit, like what what sort of animation is

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44 minutes, 8 seconds

going to appear, how are we going to track that um long term? But I would say this probably has most of the functionality gap that I want. Um obviously there are just a couple of

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44 minutes, 16 seconds

issues like the fact that the text is the same color and uh the fact that this reminder here is like I can't change it.

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44 minutes, 22 seconds

just says daily at 9:00 a.m. So, what I'm going to do is I'm going to do the same thing I just did a moment ago where uh I just use a voice transcription tool

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44 minutes, 29 seconds

to like talk Claude through my app. Uh it's just this time I'm going to change the you know make minor changes wherever appropriate. We're not actually building the whole thing now.

44:38

44 minutes, 38 seconds

Excellent job. This looks fantastic. The first thing I want to change is I need some sort of testing view for a developer so that I can modify the day

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44 minutes, 47 seconds

of the challenge or at least trigger the event that occurs when we hit let's say 3 days out of the 3-day kickstart. Right now I just sort of have to trust that

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44 minutes, 55 seconds

it's working, but I'd like to ideally really have it work. The second major issue is right now the font color that

45:02

45 minutes, 2 seconds

you're using is the same as the background color for a lot of these icons. Uh the check mark icon for instance is invisible because of that. A

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45 minutes, 10 seconds

lot of the text in the um history and then manage pages is also invisible. Um there's also no way right now to create your own challenge which is unfortunate.

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45 minutes, 19 seconds

We just sort of have to to go based off of the the pre-existing challenges. I'd like us to be able to do that. Also double check the stats. Right now I'm

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45 minutes, 27 seconds

seeing exercise is 3% checked for instance after one day of doing it.

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45 minutes, 32 seconds

Mathematically it doesn't look like a 7-day challenge. It looks like a 30-day challenge, but uh on the consistency

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45 minutes, 38 seconds

end, it's saying uh 7 days. Some of the uh text kind of rolls or wraps over on the

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45 minutes, 46 seconds

history page. For instance, the current streak, the best streak, um just because the text itself is a little bit longer.

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45 minutes, 52 seconds

Uh you know, on mobile, it's most likely going to wrap. And then when it wraps, it's also left aligned, which I think just looks kind of weird. It doesn't look as good as it probably could.

46:01

46 minutes, 1 second

And then on the manage page, um I like all the icons. Everything there is really cute, but the um habit type selectors have the same font color problem. Same thing with the habit name.

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46 minutes, 10 seconds

So, I think that's just a widespread issue. Finally, at the very bottom, you have the ability to set a daily reminder, which I think is valuable, but uh we would want to uh be able to

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46 minutes, 20 seconds

customize the the date and time of the reminder. And also, ideally, we want to be able to specify different reminder times for different apps uh habits

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46 minutes, 27 seconds

rather. Right now, we only have one reminder time, 9:00 a.m., across all of our apps. And that'll help with uh you know having the user have more control over when they want to be notified.

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46 minutes, 37 seconds

Aside from that, I liked our onboarding.

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Um I think we could do with one more screen that just very clearly explains how the app works. The core functionality is you can create an app

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46 minutes, 47 seconds

and then you can track it. Uh you know hit milestones and kickstarts and then track your progress and then use push notifications so that you know if you

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46 minutes, 55 seconds

want to be reminded in a particular point in time to assist you uh you can.

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46 minutes, 59 seconds

I'd say that's probably the 8020. Okay, with all that in mind, go through everything top to bottom, implement those changes, and then just open it up

47:06

47 minutes, 6 seconds

in another Chrome tab. I'll uh I'll test it. I'm now just going to dump that in and then cut to whenever it's done with all of its changes. Okay, after a couple

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47 minutes, 14 seconds

of modifications, just making this more visible for you guys. You can see that it's opened a new tab. It's done those font color fixes, the stat card fixes,

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47 minutes, 22 seconds

the per habit reminders, custom challenge creation, and then it's also done some developer tool stuff for us, which is going to enable us to test the

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47 minutes, 29 seconds

app like I was talking about. So, just opening this back up in this window here. So, we could see we now have a better onboarding screen. So, here's how

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47 minutes, 37 seconds

it works. We create habits, add daily check-ins or counted goals, like drink eight glasses of water, take challenges, push yourself with three, seven, or 30-day streaks, track progress to your

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47 minutes, 45 seconds

streaks, consistency charts, and activity history. Stay on track, set personalized reminders for each habit at the time that works for you. I see it's implemented a skip button. I think it

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47 minutes, 53 seconds

did this as a developer just for me. So, what I'll do here is I'll say read uh walk and then I don't know, exercise,

48:00

48 minutes

let's say. Actually, why don't we do um healthy meal? That way, we test two different types. Now, we're going to start this 3-day challenge. Complete all

48:08

48 minutes, 8 seconds

your habits for 3 days to earn your first achievement. Ready? Start the challenge. And now, we actually have this 3-day kickstart just as we did before. I'm realizing that I don't think

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48 minutes, 16 seconds

you guys could actually hear my uh chime before, but rest assured, this thing is chiming. Uh it's actually very delightful. And then when it goes all done for today, we we still have that

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48 minutes, 24 seconds

confetti, which is nice. And I can trigger it over and over and over again cuz I'm addicted to that paper, baby.

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48 minutes, 30 seconds

All right, so going to history here. Um it looks like we minimized the streak, the best, the 30-day average now sort of exists all in one line, which is quite

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48 minutes, 38 seconds

nice. We also have obviously eliminated the font issues. And then we've uh implemented some brief tracking for per habit consistency. You can see here we

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48 minutes, 45 seconds

have 14% of the 7-day, which makes sense cuz we're done 17th. And there's a third 3% of 30-day as well. And this updates,

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48 minutes, 53 seconds

too, which is nice. Under manage, we still have those icons. I think we should probably add some functionality, not just to use these basic icons, but as you guys see, it's already better

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49 minutes, 1 second

than like a, you know, over the the counter basically habit tracker app, despite the fact that visually it might not be as interesting. And then I'm actually just going to write morning

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49 minutes, 9 seconds

run. Okay. And then you can see there's also the ability to create a counted habit with I don't know five, let's say.

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49 minutes, 17 seconds

Cool. Both of these are pills for some reason. We have the challenges that they're connected to. But now on the right hand side, we have this little chime, which is cool. So I'm going to go

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49 minutes, 25 seconds

chime and I'll say, "Hey, I want you to remind me every day at uh I don't know, let's just do 12 p.m. Let's say." Now, ideally, we'd have a little bit more

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49 minutes, 34 seconds

like feedback here to set uh that morning run. I'm noticing that we don't have that. So, I'm going to probably want to change that. Then down over here, we have simulate full challenge

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49 minutes, 42 seconds

completion, force complete challenge, reset onboarding, clear all data. So, this is basically like how we reset it from scratch. So, I'm just going to pretend that we're now done our 3-day

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49 minutes, 50 seconds

kickstart challenge. Okay, cool. And we can see we have the beautiful crown, uh, little trophy, and then 3-day kickstart, 1 days completed, three habits track,

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49 minutes, 58 seconds

continue. And now it jumps us over to the create a new challenge section. So, I don't know, new challenge. And then we

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50 minutes, 7 seconds

want these habits. We're going to start the challenge. We're now basically doing another one. If I go back to today, you can see the new challenge is sort of up at the top. All right, that looks pretty

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50 minutes, 15 seconds

reasonable. Um, I like this. I think I think there are just some weird issues with the um editing of that. Like some of the buttons and and check marks are

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50 minutes, 22 seconds

kind of odd. And then I want this to be a little bit clear. So, what I think I'm going to do is I'm just going to do our last major change, you know, once we're done with this last major change. Sorry.

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50 minutes, 31 seconds

I think that's the way the app's supposed to look. Um, I'm actually just going to like try try running it live then on my phone. I'll show you guys that and then after we're done with that, we'll uh actually push it out so

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that you know I can use it uh a little bit broader and then we'll consistently add functionality to it like databases and so on and so forth to turn it into something that you can actually get on

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50 minutes, 47 seconds

the app store. So, taking a look at the app a third time, the new challenge section at the top of the today page looks to be indented a little weirdly.

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50 minutes, 57 seconds

There may be a div behind it. There may be an element or something like a box, but I can't see it. the background color might be the same. So, just fix that.

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51 minutes, 5 seconds

Um, everything else on this page looks pretty good. I don't see any major issues. Oh, uh, some of the counted

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51 minutes, 14 seconds

habits look strange. For instance, healthy meal, which is uh one of the defaults, has a three out of three under

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51 minutes, 23 seconds

it, but because it's underneath it and the icon to the left is centered, uh it looks strange compared to the rest of

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51 minutes, 32 seconds

the habits. So, I'd like you to to fix that. Just have a layout where everything's just on one line.

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51 minutes, 38 seconds

History looks pretty good. I think the cards at the top of the page, the three for streak best and 7-day average, they're just a little tight vertically.

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51 minutes, 48 seconds

So, just add a little bit more vertical spacing between the emojis, the numbers, and then the U descriptor.

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51 minutes, 54 seconds

And then on the manage page, uh add five more icons. So, we should have one more row. And then for the placeholder text, morning run, capitalize the R.

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52 minutes, 6 seconds

And then for the reminders, right now there's no way to confirm the reminder.

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52 minutes, 12 seconds

I think the user won't know whether or not the reminders are set for specific times unless we have a confirmation button. It also looks like the

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52 minutes, 22 seconds

reminder div or section cuts off strangely. I think because the borders aren't rounded. Um just find a way to

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52 minutes, 30 seconds

combine all of that so that it's seamless and fluid.

Chapter 13: Finalizing the App Design and Functionality

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52 minutes, 33 seconds

And then I think the way that the new challenge edit section is right now is it's using

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52 minutes, 41 seconds

some weird checkboxes. It almost looks like it's an unfinished product or maybe the styles haven't applied. So just double double click on that when you have a chance. And I'd say that's

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52 minutes, 50 seconds

probably the 8020. So I'm going to give it another couple minutes and then circle back and then we can work on the next step. I made a couple of additional minor adjustments. It generated 17 icons

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52 minutes, 59 seconds

for instance instead of 15 and I just wanted five per row. Spilled kind of weirdly. and then hit another one up here where the set button didn't do

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53 minutes, 8 seconds

anything. But with all of that done now, I'm going to go back to the app and we can do a full run through basically by resetting the onboarding. Okay, so build

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53 minutes, 16 seconds

better habits, get started. You can see we have the how it works section here, the continue. I'm going to go exercise, drink water, and then walk. And then I'm

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53 minutes, 23 seconds

going to start that challenge. Just going to tap all of these so that they they hit the the numbers. So this is going to be eight. And that other two

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53 minutes, 31 seconds

are going to be done. We also have a nice little border around this new challenge now which looks a lot sexier.

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53 minutes, 36 seconds

Uh this is now a little more vertically spaced out which is nice. And then we have those habits which are correct.

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53 minutes, 41 seconds

Under manage here we can see we have a bunch of new icons like coffee and stuff like that. So I don't know maybe we want coffee. And then let me just test this

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53 minutes, 49 seconds

out. Looks like we have the reminder feature. Okay. Although the remind at is spilling over into two lines which is kind of annoying. So I may fix that. But anyway we could set whatever time we

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53 minutes, 58 seconds

want the reminder. And then when we set it, we can see that it's now been permanently fixed. It's now 5:00 a.m.

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basically. And it looks like the UX is if I tap it again, the reminder turns off. And then if I tap it again, it turns on. So this one's reminder. Uh

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54 minutes, 11 seconds

this one here, I don't know, maybe you want like a 9:00 a.m. reminder. For this one here, maybe you want like a 12.

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54 minutes, 16 seconds

Okay. So all of that is working functionally, which is quite nice. And then I'm just going to clear all data.

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We're going to start from scratch and see if there's any change whatsoever.

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That looks pretty good to me. And then I'm just going to go down here. Um, simulating the full challenge completion and then forcing the complete challenge.

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54 minutes, 34 seconds

Cool. Awesome. And as you can see here, they fleshed this out a little more.

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54 minutes, 38 seconds

This looked a little poor before, but now we have a 3day, 7-day, 14-day, 30-day, sleep 8 hours, healthy meal, and journal. Then we can name it something

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54 minutes, 45 seconds

and then go ASDF is our current challenge. So, that looks great, but it's still on the computer. And in order to really know whether or not this app

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works the way that I want it to work, what I'm going to do now is I'm going to run this on my phone. How do you do it on your phone? Well, we need to start one of those Expo servers and then actually going to, you know, do it on my

Chapter 14: Testing the App on Your Phone

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phone and then I'm going to like pay attention to haptic feedback and and stuff like that. So, what I'm going to say is start an expo server for this in

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a new terminal window. And it's important to say new terminal window because if it does so locally here, you'll find that you can't actually take

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55 minutes, 16 seconds

the QR code um sort of uh flow. So, what I'm going to do here, it looks like it misunderstood me here. or when I say

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expo server, when I say expose server, I mean to run it on my phone um in a terminal so I could see the QR code. And

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55 minutes, 32 seconds

once I have that, just going to open up my camera here just to get it all nice and ready. Okay. And you can see it's right over here. So I'm just going to make this way bigger.

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And this is the same thing that we had earlier. So I can now just open it in Expo on my telephone.

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and then it'll go through that whole build process. Now, what you'll find is it's different to run something locally on your computer versus here. And so,

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56 minutes

what's just occurred, just to show you guys here, be 100% uh real is there's a line down at the bottom that basically

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just says uncaught in promise ID. And so, basically what that means is there's some issue with the server. And you can see when I tried reaching it, um we ran into a problem here as well where

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there's some sort of issue with the get item or something like that. When this occurs, just copy over all of the error

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56 minutes, 22 seconds

messages. Okay, paste it into clot. So, I just did that. And now you're seeing it's saying the async storage v3 isn't compatible with expo go. We're going to

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56 minutes, 31 seconds

download v2.2.0 or something like that. Okay. So, what I'm going to do now is I'm going to go back to our terminal window. Okay. And I'm just going to exit out of it. On a

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56 minutes, 39 seconds

Mac, you do so by holding control and pressing C. That's say exit. And then what you can do is you can always just like go up to see the last terminal command. So, I'm actually just going to

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56 minutes, 47 seconds

clear everything and then I'll just go up to do the last terminal command. This should rerun uh the app for me. Okay.

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56 minutes, 53 seconds

Okay. So, now what I'm going to do is I'm going to open up my camera again and then I'll do the same thing. So, now

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57 minutes, 1 second

opening it in Expo Go. And as you can see here, there's no there's no errors um you know in the log. I actually have like the build better habits right here.

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57 minutes, 9 seconds

So, I'm just doing it on my phone. I'm going to open up the iPhone mirroring app, which um I believe I mentioned is unfortunately quite small. So, you'll

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have to deal with said iPhone mirroring app being small, just kind of how it is.

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57 minutes, 21 seconds

But, as you can see, we actually have the app running locally. Um, which is quite nice. So, continue. And this is the same thing that we had before. It just looks like some of these icons are

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57 minutes, 29 seconds

a little bit cut off at the head. So, kind of odd. I'm just going to make a note of that here before I forget. Let me just say great stuff. But the icons

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57 minutes, 36 seconds

are cut off around halfway through vertically. It's like the heads are cut off or something. Okay, I'm

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57 minutes, 44 seconds

going to go back here, click continue, and yeah, we see the same issue with all of the icons. So, it's probably just a persistent problem. It also looks like there's not a lot of visual space up

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57 minutes, 52 seconds

top. For those of you guys that can't see, the uh time is basically immediately overlaying the the date.

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57 minutes, 59 seconds

Also, there's no vertical space. Um, add some sort of padding up at the top to accommodate for the fact that the iPhone has its own little bar.

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58 minutes, 11 seconds

uh the black what's this one called? The black middle window and then the um battery time etc.

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58 minutes, 21 seconds

And I'm just going to keep all that here as I continue going through the app.

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58 minutes, 24 seconds

Awesome. We're getting some great notifications and like awesome sounds as this thing comes in, which is beautiful. Ooh, and I really like that animation.

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58 minutes, 31 seconds

That's nice. Then we're going to the history. We can see the dates and times are all good. And basically what we're doing is we're just retesting this. It's just redoing it on my phone.

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58 minutes, 41 seconds

Cool. So, we have an example habit. And now, this is really cool. This didn't occur locally, obviously, because we didn't have push. But, um, now you can see it's saying, "Xigo would like to

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58 minutes, 49 seconds

send you notice." So, we're not actually going to get the notification prompt until we click that little button, which is awesome. Uh, which means it's going to be lower friction on the user POV.

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58 minutes, 59 seconds

And then, you know, naturally, most people are going to want to set some sort of reminder at some point. So, we'll allow it now. And now we can go through and say, hey, I want to set it

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59 minutes, 7 seconds

for 12, which is pretty great. Okay. And uh yeah, now I'm going to say force complete challenge. Just make sure that works. That's great. One day completed,

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59 minutes, 14 seconds

three habits tracked. Awesome. And now we could set sort of our own great new challenge, healthy meal, and

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59 minutes, 22 seconds

then start challenge. Awesome. And the final test through run is it's one thing to do it here like in front of you guys.

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59 minutes, 29 seconds

It's another thing to do it on the phone itself. And that's important because if you think about it, the way that you use a phone, it's a little bit different from the way that you use a computer. On

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59 minutes, 37 seconds

a computer, you have a mouse and so the mouse can sort of equally touch everything on the page. But me as a user, I'm going to be mostly using my thumb. So, it's actually important that

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59 minutes, 45 seconds

most of the major functionality is sort of within thumb's reach. There are other minor UX things as well to check like you want to you want to make sure that the thing scrolls the way that you

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59 minutes, 53 seconds

expect it to. uh you know there's additional functionality like long press on things and uh you know this is stuff that like you can't just get if you're

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1 hour, 1 second

doing things entirely through a code on your computer. You sort of have to play around with it a bit. So I'm just going to play around with the app a bit and I'll make any additional minor changes I

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1 hour, 8 seconds

have to. I really like this. I would say this is the 8020 and it's already significantly better habit tracking app than most of what is currently on the market. And it's pretty funny because I

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1 hour, 17 seconds

think I managed to do all this for maybe 50% of my cla uh for this uh session which is like every couple of hours. So, that's pretty crazy, right? It's a $20 a

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month subscription. I was able to build a better app than 90% of what is currently available on the app store and I was able to do it, you know, entirely myself. Just going to close out of this.

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1 hour, 34 seconds

I'm going to push these changes, do another double check, and then we're done with our first version of the app.

Chapter 15: App Development Insights

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Okay, once you've gotten the app to a place where you like it, um, there are a couple things that you should do. The first thing you should do is you should update that cloud.md again. And the

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reason why is cuz this app is no longer just like that that beta version. And this is, you know, like a full-fledged MVP. It has a lot of functionality in it. And the next time you change things

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1 hour, 56 seconds

or edit it, this cloud MD, which provides all of like the base information and is immediately loaded into every conversation that you have with Claude, um, should describe like all of the the additional functionality.

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The second thing you should do is you should push this to GitHub. For those of you guys that don't know, GitHub is a version control platform, which is basically a fancy way of saying it's

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like a place for you to store your code on the internet. And the reason why it's valuable, at least for app developers, which I guess you are now attempting to

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become a part of said class, um, is because it doesn't just store the code, it also stores any changes that you've made to the code. And turns out that's pretty valuable because when you have

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AI, you know, doing whatever the heck it wants in your codebase, if you don't have some sort of like accountable way to revert back to the previous instance, let's say you have an awesome thing

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going, you make a couple changes, the app breaks, and you're like, "What the hell? How do I fix this?" If you don't have an easy way just to roll that back and then share those rolled back updates

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with other developers in the team, you know, you can you can wind up in in quite a pickle. And so if you don't already have a G GitHub account, it's pretty easy to do. All you need to do is

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just enter your email address right over here. So I'm going to assume that you've done so. You've signed up. They might ask for your phone number, maybe your email address, uh double verification or

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something like that. They tend to be pretty secure. And then once you have it, um you can literally just go right over here and then just say, "Great, create a GitHub repo for this."

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and uh as well as a readme then push. Now, if you're not logged in to GitHub here, it'll ask you to

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basically do authentication and then connect. That'll open up a little page for you and then it'll like, you know, have you relog in again. Um once you're

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done, what this will do is this will just push this to that that database online, that sort of version controlled resource. Okay, so it's doing it right

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over here. Uh and you can see it actually called it example project, and I'm realizing I don't actually want it to be called example project. I just created that for you guys. Um, rename it

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to habit tracker and uh, rename this folder as well. And

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that way uh, you know, we'll actually know what the heck's going on. Always important to name things. Once done, you'll have the same thing um, inside of the uh, a folder on the lefth hand side.

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And if you don't like this example project, like I don't, I can already tell that's starting to get annoying.

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I'm just going to open folder and I'll just go directly to habit tracker and then don't save. And now you can see this uh window is opened. It will have

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reverted our session though. So just keep that in mind. You won't be in the same session. Then if you go to

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um your GitHub, I'm just going to go habit tracker. See if I could find it. I see one that's capitalized here. That's probably it. Yep, that's one from 2 minutes ago. You'll actually see all the

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code written on the page. So just making this really clear to you. Um, Claude always likes adding itself as uh, you know, one of the the collaborators on any project now, which I find hilarious.

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So, it's two people and Nick Sarah. So, it's GitHub, it's Claude, and it's Nick.

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Nick's doing the heavy lifting, though, trust me. Um, and scrolling down, you can see that 2 minutes ago, we actually added all these these changes alongside our habit tracker app. So, it's still

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calling an example project, which is kind of annoying. You can click here and then um, edit that at any point in time.

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You know, a couple additional prompts and it probably would have done so just fine. So, I'm just going to click commit. And by doing so, what we have now is we have like the updated version

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of this app. And uh it actually just stored that change directly to this file readme. And you know, don't worry too much about all the tiny little specifics here like the description of the change, the time of the change, specific files.

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But just know that you basically now have this online. The value there is you can share it with your team. And then as mentioned, if there are any issues, you can roll it back at any point in time,

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which is quite cool. Okay, so what do we have now? We have an app that is running and capable of running locally on my

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computer. It's also capable of running on my phone using Expo. Okay, we can't actually run it on our phone like on the

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app store yet until we push it out and actually get it running. So, just keep that in mind. Now, the issue with this is all of the data is currently stored on our phones and our computers. So,

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when I run it on my computer, for instance, as you guys saw a moment ago, um you know, it's like storing that data directly on my computer. If I sent the app to somebody else, it wouldn't be able to use that app.

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um on my phone, you know, it's running locally on my phone in so far that all the data that it's creating, the preferences of the apps and the habits and stuff like that that I'm generating, they're actually just stored on my phone

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in in like a little memory feature. We have to do if we really want to truly like internationalize and and and allow our app to be used by a bunch of other people and then also allow like

Chapter 16: Transitioning to Cloud Storage

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persistent storage that doesn't depend on the device so that if I download it on my phone and on my computer, we're both going to be like pulling the same data is I need some sort of database.

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And a database, for anybody unaware, is just very similar to your hard drive. It's just being hosted elsewhere.

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Because it's being hosted elsewhere, you need a way to talk to it. And so that's where languages like SQL and a bunch of other programming jargon and stuff like that come in handy. What's really cool

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is with AI and a couple of simple solutions, you can now communicate with these databases and set everything up almost completely autonomously. back in

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the day when I was starting uh developing apps and I developed my first one for um um 1 second copy which is a company that I ran we scaled to over

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90,000 bucks a month um basically AI generating articles for people designing the simplest thing on planet earth much simpler than what I just showed you here

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took me like 3 weeks because a good two and a half of it was me just fiddling with the database setting up my SQL schema and all that now we can do so in

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literally like 30 seconds and that's what I'm going to show you guys next taking this habit tracker and then adding uh database functionality to the back end. And once we're done with that,

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I'll weave in a little bit of AI functionality just to really complete the circle. And then we'll actually push this puppy out to the app store as well as go over some other apps that you

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could build that are increasingly exciting. All right, so now that we have the app working, right, we verified the core functionality. It's time to take it

Chapter 17: Database Integration with Supabase

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from local storage, aka storing all of the data on the same device that we are building on to cloud storage, which in

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our case is going to be storing the data on a in a database. And the database that we're going to use in particular for this app is called Superbase. Now, I

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want you guys to know that there's like a million different databases you can use. And my stack here is rather opinionated. Um, I just use the stack because I have built many apps on the

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stack myself. And typically what goes on when you end up building a lot of apps is you just kind of arrive at one specific stack or set of tools that works pretty well out of the box and

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then you just stick with it. And then it's a self-fulfilling prophecy, right?

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It's not necessarily that it's the best stack ever, but you just know how to use it. So to you it's easier. And then you get, you know, you carve out the little rivers and paths of uh uh the stack over

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and over and over and over again. And then you end up just not really wanting to use anything else. So feel free to apply the same idea here that I'm about to show you guys to like databases in

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general. You don't have to stick with the one that I'm going to show you guys how to use, but uh I really like the one that I'm showing you how to use and a lot of other vibe coders and app developers and stuff like that nowadays

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make use of it as well. Okay, so just zooming in a bit here. Um the specific platform I'm going to use is called Superbase. Now, instead of storing everything, you know, inside of the

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local storage or whatever. When you use a database, there's sort of two things that go on. And I don't know where I put my pen. So, give me a sec. I'm going to go find it. Okay. I have no idea why

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that was in my kitchen, but it was. Uh, so basically what we have right now, if you think about it logically, is we have um, you know, sort of our app over here

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on the left hand side. And basically what the app does at the moment is it communicates, okay, with whatever the device is. And so in our case, this

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device, you know, if I'm running it locally here on my computer, it'll be on my computer. Um, or, you know, if I'm running it on my phone, it'll be on my

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phone. And basically, just like, you know, you have a little hard disk with like memory and stuff like that or a little USB key or something. All of this data is partitioned and then you can you

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can send and requests and the computer keeps track of which area that the the data is and you know, it's sort of seamless. However, this is just local storage. What we need to do now is we

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need to go one step further. And basically what's going to happen is our app, okay, is going to send and receive data to our device. And then our device is also going to send and receive data

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to a database, which uh in my case I'm just going to call cloud storage. And this is going to be like our our DB. And

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our DB here is going to be called superbase.

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Now the issue with a structure like this is the second that you add that additional layer, you need to tie the

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specific user of the app with a specific table or a specific um section of the database. And so it's not enough just to

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add a database on its own. What I mean by this is you can't just add the database and then have everything work the same. You also need to add some sort of user functionality. So the database

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needs to be able to pull records for a specific user. Like they need to know that it's Nick's data for instance versus another user's data. And so actually what we're going to do on top

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of our app is we're also going to add um basically some sort of like login functionality and there's going to be like a little screen here which has like you know username and password. Um and

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then what we can do is you know our app we can use to login. That login data is then going to go to our database pull and basically verify that like okay that

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user is the right user. Um and then we can use the app as usual shuttling data to and from our device and then ultimately our our cloud storage for you

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know updates and stuff like that. And a structure like this is typically the most efficient simply because you get to take advantage of like local ondevice caching which allows your data to load

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instantly. Um but then also uh you know like updating from the database periodically. And so this is more or less what it is that uh we're going to build. And the good news is you don't

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actually have to know anything about databases or authentication or anything like that anymore. You could just have Claude do most of it for you assuming that you're using a provider that

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handles it all like Superbase. In my case, Superbase handles both the a and the login. So, what I'm going to say is I'd like to add a database to this

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project. And uh I'm just going to use my voice transcription tool. In addition to a database, I also want local caching so the user has very immediate and snappy

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experience when they use the app. Uh we're going to be using Superbase as our app. And in addition, we're going to need to set up user authentication so that you know which user is accessing

Chapter 18: Setting Up User Authentication

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which data. Help me through this process.

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And as you can see, you can just ask the model for help. It's not that big of a deal. Um, but basically what it's going to do now is it's going to change our database which is likely currently

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stored somewhere I imagine inside of our apps folder or maybe uh I don't know one of these folders here. Uh, and then it's going to replace that specific thing

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with like an API call which is basically a request to uh a service outside of our app. And because we're now extending beyond the bounds of our initial device

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uh you know the the the system needs to look a little bit different as well. We also need some uh cool features like security features to ensure uh that you

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know other people don't use my username let's say to log in and access my data unless that's something that you know I uh I specifically allow them to do you

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know bake it in or give them my my credentials and so we don't need to worry too much about knowing sort of all the different ways that you can do things um just ask claude for help

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review it make sure it makes sense for what it is that you're trying to do assuming that it does you can move forward here's what claude just told me how to do the proposed architecture of

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superbase for o which is email password and oath Those of you guys that don't know, OOTH just means being able to log in with a social provider like let's say Google. So I could do a one-click sign

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in with Google and then I could sign into my app that'll identify and authenticate me. Now then it's going to say Postgres database for habits, completions, and challenges. That's just

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built into Superbase. You don't need to worry about it. We have local first caching, which is what I was alluding to. So we're going to keep async storage as the immediate readr layer so the app

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feels instant. The idea is we want to be you want it to be really snappy. If you have to like write read and write to a database every single time you make a change, usually there's like a little bit of a lag there. It might be a few

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seconds, could be, you know, just milliseconds, but it's still like a little bit of a delayed experience for the user. And I also don't want to add any load time initially. So what's going

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to happen is we're going to cache all the data locally. So we're still going to have that layer. We're just going to add the database and periodically sync.

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Okay. So then we're going to replace the current onboarding only gate screen with a superbase authentication screen that involves sign up and sign in. And then we're going to onboard uh for new users

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as well. Now, it's also asking for preference on O methods. Email and password is simplest to start, but Superbase also supports Google, Apple,

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GitHub, OOTH out of the box. Then, it also uh is going to run me through the the process of installing Superbase. So, I'm going to say I don't yet have Superbase set up, but I'll do that now.

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Let's go with email and password. Work on everything that you can uh up until that point. Cool. And what's really interesting here is, you know, I'm

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acknowledging that I don't have the step that it asked me to have. Instead, I'm saying I'm going to do that as you do the rest of your work. And so, I'm just parallelizing. I'm batching a little

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bit. So, how do we actually go about this process? Uh, it's pretty straightforward. So, I'm just going to open up my little habit tracker app here. And then, I'm going to go Superbase. Okay, let me just make this

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big. And as you can see, Superbase here is, uh, just like any other app, you little login screen and stuff like that.

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In my case, I'm already logged in. Now, what you have to do if you want to log in is just, you know, create an account.

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It's totally free. You can then start a project. Um, after which you'll have a page that looks like this. So, I'm going to click the start a project page. Okay.

Chapter 19: Implementing AI Features

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And I just deleted all my projects here so you guys could see them for yourself.

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I think what I'll do is let's go to Nick J. Wellsorg. And here you can set a project name, which I'm just going to call habit tracker app. And then you can

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also do a database uh password right over here. So I'm just going to add my own little database password. And uh it looks like it's saying my password must be harder to guess.

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So there you go. Just going to make it a little bit longer. Okay. Okay, so now that we have the database password and then the region, uh, we can enable the

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data API automatically expose new tables. And then there's one more thing you have to do down over here where it says enable automatic RLS. Just check that button. That's pretty important.

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And then you just give it a click just like that. And now uh it's basically going to go through the process of creating the database for us. And once it creates the database, we'll have a bunch of information that we can send

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back to Claude. So that's what we're doing right now. I'm just going to save this. And it looks like it's setting it up somewhere in Oregon, which is nice.

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the location of your database is is pretty important uh because the further away it is from where users are going to be using your app, the uh more lag uh

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involved. Okay, great. So, as we see here, it's actually already doing the work in the back end, which is quite nice. So, it's installing dependencies and stuff like that. All we really have

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to do here is just wait for this to finish and then ask us for specific things from Superbase and then uh we can actually go through and like give it like our password and our API key and stuff like that. Okay, we're 3 minutes in now. The status has changed slightly.

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It's basically going through all of the database steps. You guys can't see all of them because of my fat head. So, let me make some more room. Um, the database

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is healthy. Postg rest is healthy. Off is healthy. Realtime is healthy. Storage is healthy. And then finally, the edge functions are healthy. No, you don't need to know what any of this stuff

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means. It's all good. Uh, but basically, they're just telling us that the database is now set up, and we can actually proceed with the rest of the steps. You'll see the first thing it's doing is it's giving us the ability to

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copy the project URL, publishable key, direct connection string, then some CLI setup commands. If you don't know what any of this stuff is, uh, Claude can walk you through it. But basically,

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these are all just things that are required for it to modify or manage your Superbase for you without you actually having to like click in at all these

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tables. So, going back to Claude Code, you can see that it's now saying that it's it's done. It's given us all this information and then your next steps are

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to create a superbase project at whatever. Um, so, okay, I have the Superbase project ready to go.

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Uh, I'm just going to paste in all this information. So I'm going to copy the project URL. Then I'm going to copy the publishable key and I'm also going to

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copy the direct connection string. Then finally I'll copy the CLI setup commands as well. So this is giving it literally everything that it needs in order to

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manage as much of this process on its own as it can. And the reason why is because um you know as mentioned cloud a lot of the time is trained off of old

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data where you know a human being had to actually manually do things like run the schema. A lot of the time you can just give it full access to superbase to have it do it for you. um not to mention like

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other platforms as well. So that's what we're doing here. We're basically saying, "Hey man, here's all of the stuff you need in order to do it for me." And now it's going through and what

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it'll do is it'll set up that schema um using, you know, this SQL editor. And I'll show you guys what that looks like after. But basically, this is just how you set up the database. This is how you

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set up the rows, the columns, the headers. It's how you know that you're going to have like a field called data type, let's say, and that's going to be

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like habit, uh data type name or whatever, and that's going to be like, I don't know, morning meditation or something.

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Okay, so going back over here, it's saying that we need to do a Superbase login step interactively. Okay, so I'm going to say run login script all

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authenticate. Now, if it gives you an error like it just did to me, basically saying you can't do it within your non-TY environment. That's just a byproduct of the current terminal type

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that we have, you can also just open up a new terminal like I did here and then just run that same command, superbase login. What it's going to do is it's

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going to give you a login link. So then you can copy that and then what you have is you have a little copy paste window.

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So you can copy it, go back to the terminal, just paste it in. So as I just did now we've authenticated. We're basically in uh Superbase. So now I can

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go back here and I can say okay, logged in. And now because we share access across the entire um basically across

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all of the terminal instances of my computer, we can now download and kind of proceed with that. So now it's installing it and now it's applying the

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schema. For those of you guys that are curious, what does the schema actually look like? Well, the schema is just like the the the the the set of tables uh in

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our database. So, you can see here that we now basically have three. We have challenges. Challenges contain an ID and a user ID, a name, habit ID, start date,

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duration date is complete, reward claimed. Okay. And then if we go down, we also have completions. Then we have habits. And actually looks like we have a fourth, which is profiles. Profiles is

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where we're going to store some of that user data. So what's really interesting uh just historically speaking is like you used to have to come up with all this stuff and uh you used to also have

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like really tight associations between things like the user ids uh in each table so that you knew that the habits that belong to a specific user ID um you

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know connected to the specific completions and challenges and you used to have to run all this like almost math every single time that you set up one of these apps. Uh this just does it

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all basically entirely automatically for you just based off of what is logical.

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Like it's logical for instance that you know your habits and your challenges need a name. Uh you know it's logical that they need an emoji because that's just how our app's built. Logical that

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they might have a type between two or three different types, right? These are all things that just like standardize and operationalize the data within our app. It's just now this is the back end

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and uh you know we're doing this on like a a cloud database as opposed to locally. So now that we have all this stuff, all we have to do is just disable email confirmation. So I'm going to go

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to the superbase dashboard and uh right over here it'll basically say confirm email and I'll say no. And the reason why we're doing this is because if not,

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the users have to go through an additional step in order to log in, which is like they'll create an account and then they'll say, "Okay, go confirm your email." And then we'll have to go

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back and forth that. So, this is just eliminating the step to make it a lot easier for us. Okay, great. Let's test the full flow. And with all that said,

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let me show you guys what it actually looks like now. Now, I'm going to run this locally on my computer and then I'm going to do the app stuff that we were doing earlier where, you know, I test it

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via Expo and then ultimately I test it on my phone. And you can see it's finding a couple of crashes before it launches. This sort of thing is pretty normal. It'll selfheal and self anneal

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as needed. So we'll just run this a couple of times until we get the onboarding.

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Okay. And now if I make this smaller so it's more like a mobile app. And then zoom in. I think 150 is probably pretty good. Just do that. Make it a little bit bigger.

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You can now see that we have the whole signup flow. So I'm going to go sign up down at the bottom. Just delete this other one here. And then I'm just going to add in my email address.

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And then we'll just add in some silly password and click sign up. So what happens after we sign up? Well, obviously we need to go to onboarding,

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right? So now we're saying build better habits. So I'm going to get started. And as you can see, we've now basically just wrapped the app in a login flow. It's the same core functionality I had

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before. It's just now it's like wrapped where in order to enter you have a login flow and then the back end you have the database. So I'm going to click continue and I'm going to say I want to read, I

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want to walk, I want to do journals. And then I'm going to continue start challenge. Uh, and now maybe I tap each of these three, you know, we have this cool little thing. We have our history.

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The history is actually being pulled from our database now. And then we can also manage it. I want to add something new. Okay. And then maybe uh I don't know, I want to set a set a reminder

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time like I did before. Okay. So, obviously we can't see sort of the way that the backend works here because we're just doing this locally in our computer, right? If you actually want to

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see the way that this is, you have to go to uh the database and then you can actually see like specific tables. So, let me show you what that looks like.

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Okay. The first thing is if you go to authentication on the lefth hand side and then click on users, you can actually see the uh data that we just created a moment ago. So I just created

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an account using my email address. You can see that we were just supplied a UID or a user ID. There are a couple of additional fields here which just

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creates for you like phone provider type uh and stuff like that. And then also under provider there's email. Uh as talked about you guys can also sign in using Google and stuff like that. Just

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ask cloud to do that for you. You can also add a user locally here. So, this is sort of like your admin, you know, dashboard, at least for now. Um, we can

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actually just set up a new user. Let's say you have somebody being like, "Hey, I'm having an issue creating an account.

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Can you create an account for me?" You can actually do it entirely. You just create a new user. You put their email address and password, autoconfirm them, and then just give that information to them. They'll be able to log in. So, all

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the stuff that we just did in our app, like it is actually showing up here on a website essentially. Um, sort of verifying the fact that our app is now connected, but it's not just connected

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to our website, right? As I showed you guys earlier, I don't know why I'm not using my pen for this. Um, we're also communicating locally with our uh, you

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know, with our device. So, that's why I'm not using it cuz my arrows are so much more uh, uh, janky. So, you know,

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this is our app. This is our device as mentioned. And now our device is sort of communicating with the DB, which in our case is going to be superbase. We've

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just verified basically that the app can create sort of like a little off page.

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Okay, this information can then trickle back to Superbase and then we can, you know, have a couple of rows or whatever with like different different uh users.

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What we have to verify now is basically how about all the data within that user and that's pretty easy to see as well.

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All you need to do is go up here to where it says uh table editor and then you can actually see the individual or independent table. So challenges, completions, habits, profiles. Probably

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the simplest one for you guys to see is habits because we just created some habits. And you can see that for a specific user ID, okay, which is 3F56A

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whatever. If I click this, you could see that this is actually uh Nicholas Sarrive. Okay, it's me. For that specific user, they have four things.

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They have a a habit called read, another habit called walk, another habit called journal, another habit called ASDF. And if I go back to my app, you can see this is actually like live, right? This is

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100% live. It's like being tracked right now. Okay, you can see the reminder hour for one of these is set as 12. What do I mean by that? I can go to manage. I can

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scroll down. And I can see that information is reflected inside of my database. Right. Um I can also just remove this. Maybe delete it. Okay. Then

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I can go back here and I can give this a quick little flicker. And you can see now what happened. We went from four records in our database to three. Why?

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Because we just deleted. And when we delete it, there's a line of code basically that cloud set up where it's like, hey, when you delete it, I want you to delete it from the database as well. So how neat is that? You can also,

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you know, check to see what all of these do. So reminder enabled, false, false, false. So maybe I want to go true, true, true. Okay, we're I'm actually just

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going to set all these. I'm going to refresh. If I go down to this reminder enabled now, just to really prove it to you guys, you can see that they're all

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true. And then I can change the hour. It looks like we also have the ability to change a minute, which is kind of neat.

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U profiles here. This is basically all of the additional information for the user. So in this case, they just have onboarding complete, and that's additional information aside from my

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email address and stuff like that. You can imagine how you can make this as complicated as you want. Completion. So this is basically every single time I've completed a habit. It stores the exact

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date, the count, and the habit ID. So you know, in 3 months from now, I'll be able to look back inside of my uh database, and I'll see, hey, you know, I've actually finished three habits on

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this day. What are the IDs of those habits? Well, it's 1 177817. And then the app can basically go and it can look for that specific ID and say, okay, so

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he did three of read, then walk, then journal, which is pretty neat. So, this isn't a comprehensive database walkthrough by any means, but I want you guys to know that more or less all

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databases work in the same way nowadays, or at least like the SQL databases, which is the specific type that we're using and the one that I'd recommend for most apps. Um, you're basically just

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going to have, you know, the database, which you're going to set up using this little um what's called a schema editor, which AI will do for you. It'll set up multiple of these like data structures.

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So, habits, completions, challenges, these are called tables. And then what they'll do is they'll associate that to different user ids using this

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authentication um page which is where you you know have like the the specific users that signed up and you can manage all this stuff using um you know like in in our case superbase but usually no

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matter what app you're using or what sort of database you're using there's going to be some sort of interface where you can do things like you know create new users on the fly and so on and so forth. Now it's not enough just to do

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this and test it obviously on my computer. I also have to test it on expo and then I want to test it on mobile as well. And the reason why is because I want to verify that the stuff that I

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just showed you that you know me creating a habit or signing up with a new user or whatever reflects not just through like the little Chrome tab but

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also through you know my my my phone as well as you know that Expo server. And so what I'm going to do offscreen here is I'm going to go through another two levels of testing. I'm going to make

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sure that it works on my computer the exact same way it works on Expo and then ultimately like on mobile because I want to see if there are any weird sort of interactions, if the haptics are off, if

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the icons get cut off halfway through and stuff like that. Unfortunately, this testing loop is mandatory and you're just going to have to do this every time you design an app. There are some

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promising approaches where Claude can actually test things for you on the computer. Um, but I've yet to find one that is automated that also allows it to test it on your phone. So, until we get

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to that point, uh, you know, you're going to have to do some of that testing manually. And I'd always allocate an additional 10 or 15 minutes. Anytime you do encounter an error, you know, I'd

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recommend just voice transcribing what the error is, speaking in plain natural language to clot, and it'll probably be able to do a good job. But that's what I'm going to do now. I'm going to go through the process of setting up that

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Expo server like I talked about and then I'm also editing it on my phone. All right, so I've bug handled, I've fixed everything, and I've verified that it's

Chapter 20: Enhancing User Experience

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good to go. I mean, we could publish our app right now, but as promised, I want to take you guys through higher and higher levels of functionality through this course. So, before we finish up and

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wrap up with this habit tracker, um, I actually want to add two additional features. And these are both features which are going to rely on third-party

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services or APIs, specifically AI services or APIs uh to basically do things for our users in the background.

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And I did some reflection on it and there are two here that I really want.

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The first is I want some form of smart coaching or nudges. And you guys may see stuff like this get rolled out to more apps over the coming weeks and months.

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I've seen this on like my Whoop band app for instance. I've seen smart coaching in a couple of other places. But basically what I want is I want

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periodically to analyze the streak and consistency data in our database to send personalized motivational messages or

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suggest when to adjust their goals. So, an example might be, you know, once every day or week, whatever the cadence

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is that we figure out with Claude, uh, you know, we're going to analyze the back end and then more or less push a notification. And the notification might

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say, "Hey, you've nailed sleep for 14 days. Congrats, but we're noticing that water intake is low. Here's a quick hack that you could use, let's say, to

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improve your water intake." Okay, so this smart coaching and sort of nudge feature is sort of one of the two I want to implement. And then there's one other

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one as well uh called reflection summaries which uh basically on a weekly maybe monthly basis I want to AI

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generate reports that go into detail uh basically giving people significantly more context about how far they uh they've gone. So for instance you're the

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most consistent with meditation at 92% but exercise dropped off midweek. Uh you know you scored 19% out of your goal of 33% on X Y and Z and so on and so forth.

Chapter 21: Smart Coaching Implementation

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Okay. So anyway, whatever the features are aren't super important. I'm just going to show you guys how how you can implement them now. And it's fairly straightforward. Um, basically so that

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you know right off the bat, you know what we're doing here is we're introducing another service aka another layer to things. And so in the in the case of our AI coaching architecture,

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what we're going to be doing is we're going to have our mobile app which is sort of over here. Okay. And uh again you know I just I want you to to understand that this mobile app so I can

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make my pen work for once is actually connected to you know in this case our phone you know pay don't pay attention

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to that specific diagram but basically this application is running on our phone okay and uh basically our phone you know is going to load up the app and when the

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app let's say wants to do this I don't know maybe daily or weekly depends on whatever claude thinks makes more sense for a database architecture which I'll

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remind you guys to in a second. Um, you know, on a daily or weekly cadence, our mobile app is going to fire off a a function call, basically a request to a

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service. And this is going to be managed by a tool called Superbase Edge Functions, which are basically simple uh

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serverless backend tools that allow us to on whatever schedule we want just fire something really quickly and

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extemporaneously like very very um transiently uh over to another service.

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And so this is sort of our back end here. This is our catch hall. And the alternative to this is you setting up an always on server which is just constantly waiting you know 24 hours of

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the day. What this is this is a server that basically turns on very briefly for like 1 second fires our function and then turns off. And the reason why uh

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you know this this edge function and sort of serverless function exists is simply because it's more compute efficient and then it saves us a lot of money. Like you wouldn't be able to run

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this anywhere near as cheaply if you had a server that was always on. Okay. At least for our level of scale. this is um the way we're going to do it. So we're going to use these superbase edge functions and then basically what

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happens is okay you know like um you know it's on our phone the mobile app is initialized everything set up and then daily or weekly we'll fire off one of these superbase edge functions okay

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these functions are first just going to read our database so it's going to read in our case maybe the habits the completions the streaks compile a big package okay and then we're going to

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send it to an AI and so there's sort of a two-way graph here right this is going to send a request this is going to get all of the data so the habit data

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completion data streak data send it back over here we're It's going to like dump all that stuff to an AI model, in our case, cloud API. And we're going to do so alongside some sort of, I don't know,

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prompt. And the prompt is going to be really simple and really straightforward. I'm basically going to say like, hey, here is an example of a a

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a coaching message. Your job is to come up with a bunch of coaching messages, go through the user's data and stuff like that, and then do so. Understand that this is going to occur on like a daily, weekly, or monthly basis, whatever.

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Then, um, you know, what's going to happen is we're going to return that, okay, right over here. We're then going to dump that in um our database except now this is going to be like I don't

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know some sort of like coach message feature. After that that is going to come back to our superbase edge function which is going to deliver it to our

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mobile app. And then finally our mobile app is going to send um in essence some form of like push notification as well as maybe some persistent notification um

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in you know on our phone. So that is in general sort of like what the highle architecture of this is going to look like. And I just want you guys to know this is just more or less how all tools

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and apps and services that you integrate within an uh you know like a phone app actually work like you're always going to have like these core features. You're

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going to have the app, you're going to have the the server, okay? You're going to have the database and then you're going to have whatever API it is that you're reaching out to. In our case, we're just reaching out to claude API.

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But uh you know, as you'll see as we develop, you know, our calorie tracker app after this one and and you know, other additional functionality, this is just going to be the same loop over and

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over and over again. starts with phone, goes like this, goes back and then basically just continu continuously goes back and forth like this. So don't worry

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too much about the specifics and sort of the technical features. Just understand that this is the flow uh of you know in our case our AI coaching architecture

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app and then also the uh the reflection summary stuff. So where do we go from here? Uh it's actually fairly straightforward. I'm just going to um

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actually feed in both of these to AI requesting that they do them and I'm even going to feed in a little screenshot of the AI coaching architecture. But I want you to know Claude probably would have come up with

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basically the same thing just cuz that's more or less how all of them work. So let's head over here to our anti-gravity and then I'm just going to paste this in. So we'll say smart coaching and

Chapter 22: Finalizing App Security Measures

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nudges. We'll also say reflection summaries.

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Then I'm just going to use a voice transcript tool to say I'd like to implement both of these features into our application. I want you to use Claude as the backend and then send the

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request via Superbase. Um use pretty smart models. Let's use the sonnet models for Claude. Um, and then you'll

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also have to update the database and do everything like that to ensure that it works alongside the the flow diagram that I'm attaching.

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Okay. And I'm just going to zoom in so you guys could see it a little bit better. And then just because I don't want to give it too much text and context, I'm just going to zoom out a

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bit. And then I'll just screenshot. And you can feed screenshots in. Um, one thing that we'll do with the next app that we create is I'm going to show you

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guys how you can take screenshots of another app and use that to inspire your design. Um, so we're not going to duplicate or copy the design of an an

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app, but we can definitely use, I don't know, the Apple design schema or like the Microsoft design schema or something just as inspiration to get like the

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padding, the corner radius, the little rounded circles, the outlines, all that stuff that like makes a design a design

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um in at least initially and from there we'll be able to customize and so on and so forth. So, uh, images and screenshots and stuff like that are a very important part of any sort of design loop nowadays

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with AI. And you guys will see how to do that on the next app. What it's doing now is loading the superbase skill and then checking um the docs for edge

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functions. So, edge functions are very particular type of tool. Claude is going to have to actually read up on what the tool is and how it works before it implements it. For those of you guys

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that don't know, if you just hold control and then press O over here, and I think this works on both Mac and PC, you can actually see a much more detailed breakdown of everything that's

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sort of going on under the hood. So, for instance, basically what we're seeing up here is just a brief summary. But, um, if you hold control and press O, you can actually see like this is loading all of

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this documentation into context right now. And this is all like edge function documentation. So, you know, here's how to test with curl, here's deployment, here's deploy function. I don't want to

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know how all this stuff works because it's obviously extraordinarily in-depth and kind of confusing for me. All I like

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knowing is basically that clot has all the context that it needs. So every now and then I'll just hold control and press O, open it up and and take a peek at sort of what's going on under the

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hood. On the bottom right hand corner here, you can see that we're using a fair number of tokens for this. So I just zoom in a bit. Um we're from 20,000

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up to about 51,000 or so. And so what that means is, you know, 20,000 is basically the floor for most of the time. Like when you start a new cloud

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chat, you'll already be at about 20,000 tokens because there's a fair amount of context that just naturally gets loaded in. You know, it'll like read through your files. It'll read that like

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claude.md initialization thing. It'll it has some built-in tools and stuff like that that are already part of the prompt that you just can't see. Um, but

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everything from here on out is actually just like claude consuming our token usage. So understand that uh you know these features are are not free.

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Basically what we are currently doing is we are translating or converting two currencies uh you know tokens and money

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to feature. And the whole idea with SAS as I think we continue to get further and further into you know our AI future

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is that there will soon come a point where you know nobody will want to use like a templated SAS at all. I think what we're probably going to do is most

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of these apps are actually going to be like headless or uh like backends that any user can connect a front end to. and you know use whatever functionality they

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want. So in the case of a habit tracker app for instance you know we might deliver an app right out of the box to a bunch of users and they'll they'll happily pay money for it but uh you know

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maybe they want it to like look a certain way and then they'll use their AI to make some minor adjustments to the app uh and maybe that's like an additional plan or feature or something like that. So that's basically what

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we're doing right now. We're just taking like out of the box functionality and we're just making it better, right? sort of uh an example I was given maker school which is my um AI automation

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community where I show people basically how to monetize all the skills that I'm providing you here is you know back in the day marketing teams would uh build

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these things called lead magnets and lead magnets are usually just little PDFs or you know three to five page documents that like teach people how to

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do something and the thing about a lead magnet is lead magnets were also always very templated and so you know you just wrote it once and the idea was well you'll just send it to a bunch of people

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anytime somebody needs help with But, you know, because they're very templated, they don't solve everybody's problem super specifically. They just provide like a general walkthrough of

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how to do something. Imagine though if you could customize that walkthrough to like a person's reading level, to their ability, to their experience, and so on and so forth. And that's basically what

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AI SAS apps allow us to do. And so, just like lead magnets nowadays in uh insert current year, my strategist told me not to include the year in my YouTube videos

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anymore, which I would agree. Um, but just like, you know, lead magnets don't really work super well anymore because people are just like, "Well, can't I

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just ask AI and it'll give me more customized information?" So, too, I think do like out of the box SAS apps just slowly lose more and more of their

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um value when we when we get into an AI age. So, yeah. Anyway, to make a long story short, I don't want to get too philosophical here. I just um I basically did a currency conversion on

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the cost of 40,000 tokens on my cloud subscription over to like a customized feature in an app that I really like.

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Okay, so it's saying clean. Here's a summary of everything that was built. So what was created database coaching messages reflection summaries superbase edge functions generate coaching

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generate reflection app side you know we have context and and and app layout and so on and so forth. Now you'll see that it is asking me for something to set the

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entropic API key. So I basically need to do some things. Um it's it's asking me to run the new table. So what I'm going to do is I'll just say do everything for me that you can without the API key.

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then walk me through the API key as well. Now I already know what the API key bit is since we're using another

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service. Okay, anytime you use another service, you're always going to be um you need to authenticate, right? And so what's happening here is when the Superbase Edge function, this goes to

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our database and then like collects a bunch of information. Database sends a request back. Remember um this is our actual device over here. And basically what's occurring is the the message is

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going from our phone to the mobile app to Superbase down here back up here. And then in order to actually access, you know, Claude, we need some form of O.

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And so you'll you'll always need an API key of some kind, which solves that problem. You can think of it as just like a password that allows you to sign in. Uh and that's that's kind of what

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1 hour, 38 minutes, 32 seconds

we're doing here. So anyway, it's it's done and created everything that it can.

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All we need to do is now set the API key. And you'll see it actually gives us um the steps. You just go to platform.cloud.com cloud.com and you know this feature is not going to be free right this is obviously something

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we are paying for it's an API it's an app right um but anyway go here create a new API key paste it into the above now if you don't already have an account

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you're going to have to create one in my case I already have an account so I'm just going to go platform make a left click and then I'm just going to sign into my uh to my email address now that

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1 hour, 39 minutes

I'm signed in you'll see I have a bunch of pre-existing API keys already but uh just to show you guys how this works I'm actually going to create a totally new

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not expiring key and I'll go add. Okay, so that's the whole API key right over there. And you know, now I have it copied. And what I need to do is I just

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need to go to anti-gravity. And it'll tell you not to um you know, actually paste it directly in plain text. And in general, like you you shouldn't I'm just

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being lazy here because I just want to speed this thing up. Um but in general, instead what you should do is you should go to this env here. And then you should just paste the key in sort of like this.

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Enthropic API key equals and then just paste paste it in here. And then you save the file.

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And the reason why is because this is basically like your password manager.

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1 hour, 39 minutes, 41 seconds

Just contains all your passwords. Um when you insert it into a chat like I just did here. Basically what's happening is um Claude stores all of our

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1 hour, 39 minutes, 49 seconds

conversation history locally on our computer. So this is now going to be stored in multiple places. It's not just going to be stored in this little password uh manager file which it'll

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automatically add things to. It'll add it to everywhere. It'll add it to like this chat. It'll add it to this password manager. It'll add it to our super basease and so on and so forth. Anyway,

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now that it's done, I'm just going to say, "Okay, let's test on computer." And then, you know, we're just going to go through that three-step loop where we

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start with a computer and then we go to our Expo and then after Expo, we go to um, you know, our phone and then just

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kind of loop back and forth. See here, I have my email address sign in. So, I'm then going to do this. I don't actually

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remember my password. We'll see if that worked. No, it didn't work. Okay. So, why don't I just go Oh, maybe it's this one. How am I already forgetting my

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password? That's so silly. Let's go sign up. I'm just going to um see if I can create a new account here. Looks like the user is already registered. So, I'm

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I'm just going to sign up with a different email. You can imagine how one cool feature would be a forgot password feature. So, I'll say implement a forgot

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password feature. Um and I'll just have that churning in the background while I'm there. And then here I'm just going to sign up.

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Cool. That looks like I'm now signed up. I can now get started. We're logged in.

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How it works. Create the habits. So on and so forth. Good. Walk, journal, sleep 8 hours. Good. Start challenge. Okay.

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And you can see here probably just as part of a test, although I will verify.

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You can see it says get today's coaching nudge. So this will basically look through our behavior and then actually fire it off. And I'll and I'll take a look at that just in one second. I just

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want to make sure that everything else is good. We have a weekly reflection feature down over here.

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Okay, cool. We have everything that we need. So, I'm just going to get today's coaching nudge. Click that button and see what happens because that's kind of the core piece of functionality. And

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ideally, instead of just getting the coaching nudge, what we realistically want is, you know, we want the uh we want the thing just to pop up as a

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notification outside of the app and then we also want it to pop up as sort of the AI coach inside of the app. So, I'm going to say is great. I don't just want

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the user to manually generate the coaching nudges, though. I'd like you to come up with some automatic cadence and then populate it in the app. The idea is every time the user comes onto the app, they have some form of coaching.

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Additionally, I want some sort of push notification so that, you know, if I'm out and about doing my thing, um, you know, and it's 3 p.m. or something, it

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fires off and then sends me a push saying, "Hey, Nick, I noticed you've been crushing it on X recently, but have you considered why? Here's a quick and

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easy way to do this." Awesome. And my voice transcript tool here did a pretty good job of converting my requests and what I wanted directly in. And I see now

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it's saying we've also automatically implemented the forgotten password feature, which is kind of neat. Okay, so yeah, that's on the coaching nudge. And it looks like, you know, I can only

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click on this once because it's not it's not repopulating. So that's another feature I'm going to want to fix. And this generate weekly reflection feature

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as well. If I give this button a click, I think I can just do this more or less every week. Uh, I don't actually currently have, you know, built-in. I

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don't have a lot of data, basically. So, the probability of this being able to like actually give me something good or meaningful is pretty low, but let's read it regardless. The past week was a tough

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one with an overall completion rate of just 14% across your top three habits, journal, walk, and sleep. The one bright spot was Friday, which is today, which

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where you managed to complete all three habits in a single day, which shows you absolutely have the capacity to do this when the conditions are right. the rest of the week, Monday. Okay. And it just

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updated the app, unfortunately, which is why we can't see it. But you'll notice that it just cached it, right? So, we're pulling up the exact same thing Monday through Thursday in the weekend. So, zero completions across the boards. The

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main challenge right now is to consistency beyond that single good day for the coming week. Try to reverse engineer it. Mid Friday work for you, whether it was your schedule, energy, or mindset, and see if you can

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intentionally recreate those same conditions on at least two or three other days. Okay, cool. Looks great. You can see now we no longer have the little

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coaching feature thing right up here, which is nice. Um, this looks pretty good as well. Awesome. So, I'm just going to generate that weekly

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reflection. See if the same thing pops up. Good. So, this is illustrating to me that our like reflections and nudges are sort of in our database already. Right.

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So, I'm just going to let this continue on until we get uh everything with the nudging feature and stuff like that exactly how I want. And then I'll circle back and uh show you guys. Also, I'm

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realizing there's this slight little black border around this div. And uh that just looks silly considering everything else does not have a black border. So just while we're editing all

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of this stuff, I'll also say I've noticed there's a black border around the top bar.

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The one that says 3day kickstart. Uh it doesn't look like any other div has a black border. So just fix this and ensure that the design is uniform across the app.

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Now you notice what I'm doing here is I'm starting to queue up messages. And the reason why is because I don't have to wait for cloud to finish the previous thing in order to start on something

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new. um especially if it's like kind of unrelated. And so what what it was doing over here was it was doing like AI coaching features. What it's doing over here is just doing design features. And

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so I I'm happil I'm happy to like ceue up as many messages as it takes. Um doing it in this way is just a lot cleaner for me and a lot simpler. And you see that, you know, just while I was

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explaining this to you, it's gone ahead and actually edited that. Um so I think that looks way cleaner than it did before. So let's actually just read through what else we have here. So auto

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coaching on app open. Cool. So it's every time you open the app, right? And then if there's um a recent coaching message within 12 hours, it'll instantly do it. There's daily push notice. Great.

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I'm noticing right now because we're testing and I don't have any pre-existing um app or uh habit tracking

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or usage. The reflections and the nudges are all very simple. I'd like to test this out with simulated history. Go through and send me 10 examples of coaching u nudges and then reflections.

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And I'm asking this because I just want to see, you know, if we were to run this 10 times with a a big diversity of different activities and stuff like that in the background, what sort of messages

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would we actually see? And so, um, you know, that's how I'm going to modify the prompt that the AI is using to call the other AI, I think, uh, is probably the

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simplest way to put it. Okay. And I should also organize this a little bit simpler. It'll realistically look like this, which will change the UX and stuff. So, just important to do. Okay.

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And you can see what it's done sort of in the background. If you can read computer speak, um it's just sent 10.

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Okay. And so that's what all of these are. So I'm just going to take a quick peek at these and make sure they're not total trash. Um the first is crushing meditation water dropping off. So you'll

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see there'll be like a notification with a header sort of at the top of your your your push page saying your meditation streak is genuinely impressive. 14 days at 100% this week shows real discipline,

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but hydration is great. Quietly working against you with only two out of the last 7 days completed. No streak to speak of. One thing I don't like is I don't like these M dashes because that's

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very AI. So, I'm just going to say this looks great, but modify the prompt so um Claude does not output any M dashes. M dashes are very typically AI. And the

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more m dashes we have in these messages, the less humanlike they'll seem.

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Okay, so I'm going to do that while it's outputting the previous response just because that's obviously something that I can do reasonably quickly. And the idea is these next ones should not have

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um any M dashes. Okay. And then what else we got? So, let's see. Solid week, meditation star, gradual improvement trend, perfect week. Let's also add some

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emojis to these titles specifically. And I just want the titles to have emojis. I don't want anything else to have emojis. And

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it looks like, you know, if we did a bunch of quantity habit apps, your strongest performance this month was read 30 pages that you completed 21 out

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of 30 days at a 70% consistency rate, showing you can build real momentum when a habit clicks. your 10,000 steps have it also held its own at 60% giving you a

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solid physical foundation to build on and then we also have a bunch of that need attention and stuff like that. So like if you think about the purpose of um like a review versus a nudge, like a

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review or reflection is like a lot longer, right? It's kind of a retrospective and I done whereas a nudge is a lot shorter and it's simpler and

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there's a brief little title and then like a oneline message. And so I think this is a good combination of both. The retrospectives, the the reflections and the reviews, these are very long, but

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then those little nudges are just, you know, two sentences. Okay. So, I'm just going to scroll all the way down. And you can see we have a couple of emojis

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in the nudges now, which is cool. Rerun this, but just do it three times so we could see how they look.

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And now what I'm doing here is I'm just having it actually show me some examples. I always want to see some examples of these outputs before I like 100% hardcode this into the app and then go through the whole testing procedure.

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Oh, also speaking of testing, come up with a way that I can quickly generate or test these because I'm going through a endto-end test right now.

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Okay, and now we're taking a look at the nudges. Your meditation habit is genuinely impressive. 14 days straight and perfect consistency this week. So, you clearly know how to lock something in. Notice how there are no M dashes,

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right? They just have these dashes, which are just sort of like the way it's presenting the information to me. Um, after I asked it to give me like a

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little tool, what it's doing is adding buttons to simulate 30 days of habit history and then trigger the coaching and reflection generation just so I could see the the UX as well, cuz that's

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pretty important to me. I want to be able to see sort of the way it's going to look in in a real app. Okay. And it looks like it's just done this. So, I'm going to head over to the app as well.

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And I'm just going to give this a little refresh. Okay. And you can see it actually automates the process of getting the coaching nudge, it looks like. And I think that just triggered a

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push notification. Although I can't see it cuz I am local. That's why you always have to test things in your computer as well. So I can see I can generate a

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coaching nudge and I can't see the coaching nudge which is unfortunate.

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Generate the weekly reflection. Probably not going to be able to see that either. Nope. How about a monthly reflection?

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Okay, so it looks like we have daily coaching nudges, weekly reflections, then monthly reflections which is interesting. Because I'm running this locally in Chrome, I can't actually see

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these come up as notifications. Anyway, we could simulate them. uh brief little simulated modal or toast or something like that. For those of you that don't

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know, a modal is just something that pops up on your app screen really briefly. So, it's like a full page sort of like thing and then can disappear.

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And a toast is a much smaller version of that. A toast is basically a notification. If you guys have ever seen uh those little like rectangular bars that like pop up when you get a text

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message, what that's called is a toast notification. So, what we're doing here is I'm basically having the app simulate one of these so that I could see it

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within my Chrome uh u sort of local run, even if it's not something that I have the ability to get real notifications in

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right now as a test. I just want to be able to simulate and then make sure it 100% works because the cost of doing this and then verifying whether or not

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it looks and and reads the way that I want it to read in this sort of simulated aspect is way lower than the cost of me doing it wrong the entire way

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through the testing loop from Chrome to Expo to my mobile. Okay. Okay. So, what I'm going to do here is I'm just going to refresh this cuz it just gave me the

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the the final button. And I want to simulate 30 days of habits where some are mixed, some are 50%. So, if we go

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back here, you can see that I have a couple of tracked days. Now, these are my 30 days of previous habits. And now I

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think I can simulate some form of notification. You can see edge function returning on to see see what I was saying here. So, this is actually simulating a notification, but uh the

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function failed. And so there's actually no there's nothing happening underneath if I had waited the entire way through to my phone. You know, I would have had

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to like go through this whole step, this whole process multiple times. Um, so I just basically saved myself a little bit of time there, which is quite nice. And that's what the push notification toast is kind of going to look like.

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Okay. All right. So this is going to uh do some real user JWT stuff. It looks like it was just a minor authentication error, and that's all good. Uh, we'll

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wait for that to finish and then circle back. And now that it's done, you can see I reimulated that and we had a really nice push notification right there. And uh you know, we got a little

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like robot coach up top, little toast scrolls down. You've hit all three habits today, but your 30-day consistency sitting at just 3% tells me

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today might be one of those fresh dot dot dot days. You know, if you're on your phone, you give it a click, it opens up the actual notice, and then you're you're in the app, which is quite nice. So, that's a lot better. And now I

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can actually go through the rest of the testing loop, which is starting up at a little expo and then working my way through that. So, looks great. open up a

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fresh terminal for me. Um, and then run the Expo server. I'll take a QR

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code pick and then use that to open on my phone. Make sure it's a fresh terminal, eg not inside of this thread, but actually on my computer terminal.

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Okay, I just have to add that extra context because I've noticed the last two times that I've done this, it's repeatedly launched a terminal inside of

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it. Okay, so this looks pretty good to me. I'm just going to zoom way in. Let's go Y. And what this is doing, if I just zoom in, this will go through the

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process of what looks like installing a tunnel. Okay, now after we sorted that

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out, I'm now opening up the Expo app just like I did before. And understand that building and doing it on your phone, it's a very different process

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than building and doing it on your computer, of course. And so it may not work right off the get- go. And I'm already seeing a couple of errors which I will show you guys u right now by

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running this iPhone mirroring app just so you guys can see it with me. I'm not going to do the debug after this just because hopefully you guys understand the debugs tend to be pretty similar.

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But it looks like we still have the icon um getting cut off error. So that's one thing that we need to fix. So I'm actually just going to open up a little um window here. And this window is going

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uh a little transcript window here. And then I'm just going to fire off all the changes.

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Okay. So, the first problem is the icons are all cut off uh halfway through for

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whatever reason. It's like the heads of the icons are all cut off.

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I think this is a mobile specific issue. I'm not seeing it happen anywhere else.

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It's not on the local Chrome when we run it. It's always just on mobile. I see this with the little star uh emoji

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initially. I see it with the confetti emoji on the uh today page. I'm seeing

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it on and off across the other pages as well. So, this is telling me there's just some issue with how the icons and emojis are sort of being formatted.

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Now, while I was testing, I kept on getting this error here. Commander failed to start tunnel. And basically what ended up happening was after you

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make a certain number of servers with a particular service, um, it suspends you and basically says, "Hey, can you verify your email? We want to make sure you're not spammers." And so I I kept on like

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banging my head against the wall. I was like, "Hey, what's going on? What's going on? What's going on? What's going on?" The um debug loop of that was literally me just copying this in

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and then pasting this right here. And after I did this three or four times, I just naturally and sort of inherently found the issue. So that's what I did up

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here. And then it ended up saying, "Hey, found it. your Enro accounts needs email verification. So now I'm going to open this up. We just have to verify my

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email. So I don't I don't actually know if I have an account and I'm just going to go back here and then probably set one up. Let me see.

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Okay. And then after verifying my email, I have the thread back up. So I can then just take a screenshot of this or rather open this up in my camera and just repeat the exercise.

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So, I leave this part of the video in not because I think it's going to win me any engagement points, but just because little weird stuff like this happens when you program and develop things.

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Just is how it is. I think most people would like to eliminate that from their videos and tutorials so that you think it's a lot easier than it is. But just

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like little pieces of friction and stuff like that. From time to time, this will consume your tokens. This will cause your rate limit to stop and stuff like that. The most important thing is not

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like will I run into a bug because I'll be honest, you are going to run into a bug. The thing to remember is when I run

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into a bug, keep a cool head. It's not the end of the world. I am literally talking to intelligent sand right now.

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So, it would be kind of unreasonable to expect that there wouldn't be some sort of problem with some server or whatnot.

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Anyway, I now have it open on my actual phone. So, I'm just going to run through a testing loop on on my phone because um I just finished up with this and uh I'll

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let you know how the push notifications and everything like that go. Just not going to leave this in because you're just going to be staring at me clicking on the screen. And it looks like everything is working good on the phone

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end. We get the push notifications. Uh although I did have to enable that manually in the settings. I also have the ability to swipe them away and you

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know they sort of come in at odd intervals. Uh I have the ability to trigger them. So, for instance, I triggered one on my phone to occur right now. And then I saw it pop up at the

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top. Really, the only last thing to test if you wanted to make sure that this worked 100% of the time would be um some sort of like scheduling. So, you need to

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ask Claude to set one to show up on your phone, let's say, at a particular time.

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So, maybe in my case, that would be, I don't know, like 12 or or 123, let's say, 1 minute from now. And I just have to sit there sort of staring at my phone to make sure that, you know, that works.

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And assuming that the notification pops up, awesome. Everything's good to go. If not, there are a couple of things that you're going to have to uh uh do before

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then, obviously. Okay, so that's that for actually building in AI functionality and API calls into the app. Hopefully, you guys see it's not

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actually all that difficult. It's more or less the exact same thing that we were doing before, just talking Claude and coaching it through, solving some problems for us. Uh what I want to talk

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about now is you know before we actually deploy this and then push this to you know a place like the app store there's just one final step and that final step

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is one that I think like 99% of people in our vibecoded economy nowadays are going to skip because it takes some time and it's not very sexy. Uh but it's it's

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that 1% I think of people that do that ultimately have apps that that end up going somewhere making them a fair amount of money and you know being used

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by by thousands or millions. And that step is security. You know, if your app is uh security dumpster fire, you may

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have some people start to use it, but when all of their data gets, you know, horrifically leaked, when their usernames and passwords or whatever go into the the the dark web, ether,

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whatever the hell, you know, essentially when they get compromised, um you'll completely murder any sort of reputation that you might have built uh with setup.

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But more importantly, you're also exposing yourself to massive like fiduciary uh liabilities. you're exposing yourself to issues like people

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leaking your API token and then running up your and then other users accounts to millions if not tens of millions of dollars. We've seen this sort of thing

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happen actually at scale here and these big providers are only sort of like loosely able to uh reimburse you or subsidize you for some of that token

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usage because at the end of the day it is your fault. Um there's a lot more as well if you think about like government regulations surrounding particular types

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of sensitive data. Now, in our case, we're dealing with a habit tracker, but if you're dealing with somebody's like personal health data or, you know, personal financial data, or if you're

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using maybe some open authentication connector that uh connected to something that allowed people to sign into their bank accounts, you know, if that info

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gets leaked, you can be on the hook for quite the pretty penny. And so, I'm not a lawyer, and this isn't legal advice.

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I'm also not a computer security specialist or anything like that, although I think that would be a pretty cool career now that I uh now that I come to to think of it. What I am is I'm

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just a guy that's interested in getting the biggest bang for my buck. And the way that you do that, at least with AI, for vibecoded apps, especially mobile

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apps, is you just have a prompt that you feed Claude that methodically and consistently checks all of the major

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lowhanging fruit in your security. The reality is us as, you know, solo devs or even working within a team are very unlikely to make something 100% secure.

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And you best bet, even if you think it's 100% secure, it's not. there's probably still some sort of loophole or problem.

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What we can do though is we can eliminate, you know, the 8020. We can eliminate all of the lowhanging fruit that, you know, attackers and then also

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uh platforms typically use in order to uh, you know, compromise your data and ultimately land you in hot water. And I

Chapter 23: Security Risks in App Development

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mean, anytime you start any business or build any sort of app or help use any sort of technology, you're always going to be facing some form of risk. So my goal is not to eliminate the risk

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completely because as mentioned, I think that's impossible. But what we can do is we can eliminate a fair amount of that risk and then just get it to the point where somebody would look at your app

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say, "Okay, that's actually pretty secure. I don't really think that's worth hacking. Let me just move on to the next one that's a little less secure." Right? So, what I'm going to do now is show you a simple and

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straightforward security prompt that you could use only on apps that have gone through all three prongs of this testing procedure. Um, you know, so, uh, Chrome,

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let's say, or whatever your browser of choice is, Expo, and then ultimately like actually testing it on your phone in Expo. Um, and once all of that is

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done and you've 100% verified that like the app is basically ready to launch, uh, you would you would run the security prompt audit like I'm showing you. And I also want you guys to know that this

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isn't the only thing you can use. It's just one of many types of security audit prompts. The most important thing is not to get it 100% correct. The most important thing is just to do it because

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by doing it, you will, as mentioned, at least make some progress towards the goal of ultimately being more secure. So what is this prompt? Let me show you.

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Okay, just zooming in so you guys could see this. This was posted in one of my older courses uh specifically in a module called security for vibecoded apps. So what I'm going to do now that

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I'm done with everything is I'm just going to scroll all the way down to the bottom. As you can see it's pretty chunky. It's pretty large and I'm just going to go back to claud and I'm going

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to feed that in. So I'm now uh before I run this sorry I'm I'm going to clear all of my conversation history. So I'm

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running with a fresh instance of cloud that doesn't have any of the context uh of our prior conversation. And now that we're down to zero tokens again, I'm just going to commandV, paste in this

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giant string, and just have it go on its way. And so basically what this thing is, if I show you guys a couple of key points,

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is it is a comprehensive list plus formatting of ways to identify and then

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solve the major low-hanging fruit, which typically involves specific vulnerability patterns like hallucinated packages, missing serverside validation,

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default open database policies, hard-coded secrets, and inconsistent O middleware uh for your mobile app. So, you know, as mentioned, you don't need

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to know what's going on under the hood here, and you're not going to be able to 100% solve all security, but you'll get pretty close. And then after what we do

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is we basically have it rank it in um a big tier list here alongside the security finding number. So you'll see there'll be a lot of security findings,

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maybe like 50 or 60 or something. And it's just going to go methodically top to bottom telling us whether or not we have a critical severity, high severity, medium severity, or low severity

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vulnerability, what sort of category it is, the location of it, and then the uh uh CWE, which I think is a particular

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acronym uh that I'm forgetting that deals with uh security. Yeah, it's called common weakness enumeration, which is basically just a list of the

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common ways that people usually get access to apps. Okay, so I'm not going to run through absolutely everything.

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What I'm going to do instead is I'm just going to show you guys what the actual output is. And you'll see that there are actually a fair number of vulnerabilities already, which kind of sucks. So, scrolling up all the way to

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the very very top of this. Okay. Um, you could see that we've passed the hard-coded secrets uh vulnerability.

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That's great. What that means is we're basically just not pasting our API key anywhere but in this file right over here or in Superbase itself. Git ignore

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coverage though is partial. And so that's something that we can fix. Public prefix leaks. Okay. So we are past that.

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Console error leaks partial. Okay.

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Startup validation is a complete and utter fail. Okay. And because it's a complete and utter fail, it's obviously something we're going to have to fix.

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Um, you know, just scrolling through here, there are very few that are 100% fails. Most of these are partials, which is good. Protected API routes, for instance, that's a major fail, and

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that's actually um kind of chunky. So, I'm glad we sorted that out. Error information leak. So there's a lot of uh info that comes out when

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you know we run like a console script or something like that. So we're going to fix that. Expensive operations. So there are some things that are pretty

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expensive and this basically says that an attacker if they just wanted to take us down could run this sort of thing over and over and over and over and over again to burn our claude tokens. there's

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some other, you know, issues and so on and so forth. But basically, once we've loaded all this stuff in, I'm going to ask it, okay, great. Run through and fix

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all of these errors end to end. After you're done, test and ensure they're 100% solved.

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Okay? And the first time that you run this and then have it, you know, solve these problems, it'll go through and it'll say, "Hey, I just solved all these problems. You're good to go." But we're

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not just going to run this once. What I've come to realize is that sometimes in solving one problem, it creates another somewhere else. And so after we

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solve this first run through, I'm actually just going to clear it and then run a totally new prompt again to see if a future version of Cloud can spot

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issues that a past version of Cloud has created. Once we're done with that, okay, after we did two, I'm like 90%

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satisfied with most of the security um alterations.

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Obviously, if you're a big CS security guy, you're probably over here shaking your head and you're like, you should be manually going through all the code yourself and so on and so forth. But, I mean, the whole idea behind AI is to massively multiply my leverage, right?

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Sure, I could 100% verify that every line in this code is written in a specific way, but even human beings make mistakes. And so, me going through all

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of that manually is unlikely to be much better than me doing a couple of passes of this. Now, if you are launching an app and scaling it to hundreds of thousands of users, of course, maybe

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that's a little bit different. And I'd recommend uh you know actually having like a whole team of people that review things and so on and so forth just to

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minimize your actual attack surface. But in my case all I'm going to do is I'm going to go back to the top of this prompt and I'm just going to paste it in

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again a second time. Okay. So I'm going to let that run one more time. I'm going to see if there are any additional issues. And uh again it has no context

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of the former fixes. So it's just going to see and and identify any new generated vulnerabilities or ones that the previous one couldn't catch. And you

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can see a lot of the previous failures are now either passes or partials. And so the get ignore coverage for instance uh that hasn't changed. The startup validation went from fail to partial.

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Severities are now low. Same thing with a couple of these other ones like schema validations and um you know unused dependencies and so on and so on and so

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forth. Same thing with the uh expensive operations as well. I'll say great work. Fix all things even if they're partial.

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Once sorted, let me know if the changes produced new vulnerabilities.

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And we'll just go through same thing top to bottom. And at this point, we've gotten basically as far as we can in app development without actually pushing this to the app store. Uh there are a

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few ways that you could test this further on your device if you wanted to like actually have a little app widget that you clicked on that immediately opened your app. Um the way that we're

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currently doing it, Expo is the freest and it's the straightest line path to doing so. It also doesn't take any time.

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Um, but if you wanted to run this, let's say on an iOS device, uh, you could sign up for test flight at this point, which is a specific like Apple developer

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program where you submit the app to a review team. The team reviews it within 24 hours and then you have the ability to give, I think up to like a,000 or 10,000 users a link to the app just to

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like test and so on and so forth. U, we're not going to do that because I think Expo covers the 8020. What we're going to do is I'm just going to move on to building a bunch more apps now with

Chapter 24: Transitioning to New App Features

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you now that we understand the the the full way through. I'll level them up, make them significantly more complex and also sexy. And then finally, we'll get to the end of the the course, I'll I'll

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actually submit the apps directly to the app store. And then after the review period, I'll I'll, you know, run them on my phone and so on and so forth just to make sure that it's 100% good. Okay,

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let's move on to our next app, which is going to take all the same sort of functionality that we built with these apps. Uh, so you know, the ability to

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add things to a database, the ability to read things from that database, the ability to do things like push notifications on your phone, the ability

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to have AI run synchronously, uh, you know, while you're using the app, as well as while you're not using the app to generate notifications and so on and

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so forth. Uh, the ability to connect with OOTH providers and also do authentication through Superbase. And then also the ability to contact and work with however many APIs you need.

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Let's take all that same functionality, but let's combine it in a much slimmer and tighter and faster and sexier form package. Uh, which I'm going to do in

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the guise of a Cal Tracker app, very similar to Cal.ai, that app that recently sold for between $50 to $100 million. So, let me run you guys through exactly how all that's going to work.

Chapter 25: Building the CalTracker App

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All right. Now that we've done one core app walkthrough and I've shown you guys the general highle process of walking

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through like an MVP all the way to an actual finished product, a live app that works on your cell phone, mind you

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without the app store bit, which I'll get into at the end of this course.

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Let's standardize it and let's create a couple of additional apps in a much more accelerated fashion. So the very first thing we have to do is, you know, we

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have to actually um standardize or outline the framework. And so what I'm going to do here is I'm just going to call this app design framework and then

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I'm just going to walk you through more or less everything that we've actually done in order to get this far. So if you think about it, the first thing we have to do is we have to do MVP ideation. And

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the MVP ideation is basically just this right over here, right? We need to come up with the one thing. Map the action to reward cycle. Add only what supports the

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loop. Ensure that there are only five to maybe seven screens total. then have some sort of like retention hook that keeps people using the app. So, we'll do

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that for our next one. But basically, what I'm saying is you have to start with the MVP ideation. If you don't at least have a rough idea of what you're building before you start building, you're probably not going to have a very

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good time. Okay? And after the MVP ideation, what we have to do is we have to build. Now, that building process a

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lot of the time ends up just being, you know, use a voice transcription tool like I'm using down over here. And then you just dump it in. Uh have Claude run

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through at least one build and then you have something that's kind of workable.

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Now I'm going to add one additional step between build and then test. Uh and that's going to be to design. Now in the habit tracker app, you know, I thought I

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just come up with like a reasonable quality design initially and I didn't really express very many opinions. But once you have the core sort of local app

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functionality built, it's very easy to take that functionality and just say, "Hey, I want you to, you know, take the same layout, but then apply different styles to it. Make it thematically

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different or I want it to look like this other app or maybe this website or I want it to have the feel of Apple or maybe, you know, Google or something like that." And basically, Cloud will

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very quickly and easily be able to spin the design in as many different ways as you want. There's actually a whole emerging class of SAS businesses that just take pre-existing app

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functionality. So apps that other people have built and then just tweak or spin the design five or 10 different ways moving around icons and stuff like that so that it's more uh you know agreeable

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to to a certain subset of users. So I'm going to show you guys how to do that.

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Anyway, after you've designed obviously we need to test and our testing protocol actually consists of three uh smaller steps. The first step is obviously we

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have to do this on our um computer. And so, you know, whether you're doing a iOS or a Chrome, sorry,

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an iOS or an Android app, you're probably going to want to test it on something like Chrome locally, at least if we're using React Native and Expo.

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Once we're done, um you can test via phone, but you can do so using a mirror.

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And that's what I was showing you guys earlier with the iPhone mirroring app.

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Similar examples exist for Android as well. And then after you're done with the mirror, I'd recommend testing on phone and that's like your real actual

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phone. And the reason why is because, you know, it's very easy to standardize testing on the computer and then just get it over and done with. And if you catch an issue here, then you can solve

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it before propagating down to the mirror and then ultimately the real phone. Um, when you do mirror testing, it's usually a lot easier again just to quickly swipe

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through um using, you know, functionality and and and uh uh usability that you already understand.

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And so I don't know using your little scroll wheel and going up and down clicking on things with the computer and stuff. You're also not kind of going between devices so you save a bit of

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time. Then finally at the end you know you actually go all the way down the stack and test with your phone because as mentioned there are a couple of key pieces of functionality you just can't

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get on the computer. Uh and that is things like push notifications for instance or like haptic feedback which you can't get on the mirror as well. And so basically after we're done with this

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test, okay, we add a database because everything up until now has just been done locally.

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that database will also include some form of O. Okay. And so in our case, you know, we're using um Superbase and and so on and so forth. That's actually

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really easy. You guys have already seen me do that once. And then after you add the database, because we've now fundamentally changed the way that the app works, we've also introduced a lot

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of opportunities for failure, right? You know, database is fundamentally going to work a little bit differently. Those are sending requests over to another server.

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uh and you know in the case of implementing like an onoff mechanism for instance there may be some friction when like you use Google signin or I don't even email and password signup so what

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you do after you add the database is you have to test again and this test again is basically going to be the exact same thing that we did over here so very straightforward we're basically just

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repeating this process and then afterwards you're going to want to do a security audit and that audit is necessary because you know back over here if you think about it we were doing

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stuff locally but after adding the database and the authentication stuff like that Now, this is technically this is technically live. This is something that's like occurring on the internet

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and anybody could use it. And after we're done with the security audit, if you think about it, the last thing we do is we do a third end toend test

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and then we deploy. And I'm using the term deployment here. Um, sort of generally this could include submitting to the app store. It could include

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pushing your your app to like, you know, a live uh production repository on GitHub or something like that if it's already up and running. uh whether

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you're implementing a whole app or a new feature, this is the exact same flow.

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Basically, you just repeat it over and over and over and over again. And so in our case, that last deploy feature, which I'm going to show you guys at the end, is going to involve bundling our app up into a nice sexy package and then

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actually submitting it to the the app store. Okay, so what am I going to do now? I'm just going to duplicate the same flow another two times to show you fundamentally different types of apps,

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apps that now, you know, use your camera apps that have like way different functionality. And uh we're just going to run through this a couple of times just to really get that muscle in. Okay.

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So, the very first thing we need to do for our next app is MVP ideation. So, what I'm going to do is I'm just going to move this down here. And I'm just going to brainstorm the next app that I

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want to build. And this is me kind of cheating. But basically, the next st uh stack I want to build is sort of like a Cal AI lookalike.

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And the reason why is because um you know, as mentioned, this app sold for $50 to $100 million. And I also think it's like a great example of using AI functionality. Essentially, the way that

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it works, and you can see the number of screens is pretty limited here. Um, this just tracks your calories on a day-to-day basis. And it's actually not that fundamentally different from our

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habit tracker. Just stay with me here for a sec. You know, we have some sort of daily log. On that daily log, we have a certain number of calories that like

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we desire eating. And so, typically, you'd set that up in some sort of onboarding screen just like you'd select your habits. Uh, you know, as the day

Chapter 26: Core Functionality of CalTracker

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goes on, you can add or track different things. That'll fill up the number of calories just like it fills up our challenge goal. Uh the only real

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additional piece of functionality is we're just going to be sending pictures of what it is that we're scanning over to AI. And uh the way that Cal AI works

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is they also hook up to like a couple of databases that have barcodes and food labels and stuff like that. I'm just going to leave that functionality out because I think the core loop really is

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that take a picture of your food and get the estimated number of calories. And you know, I should note that this core piece of functionality is reasonably accurate, but it's still plus or minus

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maybe 10 to 15% calories. there just a lot of things that AI can't know about foods like you know how deep the dish is and stuff like that. So um you know

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we're not actually expecting this to be 100% accurate with calorie trackers.

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We're just using the state-of-the-art functionality and getting about as good as these guys get with theirs. Okay. And then obviously after you take a photo you can see the calories. You can track

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your progress. Notice how they have very few screens. This says home. This says progress. I think this says groups and that says profile although it's kind of blurry for me. And then uh you know they

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also have some social sort of functionality and I'm probably going to leave that part out. Okay, so the core app loop, it's basically going to be right over here. We're going to grab

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these four and I'm going to run you guys through what that process looks like.

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And then for design, we're just going to like flip the design a little bit. It'll look a little bit different. Uh maybe even a little bit better. Who knows?

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Okay, so I'm now just going to voice dump all of this into one big prompt. So I'm going to double tap here. I'm using an app framework where I define a core

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function, then a core loop, then accessory features, then minimize the surface area, and then finally add some sort of retention hook at the end. My

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goal is to build an app similar to Cal AI. My core function will be the ability to track your calories and then take a

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screenshot of a piece of food, have that sent over to an AI image processor, and then return the probable number of

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calories as well as macronutrients. From there, we can add calories to our daily, you know, goal, uh, alongside protein,

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carbs, and then fats. The main action to reward cycle will revolve around hitting your calorie goals on a day-to-day

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basis, as well as your macro goals. Um, and also some sort of like visual stimulation every time you track and actually add something to the app

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because again, the whole idea is we want to have users return to the app over and over and over again every time they they want to log a meal. In terms of

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accessory features, we want a couple of things. We want the ability to uh track your progress. So, you need to be able to visualize this, see it sort of laid

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out in front of you in a chart type view. Uh you need to have a calendar which sort of goes back over time so you can see what previous logged entries

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were. You obviously need some sort of log in general. So, the user should be able to see everything that's going on uh historically with all of their their

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uh calorie tracked features. You should be able to have a clean interface where I take the photo uh and then send it over. And then you should have some nice

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sexy homepage that, you know, talks about all the various features that most people usually have in an app like this.

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Terms of surface area check, we're aiming for somewhere between three to four screens. You know, we're probably going to want to have a home screen, which at a glance shows most of the functionality. We're going to want the ability to take a photo of the calories.

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As mentioned, um, you're there's going to be some screen that pops up after you take the photo with the perceived nutritional information. The user should

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be able to make minor adjustments to that, maybe increase the calories or protein or whatever the heck, because sometimes um, AI calorie tracking functionality via photos is not

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accurate. And then finally, there needs to be a screen that allows them to track their progress, too. Uh, you should probably also add a settings page, uh,

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assuming that doesn't bump us over the 5 to 7 max screen limit. And then for retention hook, you're just going to want to add push notifications to the

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app to bring people back repeatedly. Um, you know, hey, have you tracked your breakfast yet? Have you tracked your lunch yet? Have you tracked your dinner yet? And so on and so forth, as well as

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periodic notifications congratulating people on streaks. So build some sort of streak functionality in as well. We'll do all of this locally to start and then

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eventually migrate this over to uh, you know, a database later. Okay. And then all we do is I'm just going to zoom in here. We're going to go new window.

Chapter 27: Designing the CalTracker Interface

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And then I'm going to open folder. And then instead of habit tracker, I'm going to call this cal tracker. As you can see, we're big on the trackers. We're just going to repeat the exercise. So,

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let me make this nice and pretty so you can see what I'm doing. Double tap on the page, open up Claude, and then this is where we're going to store all of our files and so on and so forth. Now, just

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like I did before, um I mentioned that uh you know, we we basically have Claude guide us through the expo setup. I'm

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going to build a mobile app using Claude um Expo and React Native. Should work on

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all devices. Scaffold me out the workspace in the current folder. Again, we don't actually have to do any of that

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downloading or anything like that ourselves. It'll actually just go through and do it all. And I should note, you could also just duplicate your habit tracker folder if you want to. Um and it'll just it'll scaffold it out

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very similarly to the habit tracker. You can then say like, "Hey, I want you to make some adjustments to this repo.

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Here's the new app." And so on and so forth. I'm just trying to approach it from like a token minimization standpoint. And I also want to show you guys how easy it is to like create a totally new app every time. It's very

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straightforward. Okay. So, you can see that this is what it's doing.

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Scaffolding the expo project with TypeScript, filebased routing, and crossplatform support. Uh, I'm just going to wait until all this stuff is good to go. And now I'm just going to

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paste in my giant prompt. And um, because the voice transcript tool I'm using just allows me to like kind of keep it in my clipboard. I just held command and pressed V. Now I have the

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entire app. So you can see here it uh is going to take this start scoping the functionality and then actually build me my little MVP/demo.

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I'll circle back when this is done. All right, we've now passed our TypeScript test with zero errors. That basically means that uh the code is technically

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compiling. It's technically finishing which is nice. That doesn't actually mean that the app works for sure though.

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So what we need to do is we need to actually test this locally on our computer. So I'll say uh run this locally on my computer. I want to test via Chrome.

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Looks like we also need to add our anthropic API key in settings to enable AI food photo analysis. Now, it would be nice to use totally new not expiring

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key. Uh I can't because you know you can't actually like take a look at this once you've created it. They only allow you to do so once. So, I'm just going to create a new one in my current

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workspace. And I'm just going to call it uh Cal Tracker app. Okay. We'll go add.

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We'll copy that key over. Then I'll go back over here. And then I'll say this is my key. I'm just doing it because I'm I'm going to want to actually test all

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that functionality. We're probably not going to be able to take a a photo with our camera, unfortunately. But um all good. Store this in aenv. And you know,

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you can see sometimes it'll actually say don't paste the API keys in the chat. Again, I'm just sort of skipping a step.

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You can go here, go env, and then create one. Um it just did right over here, which is nice. Okay, great. So with all that said, if I now go back to the

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actual app, you can see it's open right over here.

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So, let me see what this would actually look like. And note the design is still quite low quality, obviously, because we haven't done any um sort of customization or anything like that.

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Okay, we have it open. I'm just going to zoom in a bit more so you guys could see. 200% I think is reasonable.

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And as you see, the very first thing we have is we have this 2,000 remaining calories. We then have protein, carbs, fat, and then today's log. I can click

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this button, take a photo or choose from gallery, which is kind of nice. Okay.

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And I'm just going to grab a food uh picture. So, why don't we go plate of rice and chicken or something? I'll just

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grab uh I don't know, maybe this. That looks pretty easy, right? It's like pretty easily interpretable. And I'll just open this in a new tab. And I'm going to save this. And I'll say rice

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and chicken. JPG. So, that looks pretty good. And I'm just going to pretend, you know, for the the purpose of this example that I've actually just created

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this uh you know, like I'm actually just taking that photo with my camera. Okay.

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And we're just going to make this a little bit smaller here. I'll click take photo and it'll open up my little file finder. And now you can see we're uh coming up with an issue. This

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valvalidate path is not a function, but we do have half of the functionality done which is you know we can actually upload a upload a file which is cool. So

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I'm going to say getting this when I upload an image/take and then I'll say fix. So that part's

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pretty good. I mean you know I'm sure that'll sort it out in a second but let's take a look at the progress and see how that looks. So May 2nd to May 8th. So we actually see this is probably going to be like uh some form of

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calendar where you can see the number of calories that you've logged which is kind of cool. Then we have our settings which is where we can adjust how many calories, how many how much protein, how

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much carbs, how much fat. Then we can even enable some notifications. And you can see when I did that it said um you know should be allowed notifications. So

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if I reload this now we can actually put um notifications on. You can see here we're also getting some sort of issue.

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I'm saying this popped up when I accepted notifications. to fix.

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I'm just going to dismiss this so I can see the rest of it. We also have the ability to change our thropic API key, which is quite nice. And it doesn't really look like I can exit out of this little error message. Okay, no, I can't.

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I just need to click on that. Cool. So, while I was just testing this u me just feeding in those brief little adjustments to Claude, you know, it's saying it's actually gone ahead and

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fixed it. So, I now go to take a photo and I upload this. How's that looking?

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We're seeing failed to fetch. So now when I upload it says failed to fetch fix. We're just going to work our way

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progressively through this until we finish. We're now going to refresh. Let me see if I could take that photo. And now we actually have the analysis

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occurring which is quite nice. And this is cool. This is cool. As you guys can see here, we now have the title. Roasted chicken thigh with Spanish rice, which is nice. Number of calories in total

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650, protein 42, carbs 58, fat 26. We can actually adjust these if needed. But as you guys see, this is a very simple sort of UX, and we can we can now add

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that to the log. After we add, we get a cute little modal that says added to log. And then we have the number of calories remaining as well as what we filled it up with. That's really cool.

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So, I like I like what's gone on so far.

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Okay. And now we have a one day streak cuz we've logged and we also have today's log. Now, what I think would be really cool is if I could click on this, I could actually go into that. So, we we

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don't currently have the ability to do this. And then progress. You know, we see some of these numbers are starting to double up on each other. So, that's an issue. Uh, it would be nice if I

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could edit stuff. So, I'm going to add an edit functionality feature, which um should toggle when I give it a click.

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Also, the way I want this calendar is I literally want a calendar. I don't just want that. So, I'm just going to adjust all of these and feed that into AI.

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While I'm at it, I'm just going to test this little meal reminder toggle.

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Doesn't look like anything's happening right now. So, I'm going to make sure to make a note of that and then send it over. Hey, the app looks great so far. I like the core feedback loop. I should be

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able to click on the pop-ups under today's log to edit the entries. So, make sure there's a way to do that. I'm also noticing that the text is wrapping

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kind of weirdly. And from a design perspective, I think there are a bunch of ways we can make that better. I'd like you to comprehensively audit the design using, you know, your own local

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testing flow. Make sure that if you add things uh and log things that you could see all the text and so on and so forth.

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For the progress page, I want um an actual calendar, not just a uh you know, sort of like bar chart log. I'm also

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finding that when you add an entry, the calorie amounts sort of overlap. Uh something about the threshold looks kind of weird and now we have a bunch of text

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on top of uh other text. Uh we should also be able to edit the history. And really what I want is I want you to be able to click on a specific calendar day

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and then see all of the logged items for that day. And then finally for the settings page, when I click on meal reminders, nothing happens. So I'd also like you to comprehensively audit that.

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Additionally, add an onboarding screen so that when a user starts the app, it runs them through sort of what their goals are, asks them whether they like

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to lose weight, gain weight, maintain weight, build muscle, etc., all the standard run-of-the-mill stuff for calorie tracking apps, and then uses

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that to allow them to personalize their plan based off of uh macronutrient goals and stuff like that. Also, make sure to give them a recommended plan so that

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they're not just having to come up with all of it themselves. maybe have a way they can pump in like their their height and their weight and stuff to work out like some some calorie averages before

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you put any of those in. Actually do the research because I don't just want you to pull it out of your ass. Okay, go ahead.

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So, that's pretty comprehensive. I'm now just going to loop around and we'll see how it goes. All right, it just wrapped up here. It took 2 minutes and 56 seconds to implement this feature. Much

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faster than anything I would have been able to do on my end. Really cool seeing all this work. You can now see we have an onboarding screen that says, "What's your goal? We'll personalize your daily

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targets. lose weight, maintain weight, gain weight, or build muscles. So, I'm going to pretend it's actually me and I'm trying to gain weight. I'll click continue. And now it's actually going to

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ask us, hey, you know, use this to calculate your daily energy needs via via the Mifflin Sture equation. That's

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interesting. So, male or female, I am male. Uh, let's not talk about my age, you know. Let's say I'm 62. I am seeing that like this extends a little bit to

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the right, unfortunately. So, I'm just going to um start logging things. Hey, I'm noticing that the height and uh foot

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and inches markers unfortunately extend and sort of break the the page width on mobile. It just doesn't really work. So, fix that. Make them a lot tighter.

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And then, I don't know, let's say I'm like 175 sedentary, lightly active, moderately active, reactive. I'm moderately active. I'm going to see my plan. And you can see it's now actually

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given me my daily energy expenditure plus my daily calorie target. So, that looks pretty good. Not exactly rocket science. I can also adjust this at any point in time in settings. Let's go. So,

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now I have uh you know the same calorie tracking functionality I had earlier except I can actually click in on this and you can see the name, the calories,

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protein, carbs, fat. I'm not seeing the image and I think the image is important. Um, you know, we'll probably end up storing a fair amount of this, but I can imagine how, you know, we could easily compress the image so at

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least like there's a picture of something the the thing that we sent.

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Uh, when I click on one of the pre-existing entries, I don't see the image. Uh, this may be a bug. It may also be a storage space consideration, but ideally I want to be able to see the

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image that I took. We could do some compression to minimize the total amount of storage required. In fact, that's probably what makes the most sense.

Chapter 28: Improving User Experience

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Okay, so I'm just going to add all of this live sort of while I'm doing the rest of the check because that way we can bundle up and, you know, be be efficient. And here I can actually move

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through the various days of the month, which is really cool. Here's we can actually see like the the total number of calories that I've tracked

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in the calendar section. Um I like the little popup that occurs when you click on a day that has some some logs, but I

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also like you to show progress um of the day. So right now we have total number of calories, protein, carbs, and fat.

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But I want to show how much of that we ended up logging of the goal for that day. Okay, I'm just going to feed all of that back in. And then we'll go to

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settings. And now this looks pretty good. We actually have a little daily goal section with the notifications available on iOS, Android only. That's cool. So, if I click this, you can see

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reminders at 8, 12:30, and 6:30. That's really cool. We should also be able to adjust our reminder times so that uh you know, if somebody eats breakfast a

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little late or something like that, they can also uh customize their push notifications.

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So, I like that. That's pretty sweet. I wonder if we could also just like create more reminders in general because uh some people have like snacks and some people have other things. Alternatively,

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we could just have like an end of day log sort of deal. U you know, offering people some flexibility in an app is important. You don't want to offer people too much flexibility. I also

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don't really like the design on today's log, but that's not a big deal cuz I'm going to run you guys through a big like design uh flow later. Okay, cool. So, it

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looks like we have a bunch of uh changes here. So, I'm just going to click on this. And I'm not seeing any entry here.

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So, I'm what I'm actually going to do is I'm just going to delete this. And I'm noticing that when I click the delete button, it doesn't actually delete. when I click the delete button on an entry,

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uh, when I've full screened it, it doesn't actually delete. So, fix that.

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And, you know, this occurs for a variety of reasons. Um, but, you know, I'm unsure of whether or not that's just like a simple UX thing. Yeah, it looks like it's just a UX thing. Um, for

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whatever reason, when I clicked up in the top right hand corner, it didn't show up, but when I clicked down there, it's fine. So, now I'm just going to go and I'm going to add some more rice and chicken. And let's uh do a a meal

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analysis. And just because it's refreshing, it's sending it to the API a couple more times. So, I'm just going to log it. Okay. Now that I've logged in, I can actually click and I can actually

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see the image. So that was either a temporary problem or a problem because we kind of refactored the app, but this is what I wanted, right? I actually wanted to be able to see this. And actually, I think they compress the

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image automatically, which is quite nice. Notice that. So that's lovely.

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Okay, great. And then yeah, over here they're currently implementing the um calorie proportion feature. So it's going to show proportionally. And then we have the ability now to customize the

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meal reminder time, which is kind of cool. So breakfast, lunch, and dinner.

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That's very, very sweet. And uh yeah, you know, the goal, as mentioned, is to be opinionated, not uh so opinionated that uh you know, the user feels really

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locked in, but pretty opinionated. So you do some of the thinking for them.

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And this looks pretty solid. So you can actually see like the historical tracked logs because there's no entries for the day, we kind of failed, right? But this day where we actually had an entry that

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worked. Cool. So I like this. I think that more or less what we've done here is great. There's just one more feature that I want to add for the core loop.

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And that's where I could see some some situations where somebody might want to go back to maybe yesterday and then log.

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And I don't currently think we can cuz I don't think we can change the day.

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There's one more feature I'd like to add. I want people to be able to go back in time and then add logged entries to specific days. I think that's a pretty

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common need. Say I forgot to add stuff today and uh I sorry I forgot to add stuff yesterday and you know it's the morning. I want to be able to go back and add everything for the previous day.

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So add some way to change the day that we're currently on on the today page so that we can kind of like go back and forth and rather than call it today,

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call it like dashboard or something like that so that the user has the ability to kind of modify the date. Um they'll be able to do that and then they should

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also be able to actually add entries on the progress page when I click in a specific day because that might be a little bit faster uh for some use cases.

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Finally, provide the ability to delete uh different reminders and then customize what they're called in addition to their time. So, for instance, if somebody only eats two meals a day, maybe only breakfast and

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lunch, he should be able to delete the dinner entry. Cool. Looking pretty good.

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If I go back to Chrome here, you can see that we now have the additional functionality of being able to change the day. And at any point in time, I can click this button and h and jump back to

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today, which is really cool. So, you know, I could go back to the sixth. I could log the exact same meal here. And you'll see that like the calorie consistency is very similar. 620, 38,

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52, 26. I think fat might have been 28 before, but this is actually hyper hyper consistent. So, now I can add that to my log. And you'll see I now have that same

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meal, that roasted chicken uh on the 6th as well as today. And if I go to progress, I now have two logged entries, one on the 6th and one today, which is

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quite nice. And I can also edit this at any point in time. Maybe I want this to be 650 calories. I can save that. And now we actually have like some mild caloric differences. Although

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mathematically this doesn't really make sense cuz you can't just add calories out of nowhere. Realistically, the protein, carbs, and fat change, too.

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Still, I want to give people the granularity and the ability to do that.

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Now, we have our reminder set up so I can minus out the thing. And maybe I just want breakfast and lunch and I want lunch to occur at 2:30. Uh, that looks pretty good to me. No major issues. And

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let's just double check that we can actually rerun the onboarding to update my goals. Looks like we can. Just want to maintain the weight. This looks like it fixed itself and I can actually see

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my plan. Let's go. And when I do that, when I rerun the onboarding, what's really important is it doesn't just re like jig the whole app, right? I still have the logged entries from before. Um,

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it's just these logged entries have been updated with the new calorie totals. All in all, this is looking pretty good and I'm liking where this is going. I'm still obviously going to need to run

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through the testing of, you know, like the ability to take photos and stuff like that on my phone. What I want to do now is I want to significantly upgrade the design cuz to me, this design is actually really stuffy and, you know,

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the dark mode apps are are kind of annoying nowadays. I I don't know. I I like the Cal AI uh vibe. Now, what I'm going to do is, you know, I don't just

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want to copy this mercilessly, but I really do like the way the app is laid out. I think what I'll do is I'll just start with this as like my my little feature. Um, and then I'll make

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adjustments to the design after this so that you know the end result doesn't end up just being a carbon clone or carbon copy. What I'll say is, I love the

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design of the app that I just screenshotted over to you. I want you to start by emulating that design. Right now, the design is pretty weak. I'd like you to upgrade it so it more or less

Chapter 29: Finalizing the App Design

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looks exactly like this app does, just without some minor logo things rather than call it Cal AI, call it Cal Tracker. After we're done uh modifying

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the design and at least like building in like a reasonable uh library of components, etc., uh I'll modify the design so that the end result looks a little different.

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Now, I'm just going to have that go. And what's really cool is the design at the end of the day ends up being a very, you know, once you have the core functionality, all we're really doing is we're just changing the styles. We're

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changing the fonts a little bit. We're changing the colors in the background and stuff like that. So, this part's much much easier. And uh personally, I think this Claude is significantly

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better at doing this. You can also use like built-in uh libraries and tools like claw design in order to do that for you. Although I think in our case it's much easier for me just to proceed uh

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manually just prompting it and going back and forth. What I think I'm probably going to want to do is this is sort of like a black and white color scheme. I think I'm going to want to go

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uh not black but like kind of like dark blue gray instead. So give it sort of like a cool gray feature. Um, I think I'm also going to modify the font. So,

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we have some Sarah fonts because while I like this sans Sarah font, I think a Sarah font would be really cool. And then I like how we have these emojis,

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but I'm noticing these emojis are sort of all over the place color-wise. I think I'm just going to do some sort of like monochrome, dark gray, dark blue emoji. I think that's going to look

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really clean. Um, this is cool. I mean, obviously, it's not identifying the specific elements in the in the photo like through our app. Um, but you know,

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we can we can scan the food. We can take a photo of the food and then we can upload it and that's clean. I like this sort of modal or layout. You know, the thing about app design is it everything

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tends to be sort of rounded corners and that's just because usually the phones that you're on have rounded corners and you know, in the case of an iPhone anyway, um everything is sort of beveled

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and and has that clean look. But you can also experiment with a rounded corner nature of it and the color scheme. So, I think that's probably what I'm going to do. I think I'm going to make the

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corners significantly less round because I want to give like a higher lux feel.

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And then I'm also just going to make all of the the icons monochrome as well, which should look a lot easier, a lot clearer. And then I see we have a couple

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of additional buttons here which we didn't have before. But um you know with the with the graphs and stuff like that, I think I'm just going to stick to like

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that nice cool cool blue um color palette. And there's like um a feature here.

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Well, not a feature, but uh a little tool here called the the UI colors app generate which you can use to very quickly scaffold out some reasonable

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colors. So for instance, this is like a a color skin or scale for what looks to be like a lily sort of color. What you can also do is at any point in time you

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can just press spacebar and then you can customize it and change it. And so what I think I'm going to do is I'm going to go until this is like kind of like dark, you know, blueish.

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Maybe I'm going to move this just a tad so it's kind of on like the I don't want to do like indigo, but maybe something like this. That looks pretty cool to me.

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And you know, you can actually go through this whole website, let's say, and you can see how these colors work.

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Although, I guess you need to upgrade to Pro for that. Uh, and you know, this isn't a pitch for this particular service or website. I think I'll just stick with cards. But once I have this

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now, what I can do is I can actually just like copy all of these colors.

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This thing stops opening every two seconds. I can actually copy all these colors. Okay. And then I can paste that in and I can say, great,

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update color scheme. So, it looks like this. Also make sure all icons, emojis

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are monochrome, eg they're of the same color as that palette. Important we

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stick to that palette from now on. Also apply high-end lux style serif fonts

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rather than sans serif. Let's do serif fonts for the um you know display display/headings rather than sans serif

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and focus on reducing the corner rounding just a tad to make it

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feel higher end. Okay. Now for anybody that doesn't know a saraf font is just a font that um is sort of like a little old timey like this is a sans saraf

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font. sort of a new font, but a serif font, if I just exit out of this, is

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just a little twang essentially. You see how this s um sans serif would have just been, you know, something that look kind

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of like this. Well, a sand serif is well, a serif is one that sort of has like these little things at the ends.

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And so, you know, design has sort of gone back and forth between serif and sans serif a bunch of times, but I like the idea of sort of a a sans or sorry, a

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serif font for um headings just because I think it makes the app look kind of unique. Okay, so anyway, they've opened this up now and this is sort of our cal tracker. And you can see here the design

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is still lacking. It's not exactly what we wanted, but we're going to continue modifying this until it gets higher and higher end. Um, but I am liking what's going on here. We sort of have these little charts that are popping up. Okay.

Chapter 30: Testing and Debugging the App

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So, it's looking a lot cleaner and uh you can actually see the little calendar view on the main window now, not just on uh you know the progress page. Now,

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we're going to modify all of the colors and stuff so that it looks a lot cleaner. And what I like about this is notice how we can actually see um the

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entire header here and it only gets truncated when we mouse over it here.

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What I'd like is I'd actually like this to spill over. So, I I like that we can actually see roasted chicken thigh with dirty rice. I think I'm probably going to modify the prompt a little bit as

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well so that like the outputed titles are shorter. Modify the prompt of the AI so that the outputed food titles are shorter and then fix the truncation so

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that instead it wraps. Make sure the design looks really nice on the wrap too so that uh you know if we have a longer title it still fits within the bounds of the card and nothing is cut off or looks weird.

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Okay, cool. Yeah, I'm really liking that now. And then if I go to profile, how does that look? It looks like we've lost some of the, you know, outlines or headings or cards or something like that

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because as we see here, these aren't really nicely aligned anymore. So, I think it's probably just changed the colors a little bit without fully having screenshotted all the pages on the app

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and sort of like gone through that that feature in its sense. So, I'm just going to make a couple of additional minor um style modifications and so on and so forth, and then I'll circle back when

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it's all ready. Well, it's definitely looking different. Um maybe these Sarah fonts aren't actually a good idea. Maybe it's just the specific font that I'm using cuz that looks kind of lame to me.

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I don't know. Oh, it doesn't look as high-end as I was initially planning.

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And that's okay. We can change the design anytime we want. But notice how now the background has this kind of cool gray to it. Um, the home is still kind of compressed, I would say, up top. I don't really like how compressed it is.

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I'm also noticing there's just different types of design. Do you notice how here there's no outline, but then over here there's an outline. What we need to do is we just need to fully um I want to

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say harmonize the design. I'm noticing that there are slightly different types of designs on each different page. For instance, the homepage has an outline

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around each card, whereas the profile page doesn't have an outline around each card. I want you to remove outlines around all cards and favor clean, minimalistic design over um busy design.

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Also, noticing that the homepage is very compressed and kind of stacked up top.

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I'd like you to distribute each of the elements a little bit more organically so that it doesn't look super stacked.

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And then instead of using a Sarah font like I talked about before, just pick a really good common um sans saraf font, one that is typically associated with

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high-end clean designs. Everything else looks pretty clean as it stands. Um I also like the monochrome emojis and

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icons. This is just probably the major lowhanging fruit here. Now on the entry page, I'm noticing that some of the text

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is overwritten by the div. So, just make sure that when you do a test, you actually run through every single page,

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take a screenshot of it, and um itemize anything that may be sub-optimal or sub uh you know, below par before modifying

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it on your own to make sure that it's as clean as possible. Also, I don't like the green button that uh is next to the meal reminders toggle. That looks kind

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of weird right now. So, rather than making it green, make it align with the rest of the color palette. And in general, just go through one final time and ensure everything aligns with the color palette I asked for.

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Okay, we'll go back here and continuously loop back until we get a really nice design. Looks significantly better. I want you now to go through the

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design page by page and then itemize and enumerate a list of all possible improvements you can make to make it higherend, sleeker, and more lux. Self

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loop as many times as you need to implement all of that design functionality cuz it's still looking just a little tight and rough around the

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edges. Also add way cleaner icons for the home, the progress, and the profile tabs down at the bottom. Those look pretty low-end right now. Also add some

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daily weight tracker chart that you could see over time like Cali.

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Continuing to modify the design, but we're getting pretty close. This is now looking really clean. I like the profile page. Notice how um these divs are now

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really nicely laid out. Color scheme makes perfect sense. This is like a dark blue instead of a pure black. We still have a weird green toggle here, but uh

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I'll fix that up. Progress is also looking really clean. You can see how now all of the colors sort of reflect this monochrome look. And this page here, despite that being kind of a

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compressed image, um, just looks really nice. We don't have any like weird text that's laid out and so on and so forth.

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We can also remove entries really easily just by giving these buttons quick clicks. And the actual like UX when I click this button is, uh, is nice as well. So you can see we now can modify

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the protein, carbs, and fats, and the calories are sort of up here instead. We have the ability to fix our results by basically just rerunning, saying, "Hey, something is wrong here." Once we're

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done, we get a nice, beautiful little animation. If I go back to the homepage here, I'd say this is probably the biggest issue because it's also the first place that users will see. This is

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looking a little crammed. And then I don't like how small these are right now. But I do like how we've now modified the colors and we have cute little icons. Um, so all of this stuff

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is just uh, you know, a couple of prompts away as you guys could see. Not that big of a deal. Um, this little plus button here can add a meal. I think realistically we should also have the

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ability to um, track weight. So I actually asked it, hey, you know, can you add a weight tracking feature? So, what I think we'll do after that is probably add like a weight graph right

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over here that is some sort of like line chart so that you could see your weight go up or go down over time. Um, because we're modifying it right now on like the design level, the database and stuff

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like that is really easy to to change cuz it's all local. If we were to try and make all these modifications after we had like an actual live database, every time we do this, we have to push

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to like a new schema inside of Superbase, which just take a lot of time, energy, and then also consistently deprecate any old data. So, we'd also

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have to add like a big chunk to our testing loop. So, that's kind of annoying. Not Not a fan. I'm seeing that this still isn't really getting fixed, though. Notice how this just modified it

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and kind of push it into the middle. I feel like it's probably fundamentally misunderstanding something. So, we should do some sort of screenshot testing loop here. I think what I'm

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going to do is I'm actually going to have it open it up in Chrome DevTools instead and then just test it all end to end. That way, it can go through this cycle autonomously and then fix the um

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you know, the fact that these these cards aren't fully laid out. But right before that, I'm just going to take a screenshot and I'll say make the

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protein eaten, carbs eaten, and fat eaten section,

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let's say circles much larger so that we don't have weird spacing. Right now, it's not left or right aligned. should

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be proportional to the page so that the padding etc is

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fixed left and right side. Cool. And uh you know at this point I'm really just dumping in changes as time goes on. It's

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uh a very simple and straightforward for me. Uh and I do a new change usually while it's still implementing the old change because those changes tend to be pretty separated. So now for instance I

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see this plus button. It doesn't have ability to log weight. I'll say add weight log feature to the plus button.

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so that when somebody clicks the plus button that's in there. This is looking better for sure, but we're still a little bit off. I don't like that. So,

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what I'm going to do is I'll say great, now open up in your own Chrome window and screenshot through the app.

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Enumerate all of the minor inongruencies in design. So, like spacing, margins, uh

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alignment on left and right sides, etc., and then fix each in turn. Eg the

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calories eaten and then macro nutrients section is still out of alignment. Want

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this fixed on a I don't know, you know, mobile uh respon, you know, in a mobile responsive way.

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Cool. From here on out, we can log the food, we can log the weight. That looks nice. What happens when we Okay, and you can see we're actually going through this whole process, which is very nice.

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You can see it's now modifying this actually on its own. So I'm just going to duplicate this and I'm going to have another window open over here. Okay, which is kind of my own. And uh I'm just

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going to, you know, do what I'm doing on on the lefth hand side here while it does what it's doing on the right hand side. So I don't know, maybe I want to gain weight. Have the ability to enter

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all this information in. Looks pretty clean. See my plan. And then over here we have, you know, the the three sort of

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calorie and protein, carbs, and fat uh recommendations. That looks pretty cool.

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And then what happens when I log weight now? Looks like the weight log is an issue. So when I click weight log, we can see this clearly extends a little

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bit too much over to the right hand side. And I imagine that's because this width is just fixed, which is um an issue. So I'm going go back over here and I'll say some issue with the weight

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log feature right now. Um when I click the button, the field spills all the way over to the right hand side. It doesn't really look responsive or dynamic.

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Okay, cool. And uh it's just actually going through and you know testing and so on and so forth this page right now which is kind of clean. But anyway, let me actually test that functionality.

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We'll say 175 and I'll save. And then we need to log at least two weigh-ins to see my trend. So could I log 176 now?

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And then we could kind of see this change over time.

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No, it doesn't look like I think we're going to need to actually log two in two separate days. So obviously we're going to need the ability to log the weight on a different day as well. I think

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probably what we're going to want to do is we're going to want to add the recently uploaded or recently logged and then have this be both a combination of um food and weight. For the recently

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uploaded section on today, uh make it so that it's recently logged instead of recently uploaded and that we have the ability to log or rather it has the

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ability to list both meals and also weight. The idea is right now we we should be able to log directly from the

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homepage. Instead, the ability to log weight is buried in the progress page and stuff like that. There's really no need to make this overly complex. Just

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have the ability to um you know, log weight with the plus button and then have all of that appear under recently uploaded. Also, we should be able to change the date that we're logging

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weights because we again may want to historically log our weight a couple days ago or something like that. And while we did that, you can see it actually fixed these little macro

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circles and this is now looking a lot cleaner. I'd go as far as saying like this is now, you know, about as good as you can realistically get. I think there's some minor issues like when you

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kind of scale back here and this text is still kind of um you know very very what's the term the tracking is very tight that just means like there there's

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very little space between each um element. I'm also noticing that this uh card doesn't fully you know spill over which is unfortunate just given the fact

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this is now on three lines or so. I think ideally what we'd want is we'd want the calories and then the macros to be right next to it in a case like this because then we'd actually be able to

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make the card smaller, but at the same time, you know, we it is clearly trying to respect the fact that I said, "Hey, I want you to like spill over the line as

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opposed to go all the way out." So, I think what we'll probably have to do here is we'll probably need to on really tiny displays just like really compress

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the hell out of everything. on really tiny displays. Uh we should find a way to showcase all of the information aka

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calories, uh protein, carbs, and fat under the recently log section such that the car doesn't get super tall. Right

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now, we're spilling over onto four lines like the following, which is very suboptimal and uh inefficient. Instead,

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you know, if we're on a very small display, I want you to come up with a way to showcase all that information on two lines max. And that includes both

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the title and also the um you know actual like calorie info. Okay. And I'm realizing here I just did that by accident. So now I'm filling that in.

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Notice how long this is for instance.

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Now how ugly that card is. If I click this button, can I log the weight? Okay.

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And I see it actually goes today, May 6th. So uh one more feature I'm going to do is if I'm on the uh let's say May 6th

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day and I click the plus button, the weight log should immediately jump to the May 6th day. It shouldn't default to today. Uh, it should basically default

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to whatever day the homepage is currently on. Okay, that looks pretty clean as well. And then over here, we we do have that truncation. And I guess that's sort of the way that it came up

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with things. I'm wondering if there's a way just to get like the calories down below as well. I don't want any

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truncation. Um, maybe you can truncate a little bit longer. Sorry.

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Okay. Let me make sure I 100% know what I'm about to say before I try and direct this AI because at the end of the day, I am spending tokens and whatnot. So, what exactly do I want? Well, I want this

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title, okay, can be a little bit longer, but I basically want this calorie line to be down here. And then I want all of these compressed. You can truncate the

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text, but do so significantly wider. And then combine the calorie and then macronutrient lines using a smart

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combination of abre abbreviations, compressed text, smaller text, etc. do whatever you need to in order to ensure

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that you could see maybe double the current number of characters on that top line. Um, and then I'm noticing now that

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we have the ability to log weight in addition to uh log food. The little profile picks and images are slightly

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off. They're not perfectly aligned. So, I think the padding on those cards or the spacing is uh is different. I'd like

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you to realign the logged calorie entries so that they perfectly match the

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size, width, and then layout of the weight log entries. Cool. We'll stick that in as well. Yeah, the rest of the stuff looks pretty clean. Okay, now this

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is correctly identifying sort of May 6th. If I go back to progress, you see now we actually have a chart, which is nice. So, why don't I go back here to

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May 4th, log the weight, and just pretend that I was significantly skinnier. Let's go back to progress. And now you can actually see this is this is

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rising over time which is quite nice. We can zoom out to 3 month 6 month all. Uh presumably we can't see that right now because we only do have three weight entries. Um I think the last thing we

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need is we just need some sort of x-axis. We need some sort of x-axis to the graph for weight log right now because we don't currently have that.

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Oh, and uh one more thing under profile.

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The meal reminders button is still green. The meal reminders button under profile the little toggle is still green. just make it a shade of blue

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similar to the rest of our color palette. Cool. This looks way cleaner as you can see over here. And if we go under progress, you can see we actually now have the dates logged. Although this

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is with Sarif font, so I'm going to make one minor change. Um the x-axis is with serif fonts. I want sans serif. Okay.

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And now we have the finished design of the three pages, the home, the progress, and the profile. As you guys can see, this is now populating. Um, you know, as

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I go through each of these, we have everything nicely aligned. We unfortunately were unable to fix the truncation. Like to me, this is still

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just a little bit too close. But, um, I looked at the aspect ratio and then also the total number of pixels. And most phones will be somewhere like this

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between the two. So, we should be able to see the protein, the calories, and the fat down there. Uh, if you go down to the progress page, we obviously now have the the weight progress, and everything's really clean on that end.

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And then we also can, you know, exit out or jump directly into different logged entries and stuff like that. Then we have a much nicer kind of looking flow,

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I would say, alongside the breakfast and lunch reminders. So, this looks pretty good. I mean, I'm probably not going to want to include the anthropic API key in the final product. If you think about

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it, like it's kind of like my job is like the app dev. Your requests are passed through my anthropic API key. I pay for the usage and then I just charge you a monthly subscription with like

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enough margin to make uh make me some some delta. But uh you know I'll just leave that in for now just to kind of keep the app customizable and uh see

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where we go from there. Uh oh and then finally that plus button I think is really clean. So now that we've tested all the stuff locally and I've verified the design as I like we can actually

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like you know push through and and do that next step which if we go back over here to this big excaladraw

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okay is all the way over here now which means we just tested it with our computer. Now we have to test it with

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the phone mirror. And then finally, I'm going to do a phone reel. Uh, this is obviously pretty intensive and I don't want you guys to sit around just watching me do a bunch of testing and minor edits. I think the design part was

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instructive, but from here on, I'm just going to test it on the iPhone mirroring app on my end. Then I'll circle back with any adjustments or problems that that I had. Okay, I just wanted to show

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you guys this is it is so clean. Uh, it's basically exactly what I wanted.

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And um, I just tested this with my phone and the the the haptic feedback and all like the little interactions were really really nice. But um, yeah, I mean the

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layout on this is just absolutely gorgeous. So, I think this is probably one of the nicer apps that I've put together in a pretty short period of time. Um, you know, I am noticing some

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minor asynchronies like for instance, uh, sorry, inongruencies where the lefth hand most aligned part of the height and the weight and stuff like that are not

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aligned with the text underneath. These are minor changes that I'll do, uh, that I just couldn't see until I was actually on my phone. But, um, yeah, I mean, like, this is really clean. It has more

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or less the exact functionality that I want. What's really cool is, um, unfortunately, I don't know if I could show you guys the camera feature, but the camera feature is is really cool.

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like the UX and stuff like that is because I'm using the iPhone mirroring app. Um I think this is going to bug out. Yeah. So I'm unfortunately not able to show you guys. But um very cool and

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uh you know you can take a photo of your food and when you do so it it it has the exact same UX as what we saw with uh the Chrome version of uploading a photo. Uh

Chapter 31: Integrating the Database

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the stuff like this and this minor UX change is just so awesome to see as somebody that you know put something together like this in 15 or 20 minutes.

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Cool. So now that I'm done with all of this, we sort of have to do that next step which is add a database. And then after we're done the database, we'll have to run another test, then a security audit, and then finally another

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test before deploying. Um the database is going to be pretty simple. Same idea as what we had before. We're just going to set up O on Superbase. And then I'm going to set up the whole DB stuff on my

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end. Give it the keys, and then it'll run away. Uh and you know, implement everything. Now, first before I go any further, I will say the number of tokens

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that I'm using the bottom rightand corner is getting pretty long. And it depends on what model you're using, but um typically after like 200,000 or so tokens, the performance of the model

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tends to decrease significantly and your ability to make minor kind of granular changes uh goes down. And so generally my rule of thumb is like if I'm at like

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the 250 to maybe like 300,000 mark, I usually perform what's called a compaction before going any further. Uh and that compaction is pretty easy. You

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just go slash compact. What occurs is it gets all of the history in your whole conversation and then it just like compresses it. You know, instead of

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using words like fixed, the app now explicitly requests camera or photo library permissions, it might go fixed.

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App asks for camera/photo perms before launching picker. And notice how like this still has all the

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same information as this. It's just I don't know 50% as long, maybe 60% as long. So we get to save that 40% and basically bankroll that and that

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improves the performance of um you know claw on the next iteration which is quite handy. The downside of this is it takes a fair amount of time. So, while it's doing that, I'm going to go and set

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up Superbase. Same idea as what we were doing before. I'm just going to go start a project and then um you know, we have a tracker app over here. I should be able to start a new one, I believe.

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Yeah, looks like I can for project name.

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I'm going to call this Cal Tracker. Then I'm just going to use whatever the default password is here. And I'm just going to copy that over. It's use password copy. And then remember to

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enable automatic RLS anytime you're doing this because um that's just one of the lowest hanging fruit security upgrades. This wasn't actually here before. You had to do a bunch of backend

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coding stuff in order to get it up and running. But so many uh like vibe coded apps had gotten totally leaked that uh they just decided to add that. And

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honestly, I think they should just make it a mandatory thing. There are a few instances where you know you can't really do that. But yeah, I think like they should just default to that and

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then RLS should be like opt out, not opt in. All right. And the end result is super clean. We now have this working functionally on both our database and

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then also our local. So, if I go to profile here and then sign out, you see we now have a pretty clean uh login page. I won't say this is perfect, but

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basically I went on this website here, Dribble, and then I found a couple of sign-in pages that I liked. And then I just fed one in particular, this one,

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into Claude saying, "Hey, can you build me something kind of like this for my sign-in page?" Uh, and the end result is, you know, this we got these cute little calories and weights and stuff

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like that going up and down, and it's nice and I already have an account, so when I sign in, we just jump directly into the app. And there there are a couple of issues. I've noticed like the

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onboarding flickers very briefly. Uh I'm going to fix all that right now with Claude. And that's actually what what it's doing in the background. It's fixing some of the minor usability issues now that we have a database. Um

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for instance, if uh I reload this page, you notice that it takes a second for this thing to toggle. And that's because it's actually pulling that directly from the database. So I don't want that to

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take a second. You know, I just want it to be cached. So I'm going to go through now. It just implements some additional app functionality like you guys have already seen me do. Now, what I have it doing is I'm having it loop over

Chapter 32: Enhancing App Functionality

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repeatedly and then open up every individual page in the app, which is fairly easy to do. You basically just say, "Hey, I want you to open up every individual page of the app and then

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idate small improvements that it could make to the usability and then the speed by which the app loads and then works." Um, this is sort of a a meta system

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where the AI is now doing both the ideation and the implementation for me.

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Then I basically just like take a quick peek at everything after all is said and done and verify whether or not it's good. Um, and as you can see, it's now adding styles for water and weekly cards

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and and stuff like that, too. This is a form of accessory functionality, right?

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Like save as favorite is not something that I necessarily need the app to have, but you know, I want to polish it up before I actually end up submitting this to the app store. So, I want this to

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have more or less everything that I think an app would realistically need to be on par with or on the same level as Cali. And it's all just a byproduct of token usage, right? I have this running

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relatively autonomously right now. It's been 24,000 tokens spent for almost 6 minutes. you know, if you can budget that level of token spend, you can

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basically trade your money for tokens for time and then get that time back to do whatever else you're doing. Now, while I was using this on my phone, I found that it was taking a little while

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longer than I wanted it to. Um, I noticed that every time I log something, it would take like a good 700 milliseconds to maybe a second in order

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for that log to go through. And that was just annoying to me. I don't want this to take a very long time. I understand that you should know that any calorie

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tracking or whatever that I'm doing on my Chrome is always going to occur faster than on my mobile simply because I have way more processing resources on hand on my computer than I do on my

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phone. So, generally whatever it feels like on your computer, it's going to feel twice as worse on your phone. And um when I moved over to my phone, I I

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found that. So, what I did is in addition to asking it to assist me with design and so on and so forth. Okay, right over here at the very top, I also

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asked it, hey, what and it looks like it just knocked that out, which is unfortunate. Um, hey, what sorts of changes are currently contributing to

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like the very long load times and you can see that uh it looks like most of these now are significantly faster. So,

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this one here, for instance, went from 417 to to 399. This went from 393 to 389. Uh, but it looks like, you know,

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ultimately speaking, we had some improvements. I think because we're doing a single call to the database or something like that, it's going to save something like 600 milliseconds on

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cellular. And then I also had it do a little full reload test to see how fast all the stuff is. So, um, how much

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faster is home now that we've done the improvement? It looks like we're getting it down from 355 to like 221 or

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something like that. Yeah, average time of 254 down to 219 milliseconds. So, we basically made the entire app a good like 14% uh faster. And then it's

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suggesting that on, you know, some form of like LTE or something like that, we went from 560 milliseconds, which is about half a second of pure network latency. That's kind of like what I was

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feeling uh before, to about 80 milliseconds of network latency, which is about half a second saved. So, I mean, that's that's really more or less what you need to do. uh if you really

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want it to work like super snappy on your phone, you need to trade even more tokens in order for some some optimizations like this. But the issue is because I've now kind of made a

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couple of changes to the core thing while I was testing on my phone, I have to go back and I have to test it all over again. So, what I'm going to do now is I'm going to roll it back. I'm going to test it fully end to end on Chrome

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and then I'm going to do iPhone mirroring and then I'm going to do my phone. It is annoying to have to do this over and over and over again, but there's just no other way to fully know.

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Uh my goal is, you know, I want this thing to be basically publishable ready at the end of my about. I don't want to have to send this off to some QA team or anything like that. I'm doing it all

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myself with Claude. Um, so I'm just going to give it another maybe 10 or 15 minute test and then I'll let you guys know where I'm at. Okay. And I just tested this antenna and it's looking

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really clean, but there was one tiny thing that wasn't working, which is uh the bar at the very top of the page,

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which is now black, was white. What I mean by that is those little icons like my time, uh my battery, my Wi-Fi signal, and my cell signal. Those just blended

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into the background. And it's one of those things that like you just can't know until you do the full endto-end test and actually run through everything on your phone. So now that I've fixed

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this just by saying, uh, hey, you know, my iPhone's native top bar icons are white, all this stuff. Is there any way to fix? How do I force the colors? You know, without me having done so, I never would have actually got to this point.

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That would have been like a serious clear usability problem. Now that it's fixed, though, it's it's really clean.

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And you can see that the camera's in use and, you know, everything is basically uh actually totally functional now, which is quite awesome. Also, uh, because I don't have a giant plate of

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steaming rice in front of me, what I did is I took a photo of my screen with all of that on and just confirming that that still managed to work. Although, I did for some reason get slightly lower fat

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ratings and more protein. So, yeah, it's a hell of a deal. Hey, if you want more protein in your diet, just take a photo of food on your screen instead of in

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real life. Okay, so with all that said, um we now have done more or less everything that we need to do uh to actually push our app live except for

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probably the most boring thing, which is the security audit. The good news is the security audit, as I've shown you guys with the previous app build, is very simple and it's very straightforward.

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So, all I'm going to do is I'm just going to start by clearing all of my um conversation history. So, I'll go back slashcle. Okay. And so, now we're basically starting a new conversation, which is why the tokens are down to 0%.

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And I'm going to go back to my app or my Chrome. And then I'm going to go and search up security for Vibe coded apps.

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Okay, which is the um asset that I've given people in previous courses that basically runs through this big security audit. And just like I did last time,

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I'm just going to copy this puppy, paste it in, and I'm making sure it's a new prompt. Okay, I'm just going to have it go end to end analyze everything in uh

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you know, this repo and then give me those changes. And I'm not going to have you guys be on the edge of the line for this because again I feel like you you probably understand the point. Just AI

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running through everything in a very comprehensive sort of line by line security basis. And it just ran through a final security check as well as a

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general sort of functional QA audit. Uh what I'm going to do now is I'm going to do that three test sequence starting on my computer then iPhone mirroring and then finally expo on my phone. And

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that's that. Now that we've done everything involved in creating an app that has AI functionality, why don't we wrap this up with a third app? This

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one's sort of going to be a mixture between the habit tracker and then the AI tracker. Very simple and straightforward. Pomodoro or time tracking app that I'm going to implement

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some cool design stuff in. In short, we're going to have something that grows as our time tracker goes up and up. And I also want you to know you don't just have to do a tracker. I've done three

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versions of trackers now. A habit one, a calorie one, and then also a a time one.

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But a tracker just happen to be one of the easiest and simplest ways for me to show you guys how all this stuff works under the hood. Uh, with all that said, let's get into it. All right. So, my next app is going to be a pomodoro app.

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For those of you guys that don't know, there's this whole idea of uh pomodoros for productivity. And a pomodoro is just a 20 to 25 minute uninterrupted work

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block where you just sit down and get things done. And the idea is this is how you train your focus and your ability to remain productive over long periods of

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time. You do a 20-minut period to take a 5minut break. You do another 20-minut period and so on. The next week you do a 25m minute period and then a 5-minute

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break and then another 25m minute period and you know you eventually just build that up. So I'm just going to call this a Pomodoro app. Okay. And just like we

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went through the whole app design framework for the Cal AI lookalike here, I'm going to go through the whole app design framework for the Pomodoro. So I'm just going to copy that over. Okay.

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Kind of stick it right over here.

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Maybe move that up so that it doesn't kind of get in the way. And then um yeah, you know, just like the one thing for calorie tracker app was the calorie tracking. Uh the one thing for the

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Pomodoro app, just logically speaking, let me just copy this to make my life easier is going to be um timers and

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it'll be like adjustable timer. And so you'll be able to set, you know, like a 20 or 25 or 30 minute timer and increase maybe the length of that over time.

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Okay, the core loop because the app is so simple, we're going to need something cool to kind of keep people coming back.

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And so the core loop in this case is going to be uh you set the timer, you you know I think I think we're going

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to start with planting a tree and then you watch it grow and then you know it like dings when you're done and then it looks awesome. Um the whole idea here is

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because the actual app is so simple. Uh we're going to add like a tremendous amount of visual polish and appeal and kind of add like some cute interactivity

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that despite not actually being like a feature is just going to keep people actually like on the app. And the whole idea is like you're going to take your phone, you're going to put it kind of on your desk, you're going to click the

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thing, and something is slowly going to grow, let's say, as you go through uh go through that pomodoro. And we see this a lot with apps nowadays. They like the

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idea of you growing and taking care of something. It's kind of cool. You plant a tree and and so on. In terms of accessory features, we'll definitely

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need a few. So, um I want the ability to like map Pomodoro or maybe log Pomodoro.

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No, uh track progress. Let's do that. So you can see your pomodoros. Um, basically have like a log of them and so on and so forth, similar to what we had

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previously. And then you should also have the ability to, I don't know, like change the thing you're growing and maybe like earn levels and and colors

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and etc., you know, some sort of award for growing your your tree longer than other people. Terms of the surface area, um, I don't think we're really going to have much more than just like the growth

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screen and then maybe like the I don't know, maybe like progress or tracking or uh log or something like that. Maybe maybe just growth and then log. I think

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that seems pretty cool. And then the retention hook is just going to be you getting checked in with. So maybe like daily check-ins about your productivity

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and your performance. Uh hey, you know, if you want to keep growing that tree, then circle back. You haven't logged a thing in a certain amount of time. Okay,

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that that seems pretty simple. And then uh I'm actually just going to walk through this whole thing left to right.

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I didn't do a very good job of that the last time. So I'm actually going to walk through this whole thing left to right with you guys. Um and kind of check everything off as we go.

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All right. So, what does that mean? That means that we have now kind of finished our MVP ideation. And so, we sort of have the MVP all laid out. All we have to do now logically is we have to uh

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actually give that to Claude. And so, the first thing I'm going to do is I'm just going to go back to my anti-gravity instance where I was setting up Google

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Play developer and then console registration and then some other stuff for the Cal Tracker app cuz I wanted to submit that to the App Store. Then, I'm just going to go up to the top here, go

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new window. That way I can just open up another window. Then here I'm going to call this Pomodoro app. Okay. And then

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I'm going to open. And now we're inside of the Pomodoro app. That's why it says so in the top lefthand corner. And just double tapping on this opening up my little claw window. It's kind of annoying how many steps you have to do.

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You could also open it um down here, but I just think this is faster. And then once we're here now, I can just say I want to create a mobile app using React

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Native and Expo. Um set up the workspace for me. assuming I intend to launch uh

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both Android and iOS versions. Okay, so now we're just going to have this run through the setup just like we did the other couple times. Build out our little

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scaffold, and then once we have the scaffold, I'll actually add all of the functionality needed. I'm building an app with a five-step framework of core function, core loop, accessory features,

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uh surface area check, and then retention hook. The app that I'm looking to build is essentially a Pomodoro style

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uh time tracker. The core function will be the ability to track time and then adjust the time that you are tracking.

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The whole idea being this app will help improve your focus. Now the loop is going to be highly visual based. What I'd like is I'd like a function where

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when you start a pomodoro you're basically planting a seed and then over the course of the pomodoro the seed grows into a full tree. This needs to be

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very visually stimulating uh look kind of cute and be in a particular visual style. I kind of want this to look almost like an like an anime, like an

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animated uh sort of TV show. When it's done, you know, I want the phone to sort of vibrate. So, I want like haptic feedback. I also want, you know, a bunch

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of visual stimulatory uh kind of functionality. For accessory features, I want the ability to track our progress, sort of like a log. Basically, as the

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user does more and more and more uh pomodoros and then gets more focused, I want the progress bar to show them improving in both their times and so on

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and so forth. I also want the trees or seeds or whatever feature we end up on uh to change. So, I want like different colors. I want the trees to change in

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nature, maybe progress through like a different series of trees. I want you to earn levels and and colors and stuff like that. For a surface area check, I

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really only think we need two main screens. Maybe a growth screen, which is where you plant the seed and then it grows and, you know, begin the Pomodoro

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session. And then some sort of log where you can see progress over time. I don't think it needs to be more complicated than that. Uh, and then the retention hook, the thing that's going to keep

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them coming back is going to be a daily check-in where if you don't come back, let's say, you know, in a 72-hour period, then your trees start withering

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and dying. And maybe you plant one tree uh for every Pomodoro session, and the whole idea is you'll eventually build a forest.

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Okay, so I just did that while this was building. Uh, so what I'm going to do now is I'm just going to press enter here and uh, you know, I'm just going to have this thing run through the build

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full end to end. Now what this is doing which is a little bit different this session than the previous ones is it's actually going through and planning and then also exploring the project state

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and then researching some animation libraries and it's doing so using uh what's called their sub aent teams feature and you can see that actually

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just exited but basically we have like a main thread which is what I'm running right now and then we have another sub agent that's operating alongside us and this one is researching react native

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animation libraries. Now, every time this finishes, it'll collapse back to the main thread uh and then, you know, just give as much context as humanly possible to uh the the the main agent

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that's sort of listening to us. Now, in addition, you can see we've also switched over to plan mode, which is a little bit different from uh you know,

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the other sort of oneshot modes that we've operated at before. And Cloud will do this from time to time. They'll typically use plan mode when it needs a little bit more information from you and it intends to ask you some questions.

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And so, you guys can see here it's asking me some questions. So, it's saying for the tree growth animation, should the tree grow in real time as the counter counts down, eg at 50% time, 50%

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grown, or should it be a pre-made animation that plays on completion? Now, we want real-time growth for sure. For data persistence, uh session history,

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forest state levels, which approach should you prefer? I want local SQLite.

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3 hours, 13 minutes, 36 seconds

That just means doing it all locally on my computer for now. What default Pomodoro duration do you want? And should users be able to customize it?

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3 hours, 13 minutes, 42 seconds

We'll do 25 minutes. And then we can actually submit all these. And you'll see most of the time it'll actually just like recommend an option. And in this case, all of these recommendations are

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what I initially wanted. But, you know, Cloud wanted to make 100% sure that if I wanted something else, we could get it done now as opposed to later after we're done with all the the token usage. Okay.

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So, now it's going to actually go through, you know, building this SVG that grows. SVG is like sort of a design and we'll see how it looks. I don't

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actually have high hopes for the design functionality yet. I think we'll probably have to go pretty in-depth there. But, you know, if it could oneshot it, uh, that would be that would be fantastic and very much preferable.

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Cool. And it's already opened up the little app here. Uh, I had it use Chrome DevTools MCP, which is a tool that allows it to take screenshots of things

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as it does the building. And that's what's going on right now. You can see it's stuck on the loading page. So, it's just going to fix that sort of while I'm uh explaining the way that this app

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works to you guys. And so, we have two pages, you know, we have a grow page and then we have a forest page. And I actually like this. I think that's split up a lot better than uh you know what I

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was considering previously. The grow page is obviously going to be where you log a session and then it's going to grow. And then the forest page is presumably just going to be a big list

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of all the things that you've uh you know all the trees that you've sort of grown before. Now it's very simple. This is not like a high quality sort of SVG

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animation at least as it stands. You can see this is sort of off to the left and so on and so forth. So uh we're just going to make a couple changes to the app. make it all small and then yeah can

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kind of continue working our way through this until we get it to a point where it like works. Then once we've got it to the point where it works I'm going to check that little built box and then

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move on to optimizing the design. All right, so just taking a look at the app.

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We have an onboarding screen here. Just uh going to make this kind of mobiley.

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This onboarding screen says plant a seed. You can see there's some sort of issue with this um chart. After you plant the seed, you'll grow your tree.

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You'll kind of earn XP to unlock new species and then you'll tend to your forest and your forest look kind of like that. So, this does look really neat.

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It's super cute. I like this. Um, couple of issues here already. One of the issues is, you know, this is obviously not big enough to encapsulate uh the

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content. Or it may be that I'm just a little too zoomed in. But regardless, like these things should not just be on one line. We should obviously have that

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spill over, right? And then I don't much like the layout of this app as it stands anyway. So, I mean, you know, if I go

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here and then make this a little taller so I can plant the seed, you can see that we planted the seed. The issue is I can't really see this grow, right? So,

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that kind of sucks. I'm just going to give up. Go back to the forest page. You can see we have these sessions, number of hours that we've logged, streak, level one, unlocked oak. This is cool.

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This is exactly what I wanted. A certain number of trees in your forest, then recent sessions. So, what we're going to have to do like just to really test the UX and stuff like that is we're we're going to need just need to add a bunch

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of entries to the database. So, I'll say add a bunch of entries to the database that I could see what this app would look like if it was fully populated.

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We'll need to do this if we are in uh to, you know, adjust the design and so on and so forth. Anyway, this still looks pretty good. I'm pretty happy with the way that is. I'm just going to go

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back to this Excalad draw and I think meaningfully mark off this build because uh you know, this has most of the functionality that I wanted. From here on out, it's just going to be sort of a design loop like we did last time.

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Slowly and consistently upgrading the design until we eventually get to a point where, you know, we we love the way the app looks. Um I'm also just going to start voice dumping all of

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these minor changes u just directly into Cloud as it continues to build. And you can see it's already started adding stuff to the database. We're at a

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certain level. If I go back to my forest here, you see, okay, it actually doesn't look like we have all of this in yet, but we do have the sessions, which is nice. And the exp. So, that's wonderful.

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I think this looks pretty solid right now. Um, forest is looking cool, too. My main issue is on the grow page, the

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timer options of 15, 25, 45, and 60 minutes do not are not currently responsive.

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Additionally, the way the page is vertically laid out is there's a tremendous amount of space up at the top that is empty until the tree grows. This makes the app look kind of ugly and

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weird uh until the tree actually goes through that that growth animation. And uh I don't want to have to scroll.

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Ideally just want everything visible. So we need to find a way to make this app much more responsive for all different types of devices and then uh and then

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solve that as well. Also, because of the lack of responsiveness, the forests are currently cut off. ag you can't actually see the tops of all

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of those trees um on the on the homepage.

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Additionally, I just I just like us to make the homepage significantly more visually engaging. Use best practices for uh mobile apps. Right now, we only

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have the level marker. We have the little fire marker and then we have the the time and then obviously the uh timer. But I think we'll need to

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significantly upgrade this both from a design perspective, but also like a font choice perspective, also like a usability perspective, uh, and so on and so forth.

Chapter 33: App Performance Insights

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Okay, so I do like this. Um, I also like the trees withering aspect. So I think this is the trees that are withering.

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That's probably what those are. Although it says two trees and I'm seeing three.

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So there may be an issue with that database there. Um, but yeah, this is pretty cute. Uh, ideally what I'd like to do is I'd like to significantly upgrade the design of all these trees

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because, you know, as it stands right now, it's just a couple of circles. I think we could do so by maybe grabbing these assets from somewhere else instead

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of just making them ourselves. Um, that'd be nice. And then I think we should be able to click into the recent sessions, too. Add functionality so we

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can click into a recent session. Right now, the recent sessions are sort of a log, and that log is fine, but we actually want to be able to click in on them and see. also add a graph of some

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kind underneath the forest section under a forest uh on the forest page, sorry.

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So that we could see that. Also, right now I'm seeing a notification that says two trees withering, but uh the actual visuals make it look like three trees

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are withering. We should identify the the withered trees a little bit better so users can see what's going on. And uh you should be able to interact with each

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of these things like aka you should be able to maybe tap or nudge on the trees to make something happen. In addition, these trees should be animated. Right

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now, they're not really animated. It's just a a static tree. And despite how cute that is, the user is definitely going to want uh it to be a lot more engaging than that. We also need a way

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to see what happens when we actually make it to the end of a Pomodoro. Right now, um you know, I'm just looking at this app on a static basis. I don't actually know what the tree looks like

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as it grows. So, ideally, we'd want a way to trigger all the animation states that I could take a look and then maybe help you modify them more granularly.

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Uh, and then I also want to wait just to be able to trigger it. Uh, finished dumping all that stuff in while it's continuing making other changes. You can

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see it's now adjusting uh the little exp bar at the top. That's cool. And then it's also changing the the vertical spacing. Uh, it's making the forest sort

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of like fixed, which is nice, but obviously we are still cutting off that top. I think I'm actually just going to like significantly change this design.

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Um, I mean like as much as I like the way that it's laid out, I don't really like the colors and I think we're going to have to like really change the fonts in order to make this nice. So, why

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don't I go over to our lovely friend Dribble here? And then why don't we see I don't know, tracker app. We do that.

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Is there like a time tracker app or something like that? Anyway, I like this. This looks really cool.

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I'm not seeing a time tracker app specifically.

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Uh, some of these are all right. Oh, this one literally says Pomodoro. That's kind of cute. Uh, we might be able to make some use of that.

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Yeah, these are time tracker apps for sure. Just zooming in here. I like this.

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I like this design. Good mood, new day, fresh start. How well did you sleep today? It's kind of a sleep tracker, but do you notice how it's like it's got this nice sort of warm color scheme to

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it. And then also we have these um little graphs. I think this would be great from a graph perspective.

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This is the Pomodoro tracker, which is really interesting. Uh I think this is probably a little too much. What's the next page look like? Okay, it's just a

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mockup. Hey, how about this one? Okay, we actually have the time tracker, tracking history, ongoing projects. I think this is probably like the best

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little design that I've seen so far. So, let me just take a look at a couple other ones. Maybe just tracker app instead of time tracker app. That should

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at least show us more. Moody area. That looks pretty cute. I like that.

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Just taking a look at all the possible options here. You know, I want it to be fun and sort of like I don't know like like youthful, like kidlike almost.

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I like this. This is very cute, too. You see with the sort of outlines here?

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That's super cute. Although, I I'm realizing this has more to do with the pictures than it has to do with anything else. What we could also do is we could

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have GPT image generate a couple of designs for an app like this and then feed in those designs to Claude and have it try and reproduce and recreate it.

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That might be not a not a bad idea. Why don't I try that? So, I'm going to I'm just going to use a slashbtw feature here to allow me to ask Claude questions

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about this app while it works. So, I'll say describe this app in a prompt feeding into an image generator um so it can come up with a

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an indepth design uh you know set of designs. Let's try that. Now, uh GPT

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image 2 is one of the better AI image generators right now. So, this can actually generate total like mockups for mobile apps and stuff. And the idea is

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we're going to take this this mockup idea, feed it to GP2, GPT2 image or image 2, and then uh once it has this, we should be good. Okay, so I now just

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fed all of this in to GPT image, and we'll see how it does. Awesome. We now have these designs, and these look a lot cleaner. And this is more or less what I wanted. I wanted something that had, you

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know, fantastic textures. Uh I love how that it's actually somewhat similar to our app right now, actually, with a

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little field and stuff. Um I want to see this thing grow, right? I think that's really important to me. This is the forest with a bunch of information. And

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you can see it's actually kind of mimicked the structure and layout. Uh, which is telling me that really the thing that we're lacking is just like the visual assets right now. The the

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actual forest bit looks good. I think it's generated multiple of these. Okay. Yes, it has generated multiple of these.

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Um, this this is really the main screen I think that we need to focus on. And then we could see it sort of grow in the background here too. I am curious how

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we're actually going to implement something like that because obviously we're going to need to design like really high quality assets. Um I think what we could probably do is we could

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just generate each of these trees in a sequence and then maybe extract them uh and then add some sort of animation

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where like the the little things sort of move around and the leaves rustle and so on and so forth. Oh, and this is going to be the onboarding screen which I

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think is going to look fantastic. The other pages here also look really clean.

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Oh, that's beautiful. Okay, great. So, I think what we can do is we could probably just feed in these directly and say, you know, I want you to build the

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onboarding just like this. So, I'm scroll down here and I'll say these are the four onboarding screens. I want you to design the app onboarding screens to

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look exactly like this. I want you to standardize the design, the colors, and everything so that it follows uh this

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structure. And I want you to screenshot loop over and over and over again until you get it down so that it's just about pixel perfect. There are minor changes

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in position of certain buttons etc. So it doesn't need to be exactly pixel perfect with the placement of elements but it should look the exact same outside of that. Okay. And then I think

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what I'm going to do now that it's generated this I'll say this looks excellent. I want you to isolate all of the assets

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all of the let's say image assets in these mockups. So the trees, the backgrounds etc and give them to me

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as images so I can import them into the app. Okay. So now what it's going to do is actually like extract the asset. So

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it should extract this picture for instance. It should extract these for instance. It should extract this image and so on and so forth.

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Uh I can't actually do this, you know, I can't actually proceed until we get those images. So what I'm going to do is I'm just going to create a new folder here and I'll call this images.

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You can find the images for each section in images.

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And then I'm just going to upload all the images there. And this is going to be kind of hit or miss. I mean, um, you know, obviously some of this design may

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not necessarily be reproducible procedurally, right? Like for instance, this little button background, you know, it's kind of like patchy, almost like a

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watercolor. Same thing with this. And I mean, where are we going to get an animation that looks like that, right?

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I'm not entirely sure, but uh I think we can probably get most of it, which is what's important. There are also minor kind of issues with this. For instance, this is like an AI generated artifact.

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These images of these trees aren't all the same, which is kind of funny because obviously we're generating them with AI, but um when they're in our app, they'll

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all be the same. And then we need to make sure these number of trees actually matches. I like this. This is a lot clearer than whatever Claude just generated before.

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Anyway, we'll see how it goes. It's um now actually finalizing the image call.

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Cool. And we actually have some of these already, which is nice. Um I think Claude is going to have to know to cut this image, you know, in order to get

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the right section of the trees and stuff. So that's going to be a little tough, but I think we could still do this. I'm going to save uh we'll call

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this, I don't know, trees or something. I'm going to go back here and in images, I'm just going to drag in this. Oh, you know

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what? I don't think we can actually do that, unfortunately. Uh, we have to open the folder in Finder first. And then we can go here and drag this into images.

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Then I don't know why that's not allowing me to actually feed this in images. There we go.

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Some of these images feature um multiple mockups. So you'll have to cut, crop,

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divide the image as needed. Before proceeding, perform these C cut crops

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divisions so that you end up with a straightforward list of assets for the

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app. And I'm just going to copy all these in sort of one by one. Oh, this looks great. Look at this. This is fantastic. So, I guess this grows into this. Oh, not necessarily.

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It looks like we have different colors here, which I think is important, too. So anyway, we'll download this as well.

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And then rather than, you know, make this really complex, I think I'm just going to feed this image in like all of these images in one at a time.

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I'll go back here. This is a lot of app images. Save that.

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Why don't I save that? Why don't I save that?

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We can also I don't know about these buttons. I don't know about these buttons. I think I'm going to do everything but these buttons.

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So now go back here and then what do we need? We need all three of these. So now I'm just going to drag them all in.

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You can see for whatever reason these didn't actually make it. So there you go. Okay. I'm going to go back here and then I'll say think hard as you chop them up. Important we get this right.

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Once done with that, loop over infinitely until the design is pixel is close to pixel. Perfect. Okay, this

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should now be everything that we need in order to actually design it uh to that style, which is quite nice. App is already starting to come together.

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Notice how the fonts and the colors and everything like that look a lot better.

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We just have a background uh of this image right now, which is kind of black and white. And I think that's just because of the the way that the mockups were made. So, obviously, we're going to

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have to remove that background, but yeah, this onboarding screen is already looking a lot better. Cool. So, we have uh what looks to be pretty solid. The only issue I would say is this section

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here of like the nice blue sky. Looks a little odd, right? There's just something here with the nice blue sky slowly fading in that I don't think looks as good as it could.

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There's also some artifacting it looks like from the background removal on this image that we should remove. So, yeah, we just need to find a way to basically like make the top a little cleaner, I would say, and then it'll look great.

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Um, but all all that's pretty good so far. I mean, I like the imagery and I like the way that it looks. Okay, top corners of these images, let's say.

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Okay, bottom is now great, but the top corners of these images still look square, a bit square. Could we make the

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gradients come in a bit more, but only on the top? And I think the idea there is that will basically give us more

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circular basically. That'll basically give us like a nice little like rounded circle and then we won't even really queue into the idea that it's a uh you

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know an an image square that we're using. And yeah, I mean like that is gorgeous, right? Like that's pretty nice. I like this a lot. All right, that's looking a lot better now. Uh I

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really like this. Basically what we did is we just applied like a little gradient of this blue uh almost like a mask right around this almost like an arch. I think we could probably make it just 10% stronger. So I will do that.

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But as we go through the app here, you can see that, you know, aside from some image artifacting, we actually have something that looks pretty good.

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Granted, this is an app, right? And uh we also have dynamic resizing and stuff like that, which is pretty cool. I like that. And yeah, I mean, we have we

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basically have like a good onboarding page now. So, I'm just going to proceed in this vein and replace all the assets with the AI image generated ones and then circle back when I'm done. All

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right, perfect. This now looks exactly like how I wanted it to, very organic, and you can see it's sort of like built into the app. uh grows as we go taller

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and we don't have that weird sort of vertical line anymore which is quite nice. So it's responsive. It's basically everything that we wanted. We can now move on. So you just fixed the

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onboarding. It now looks fantastic. But the rest of the app does not look anything like the onboarding. So we'll

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need to continue rolling out the generated images and assets through the rest of the app. Continue top to bottom

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ensuring the design looks as similar as possible to the provided images. Take screenshots and loop as many times as you need to. You don't need to worry

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about the onboarding anymore since that's already done, but you do need to worry about the grow and the forest page. It should look identical to the

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images that I provided you inside of the images/folder. Okay, app is looking a lot better now. We have the seed right

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over here. We can tap on the time to change it, which is kind of handy. So, we've removed that other section, making it even more visually simple. Then, we

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can just click this little button and then our seed can grow. Um, I want to add just two additional pieces of functionality. The first is when I tap on the seed, I kind of want it to like

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jiggle around a little bit. Add functionality so that when you tap on the seed, and really any tree on the grow page, there's a slight haptics and

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then it sort of jiggles around a bit, almost like it's organic or alive. And then change it so that you put the timer

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on top of the tree when you click on the start timer.

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What I mean by that is uh basically I want the seed to be sort of where the this is right now and then this will just be kind of up top slowly counting

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down. I think that makes more sense just because having this down below doesn't make sense if this is supposed to be like the ground and you can see this is

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already growing right which is quite nice. Um looks really really sexy that way.

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All right, cool. I mean aside from that if I go back to forest and come back you see this is still operating right which is quite nice. Um, we've manually, well,

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not manually, but we've adjusted all of the trees that have grown so far. And you can see there are different types of trees. There's like evergreens. There's uh I don't know, there's there's a lot.

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When you tap on one, you can actually see the type of tree that was grown.

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Okay. And then you can also see the health of the tree, which is kind of neat. And anything that's withering, you can unwither just by tracking like time right now. Um, so the oak is withering,

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for instance. For whatever reason, all all three oaks are withering, unfortunately. But I don't know, your your oak here looks pretty healthy, which is nice. You have the ability to

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tap on this as well. Uh, and yeah, I mean, like I think this is the the 80204 set app. We've also changed the fonts and stuff, too. Um, I don't like what happened here. It looks like Okay. Yeah.

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So, we do have that organic jiggle, but for whatever reason, the timer there stopped when I kind of went back here.

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So, I'm wondering if that is a persistent issue. No, that looks fine.

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Um, we'll need to move that timer just a tiny bit higher. Move timer just a bit higher. Right now it's inter it's um on top of the sprout.

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Cool. Uh yeah. I mean I think we're just going to move that a little bit higher.

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And then I like how there's like kind of a seed here. Sorry, not a seed, a cloud.

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Higher. Still going to be on top of the tree. The next animation change.

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So I think aside from that, maybe we just need more clarification around levels and then we're done. I think that might be it.

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This just isn't really understanding that it's still on top of the Okay, that looks better. Uh, I'm just going to sit here and well, I'm not going to wait.

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I'm just going to have it trigger all of the U images so that I could see the tree and make sure that nothing kind of weird weird gets cut off. But yeah, after that, I think we're basically good

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to go. So, more or less it. And that's what this looks like. As it grows, we have some weird cut offs happening. So, we will need to fix that. But, it does

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look gorgeous. It's looking like a lot of these are cut off weirdly. Could you double check the source assets and ensure that we're not including additional trees above or below the main

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tree and we're also not cutting them off strangely? Yes, this is sort of our our dev mode. So you can see the pine and the trees and stuff like that grow. So I think when we fix this this will look

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pretty gorgeous. So now this is for instance a large sakura. Um and we can only do this by like holding it down and giving it a click. But yeah, all these

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are beautiful and the crystal tree I think is probably the coolest of all. So I'm going to take my trusty pen and now mark the design stage off. It's now time

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for me to test this end to end. So, I'm going to do that using a combination of both Chrome and then um the iPhone mirroring app and then ultimately uh

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expo just on my device. So, I mean I'm just going to uh test this end to end now and I'll do it off screen. But I have a bunch of functionality here like I want to be able to tap this. I also

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want to be able to plant the seed. I want the timer to go up as I plant the seed. I have basically this little feature where I can now long press on it

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to change the sprout. So, I'm just going to run through all that and then I'm also just going to go through the entire forest top to bottom and ensure that this functionality actually lines up. Like for instance, two trees withering.

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You know, it says three here. So, just going to go do a little end to- end pass and then circle back. All right. After a little bit of back and forth, maybe a little bit too much back and forth, uh we've now done all the testing as well.

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So, all that means for us is the last thing that we need to do, well, not last thing. I guess we're sort of halfway through the process, but the next thing

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we have to do is we have to add a database. Same thing that you guys have seen me do multiple times. I'm just going to do this on Superbase. I'll head over to new project and then I'm just going to call this uh Pomodoro app.

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Gonna just have this select a password for me, generate one, and then I'll copy this and then enable automatic RS as well. I'll head back over here to

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anti-gravity and then I'll say uh superbase time. I'm going to feed it in a bunch of information after this project's created. Same sort of flow.

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I'm just going to wait until all the stuff is set up. And I'm going to copy the project URL, publish, publishable key, direct connection string, etc.

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Okay. And then taking a look, they just built three different tables. Uh, sessions, trees, and then user stats.

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The uh database looks more or less like this. So, we have a bunch of ids for all the trees that get made. We have um user stats as well. So, like my level, my XP.

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We have some session information and uh so on and so forth. And then on the authentication end, um, we have confirm

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email here, which I'm just going to turn off. And I'm just doing that because it's going to be a lot easier for us.

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Uh, and now I'm again just going to test this app end to end. So I think this is local host. Is it 8081?

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Probably. Okay, cool. Cool. And now we have welcome back. Your forest is waiting for you. Um, so this looks pretty cute. I I don't think the seed is good enough. I think what we're realistically going to have to do is

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just change it. But then again, that's what the testing is for. And then I can sign in. Oh, I haven't actually created an account yet. So, let's just pretend.

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And uh okay, it's jumped us directly over to this page. We need the onboarding page. So, I'm going to move back and have it make that change. All right, we had quite a few headaches on

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that end. Unfortunately, I have yet to fully change all of the um app icons and sprites as well. I realized that some of

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the app icons I used for this just don't work. And it's not even that they were cut off weirdly. It's just when they were generated um uh sorry, like they

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they are cut off weirdly. when they were generated, they were generated in this grid pattern and then AI kind of chopped them up. So, what I have to do is I basically have to go to chat GBT and say, "Hey, can you replace these and just generate them one at a time.

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Otherwise, it just takes forever for Cloud to figure out exactly where the bounding boxes are." And honestly, I think GPT image 2, despite me having to spend maybe 10 to 20 cents to get this

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done, we'll probably just do a better job. Um, and then I can also specify I want you to remove the background and so on and so forth and whatever. So, couple of additional things I'm going to do

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there, but just because uh I'm about actually to head out to celebrate Mother's Day, I'm going to move forward to the security audit and then finally wrap it up with a final test before I

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deploy. So, how do I do my security audit? Do you guys remember there is a vibe coding security page right over here. So, I can actually just go copy

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from all the way down up to this section, security audit prompt. Now, I'm just going to feed this into AI. Same as usual. So, first thing

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to do is clear. So, actually like clear and get all this all the way up to the very top. And then now when it has zero tokens remaining, I'm just going to

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paste in that security audit. And let's just run through the entire thing end to end. And also, if it's not clear, um, you know, this isn't a process that you're only going to do once. Again,

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you're ideally going to do this a couple times, maybe two, three times run through. And don't pay attention to that Shaw cancellation message up there. I just had to deal with an internet issue

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quickly because I'm moving places. Boy, were there a lot of problems with that app. Um, I think that this is probably one of the prettier ones that I've

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designed and I really like the idea of having AI generate visuals without a background and then modifying those visuals aka uh, you know, in this case having a sprite sort of grow over time.

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There are some additional things I could have done to optimize the functioning of this particular app. Like for instance, I could have um, I don't know uh, you

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know, compressed these images so that they load a little faster on mobile.

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These are things that obviously you can do um, and if you want to do them then you know anytime you have like an app that is imageheavy or anything like

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that, just add that as sort of a dedicated step before you finish. I should also note you can automate large portions of this whole process as well.

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Uh I wanted to do it manually a few times to run you guys through, but for instance, you could just immediately pass this diagram into your cloud

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instance and then uh have it generate like a list of steps, claim, and just have it ask you questions about the app you want to make. Then it could automatically kick off a series of

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changes that first build it, then find maybe the top three app designs out there and then try and design the app like those apps. Maybe it could automate all of the Chrome testing. Can't really do that on your phone, unfortunately.

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Just kind of how it is. You would still have to manually do those steps. But then you could, I don't know, have it run through the process of creating a Superbase database and, you know, doing

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all that stuff automatically based off of best practices before finally doing that Chrome test again and then employing the security audit. And in fact, that's actually what I uh what

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I've done. I've set up a big claude.mmd file. Again, that is like your system prompt that just contains a bunch of information about my framework and what I like doing. And uh when I actually

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create an app in my new workspace, the one that uh handles my my new app designs, it's quite straightforward.

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It's literally like, hey, just ask me all the questions. And then it just asks me a bunch of questions. I answer those questions granularly and then it works together with me to to come up with a

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big plan before ultimately deploying them. So, pretty cool stuff, eh? You know, I'm just running this another time uh to catch all the security issues that we had the first time and then, you

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know, see if our solution has caused any additional security issues to crop up.

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Not a very big deal there. And it's looking like a few of these have actually flipped. Um, previously they were fine. And now I think get user versus get session is actually broken.

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So that's kind of annoying. I'm just going to run through and have it implement the changes just like I did the other time. So great work. run through and implement all changes to

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move the newly well I'll just say move any yellows or reds to greens pass.

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I'm going to feed that in because I know now that it's done everything and it has now that we're strong. Uh we can basically just do the implementation and then I'll run it through one more time.

Chapter 34: Security Audits and Final Touches

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And yeah, you can see a bunch of these things were right cuz that's what it did the last time unfortunately. Just going to wait for that to queue and then I'll circle back when it's all ready to go.

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And boy is that cool. I just went through a couple of sign up and onboarding flows and I'm just doing it one final run through. But man, the haptic feedback when you hold down on

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the little seed and then it like jiggles. Uh that's that's really sick.

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Also, despite the much higher resolution display of uh my phone versus my computer, the pictures look extremely

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clean. And uh yeah, h just happy to say that I mean this is just so much better than anything I I could have imagined. I planted a seed now for the last couple

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of minutes and I got like a cute little sprout going. The first thing in my my forest has been planted. This is one of those apps I could actually very clearly

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see myself using. And what's funny is it's it's very simple. Um, I I find that best apps typically aren't the ones that like pack the most functionality or feature, but they just have, as

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mentioned, a core kind of loop that you just want to return to over and over and over again. You know, in this case, I could either use the timer on my phone.

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And don't get me wrong, I could, but why wouldn't I click this button and just see a cute little tree blossom? I have uh accountability. There's there's a

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form of engagement here. I get push notifications saying like, "Hey, your tree is withering. Check in on X, Y, and Z." Uh, so, you know, I think I could

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probably make this significantly more complex. I could add like crazy progress trackers and stuff like that. But I'm actually totally happy with my little Pomodoro habit tracker. And uh yeah,

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with all that said, now that I've done this whole build end to end process, I think that I've completed three times now, why don't we actually push

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something to the app store? And I think because I want this to be uh you know, I want to use like the more complex app, which is probably the calendar AI app.

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Sorry, the Cal Tracker or AI app. I'm going to use that as like my push mechanism instead of this. But I want you to know that you can employ the exact same step that I'm about to show

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you or a series of steps on whatever app you want, whether it's this Pomodoro one, whether it's the habit tracker only built initially, or whether it's something else that you built on your

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end. And it's fairly straightforward. I just got to go run to Mother's Day. And I'll be back after uh giving you guys info about how to set all this stuff up.

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All right. So, I've actually compiled a full end toend guide in this Google doc right here, which walks you through everything you need to know from the

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moment that your Expo app is finished, aka, you know, ready to go, all the way up until you have a submitted app on the app store. And so, this is going to be

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the source dock that I go through as I actually do the app submission. Now, I want you guys to know right off the bat that there's no guarantee that you'll

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actually get your app onto the app store. They have, you know, relatively rigorous lists of rules and procedures.

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And you guys will see all the BS that I have to go through actually, you know, putting this app through. I've done it a couple times now, and it just never ceases to amaze me how you need to get

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your screenshots just right, and you need to have your privacy policies all set up. But whatever, that's just part of the game, right? Um, I want you guys to know we're going to we're going to go

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all the way up until the point that we submit the app. I'm going to show you guys how to do the App Store specifically, and I'm going to show you guys like an alternative that you can use if you want to push it out to the

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Play Store as well. And you don't need to know a lot of the development stuff that I'm going to be showing you guys.

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You can also and like what I recommend is take this and feed it through Claude or another AI agent um to have it actually automate a large portion of the procedure for you. So this guide has a

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bunch of prompts and stuff like that that you could use in order to you know test your app. Um let's uh let me run you guys through kind of what the process actually looks like.

Chapter 35: Preparing for App Submission

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So once you're done the expo app and I'm assuming that you've already done, you know, some form of endto-end testing, what we need to do is we need to prepare

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for production. And so for this example, I'm going to use the calorie tracker app that I developed sort of in the middle of this course. Uh simply because it's a little bit more complicated. There are a

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few more steps. So when I say prepare for production, I guess what I really mean is we need to make sure that the app is uh ready to go. We have all the

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bells and whistles and all the steps sort of in the right places. And then after we've prepared it for production, we're going to build it. and then we're actually going to push it to, you know,

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Apple. So, how do we actually do the preparation for production? Well, basically what we need to do is we need to generate a couple of things. We need

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to generate a few files like app.js n. Now, this just includes a bunch of fields that Apple uses to like, you know, call your app a specific thing.

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So, think about like the name of your app, for instance. Think about like the identifier. I don't know if you guys have ever been onto the actual app store and seen, but all apps have versions as

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well. Obviously, all apps have have icons of some kind. So, we need to generate this file. We also need to generate this EAS JSON as well, which is

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specific for Expo. Uh we need to do things like come up with the app icon, you know, the the splash screen. You know, when you push to Android, assume

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you're also doing that, you need an adaptive icon, which is uh basically some sort of foreground image on a colored background. And I think that's because, you know, dark mode and light

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mode. And so the thing is this this is going to seem pretty complicated and pretty difficult and you're like man I don't know how to do all this stuff. I just want to like develop the app and push it out. But what's really cool is

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cloud can actually do all this for you really easily. So all you really have to do is just take this and then go to whatever the workspace is that you have.

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Okay. And then just pump in this app.json and EAS.json. So it's just going to run through a quick configuration. Now I should note that

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I've already actually prepped some of this stuff um ahead of time. Uh, but it's just going to run through and then, you know, go through either create it if it hasn't already been created and then

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just walk you through the various things that you need to do in order to get it all up and running. So, you can see in my case I I actually have already set the stuff up. I have a pixel width

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of,024 by,024. My image is called public.png. Here's a bunch more data.

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Here's also a bunch of stuff about the app. So, track your calories and macros effortlessly with Cal Tracker. This is all stuff that Claude will put together for you. Basically, there are some

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permissions that you might have to, you know, adjust and kind of add runtime version policies or whatever. And then it'll just go through, it'll apply fixes. Some of the stuff is

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automatically generated for you as you develop the app. Um, but you should do this final like step basically before you proceed to make sure everything is

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100% good. So, anyway, it's gone through and it's actually fixed all this stuff, you know, with these permissions and stuff like that. Identified that because our app needs, I guess, some audio while

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it's using the camera, we need to put all this stuff in. And uh afterwards it's going to confirm your Apple ID uh assuming that you've already set it up

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and if not we're going to set it up in the in the next section. Once it's prepared what we need to do is we need to actually build for production. And what you need to understand is that apps

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sort of exists in a few stages. There's like your local test app which is sort of what we were doing here with Expo where it's running sort of like a live

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development version for us that we can adjust and tweak in real time. But then when apps actually make it onto like a a real device on the app store, um what

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happens is we we we take this big amorphous thing that we're constantly editing and changing and we basically make it really concrete. We sort of crystallize the app by building it. And

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building it just like sets a bunch of these parameters uh uh automatically so that it runs a lot faster. And so what we need to do now that we've sort of prepared our app for production is you need to build it for production. Okay.

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And the way that you do this, just moving down a little bit, it'll give you a bunch of terminal commands and stuff like that. What I'll do is I'll go back over here and I'll say, "Okay, let's

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build it." Now, I'm just going to paste this in. So, it's going to run through this step, see if EAS CLI is already

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installed, and then it'll also give you the ability to log into EAS. You might be wondering what EAS is. That's just the Expo system that we've been using up

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until now. Okay. So, I'm just going to open up a terminal, and let me just clear all this stuff. Um, now the thing is I'm already in this cal Tracker uh

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directory. But if I'm not, say, "Give me command with cd." And what this will do is it'll allow you to open any terminal window and then move to the correct

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directory. CD just stands for change directory. Uh if you're not in the directory when you run the command, then there'll be a problem. So I'm just going to assume that I'm not. And then it'll

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ask you to log in. So I'm actually already logged in as you know my whatever my account name is. If you're not logged in, um you'll have to run through that login. Again, this is something that you will have already set up at this point in the program. Why?

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because you will have already gone to, you know, EAS um and signed up for an Expo account like when we had to do the iPhone mirroring thing. Uh if you're not already signed up, you can do so just by

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heading to expo.dev/signup. Once you're done, just tell it that you are logged in. And uh you can actually go through the process of building the production

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app. And so I I'm logged in now to my account. And you can see it's now running through like the the portal ID and the build and all this stuff.

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because we're doing this for free. They have like a free tier queue which takes a lot longer to build than if you were to like pay for it. Expo is a service

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that basically just takes all of the work that we've done and then automates the process of, you know, submitting to the app store. Believe it or not, submitting to the app store used to be even more of a pain in the butt before

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Expo came around. Not to mention the testing and so on and so forth. While that's working, uh, make sure you have an Apple developer account set up. So, just head over to developer.apple.com/acount.

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then also a Google Play uh console developer account. You do that by going to play.google.com/consolezignup.

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Both of these are everchanging, so I don't really want to, you know, make this video super concrete by walking you through the actual steps, but in general, you're going to need uh you're

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going to need a couple of things. Um you're going to need to, you know, make an Apple account unless you already have one. You're going to need to create a profile here. Give it some information

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about who you are and and what you like and so on and so forth. Um give them their email address. verify your email address and so on and so forth. Uh and

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then do the same thing here. One little tip I'll give you guys for the Google Play uh console developer account is when you select your organization or

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type um I'd actually make sure that one you have a business email here not just like an an averagegmail.com. You can do

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that just by setting up like a Google uh workspace account really simply. So I can do that by just moving over to nick at leftclick.ai over here and I can click on my little developer account.

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when you do this and you go through the whole sign up, it's just a lot faster because you will have already done a lot of the verification on Google's end, like when you set up the Google

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Workspace email. Uh, so I mean that, you know, costs six or seven bucks a month to have your little Google Workspace email running, but I'm assuming you guys want to do this as a business. You should probably have a business email,

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you know, nickleclick.ai, that's the name of my agency, for instance. Once you're done with that, you'll also have to verify you have access to an Android mobile device. They

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won't actually allow you to do any of this stuff unless you have an Android, which I think kind of makes sense, right? I mean, they're not trying to make money off of you by being like, "Go buy an Android." But you do have to you

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do have to proceed through the process of like testing it on your Android phone. They want to make sure that whatever app you build is good. Okay.

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So, anyway, after you're done with all this stuff, the build uh will go through. We'll move on to the next section. So, now we built for production. I'm assuming you guys already have an Apple developer account set up and an App Store Connect sign up.

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Um, now we need to deal with privacy and compliance and then submit the build.

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So, privacy and compliance is going to vary depending on if you're going through the Apple developer um submission or the App Store submission.

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I want you guys to know that they're very similar. The shapes of them are a little different. Like um you have to do I don't know like they they put all the form fields and stuff like that

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differently if it's your Apple developer account uh uh or your app store, but like it's it's all the same thing. You got to come up with screenshots, you know, you got to come up with your

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icons. You got to, you know, come up with a big description of your app. You basically have to optimize your little landing page for the app uh and make sure your app works before you submit

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it. So what I'll do here just to save you guys a bunch of time is I'll just show you a highle overview of uh the Cal Tracker app submission for Apple and

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then it's a very similar sort of onetoone moveover if you want to do stuff for the for the app store. Now the privacy and compliance is um probably like the the most laborious step here

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because in order to set up the privacy and compliance uh these platforms actually force you to have a web accessible landing page set up with like a privacy page and then like a

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compliance kind of like policy. It's a pain in the ass to do. Um, luckily I actually already have one of these for Calracker/privacy.

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And this is what mine looks like. So this is my website here, leftclick, right? It's my agency. It's where we build things for companies using um AI technology like this. And the way that

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that this Cal Tracker privacy policy works is I literally just fed it in the entire page of like the Apple app

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submission um guidelines and I said, "Hey, I'm coming up with an app called Cal Tracker. Can you create a privacy policy and then host it on, you know, my

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my website?" And so that's what this whole page is right over here. You can see there's like a way to contact me and stuff like that. There are a couple of

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other pages as well. There's like a support page here um which you'll also have to set up. And you know the reason why I do everything using it is just so

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fast to like create a a web page like this. I literally just said, "Hey, make me a web page. Host it on my website.

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You're good to go." Right? Like that was like boom. It was 2 seconds. I didn't even have to come up with any of the text. Um I just had to feed it like the the requirement of what was asked. You know, it did things like how accurate

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are the AI nutrition estimates? Well, they're approximate. They should be used as a guide, not the the the end- all beall. How do I set up my enthropic API? Okay. Hey, here's how to do it, right?

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And it like sets up the guide and everything. So, I guess what I'm saying is like in order to proceed with the step, you are going to have to have some web accessible resource that ideally is at the same domain as your email.

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Doesn't necessarily have to be, but yeah, I mean, ideally, you would have all that stuff set up. In terms of the actual app page when you actually like go to do the submission, let me show you guys what that looks like. When you're

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done, what you can do is go to this link. Okay, that'll open up your Expo account. Um, I don't think I actually am signed in right now, which is why I'm

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getting this page. So, let me just sign in. Then, when you head over to builds, you can actually see it right here. So, iOS App Store build. Okay, give that

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button a quick click. And here is basically the whole walkthrough of everything. So, you I mean like this is like the the the finished file basically, which is referred to as a U.

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Um, so with this file here, you could actually go click submit to an app store and then you could run through these submission um steps. I'm going to copy

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that. Okay, assuming that you have everything set up, all your privacy policies and whatnot. So, what you do is you then type EAS submit platform iOS.

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And if you're not already logged in, right, you are going to have to do that login procedure. Um, and if you receive an error saying like, I don't know, your IDs aren't matched up or whatever, just

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Phoebe into Cloud Club can walk you through that process. But after you're done, you just say, "What would you like to submit?" Well, I want to select a build from EAS, which is the service

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that I just used. So, click that. And you'll see there are a couple of IDs here. Okay, these are all just different apps that I've sort of done. This is 23 hours ago. Yesterday, this is a slightly different ID. So, now I can actually

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press ID. Now, I need to log in to my Apple developer account, which I already created. So, I'll do that now.

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Because I already have like a key on my computer, it just automated the process of logging in. And now you can actually see the App Store Connects API key and everything. This is basically just

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hooked up to my App Store. And uh at the bottom, I have this big link. So, what I can do is I can open the link. That'll take me back to this page, which is now called an iOS app store submission.

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Again, this is being managed by Expo. On the lefth hand side here, you can see we have a bunch of different steps. There's builds, there's submissions, there's a couple other things. You can actually do over the air updates through Expo, which

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is pretty cool. Um, but once the actual App Store uh connect is submitted, which will take somewhere between 3 to maybe 5

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minutes or so, there'll be a little link down here that we'll click. It'll actually take us to the submission page.

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The submission page is going to look something like this. We see up here it says Cal Tracker by Nick. The reason why I have this little logo is that's just what Claude created for me. And I've

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already done this specific submission just cuz I wanted to save us some time and show you guys how straightforward it is. But basically what you're going to have to do is you're going to have to drag a bunch of screenshots for the 6.5

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in display iPhone right over here. So that's what I did with my app. And I didn't come up with any of this stuff. I just had Claude do it all for me. You'll also need to do the same thing for an

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iPad. And I think what's interesting is the iPad uh like automatically generates a bunch of other variants for you too.

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But iPhone and iPad, these are kind of annoying to do, of course, but again Claude does it all automatically now.

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It'll actually open up your your browser, take screenshots. The promotional text over here, I just copy and pasted from clot. Description, just copy and pasted from clot. Keywords.

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Here's the support URL. Okay, so you do need a support URL. Remember what I talked about earlier. That's what I pasted in there. If I copy and paste this in, you'll see I'm on that page.

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You need a marketing URL, which is just like your homepage for whatever the website is. Not a big deal. Version number should be 1.0. Set your copyright to whatever it is. I mean, in my case,

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it's my company. As you proceed and go down, you'll need to select a build. So I selected just the build that I just submitted to Expo. So uh we need to tie

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this to a particular build and you know we've submitted the build to the app store. So it's all good. Right over here you need a a required signin. So now

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this is interesting. It's a login that an actual human reviewer on the app store end will use to sign into your app and test the functionality. So right

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over here we have test e togmail.com t3stx exclamation point.414. That's like the username and the password. And then, you know, I'm saying you can basically

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just sign up or whatever to do anything you want. And then I always recommend going automatically release this version.

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Okay. Now, unfortunately, this is just one of the many pages here. You actually have a lot to go through. You have the uh app information, which is where you

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choose the name. So, I just said Cal Tracker by Nick just because, you know, this is a demo and I wanted you guys to see it. After you're done with that, you need to fill out the age ratings, all

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the content rights, the category, and everything like that. Um, once you're done with that, like I I just proceed top to bottom. You know, you have app reviews. This one's currently waiting

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for a review of somebody submitted yesterday. And then you can also see the history. U continue proceeding. We have app privacy. So, this is where you add your privacy policy. You can also add a

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user privacy choices URL. You don't need to do that. It's not required. It's optional. Something's optional. It'll tell you, and I recommend just skipping

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uh product page preview, data types, and then you literally just proceed top to bottom filling out everything that it says that you have to do. And it's

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annoying for sure. Uh if something is optional, it'll specifically tell you.

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In my case, a lot of the growth and marketing stuff are optional. Uh for pricing, you do have to select a schedule. So that's what I did here. I just made it free. And then there are a

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few other things you have to do as well, like um if there are any inapp purchases. You need to like be really clear about what they are. Then you also have to make sure that your app is not like gambling. Okay. Anyway, once all

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that stuff is done, what you do is in the top right hand corner, you'll have a little button that says submit for review. And you can just click that

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button and it'll actually be sent to Apple uh to go through that whole process on their end. Okay. So, I've done that now with the Cal Tracker. And as mentioned, there's no way I can

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guarantee that this app actually makes it onto the app store. Obviously, all we can do is we can significantly improve the chances by making sure that we have all the pages and stuff like that. If the review turns out that, you know, our

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app isn't good. Typically, the reviewers will provide at least some sort of guidance or or steps that you need to take in order to make sure that your app can make it. And it's just up to you how

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deep uh into it you want to go. What's cool, obviously, is that I have this whole guide here which walks you through it end to end. So if this video ends up insufficient for whatever reason, uh you

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can also just compare this uh and then you know sort of like diagrammatically this is more or less it you just do the same process over and over and over again every single time you want to go

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through uh you know an actual submission. And then this post-launch step down here, this is honestly mostly just like your growth and your

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marketing. And um you know earlier on the app store submission page down here where it says uh product page optimization, custom product pages,

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promo codes, game center, all this stuff. This is sort of like how you optimize the app once it's launched once you've had a couple of users. I should also note it's kind of weird to just

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like build the app and then launch it and submit it to the app store immediately. Typically, you're going to want to like do some testing. So, um there's a feature here on the App Store

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Connect called test flight which allows you before you roll out the app to push this to people uh you know within your network to give them a link that allows

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them to download your app sort of secretly. Uh, typically, you know, you actually do want to push this to a few real users to go through test your app for you, tell you if anything isn't

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working or is kind of janky, but obviously I can't really do that here. I can't go through a big like test feature period for like 2 weeks because I want to publish this video and give you guys

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that value. So, test flight is a is a very viable alternative. Um, and then yeah, make sure that after your test flight kind of distribution, you can

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then do um uh and distribution, you can then do analytics and you can see how things are going and optimize your app.

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And that's that. Thank you guys very much for joining and completing this journey of mobile app development in cloud code. Had a lot of fun putting it together for you and I absolutely love

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doing end to-end courses like this. So yeah, certainly hoping you learned something. If you guys like these sorts of courses, I also have a seminal 4-hour cloud code course that walks you guys

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through all the icons and little buttons and widgets, context management, really in-depth prompt engineering, and so on and so forth. So, now that you've sort

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of seen a top- down version of how to build things with cloud code, you know, sort of playing around with and learning by doing, if you want a much more bottom

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up and comprehensive curriculum, definitely check that out. That'll be down below in the description. I also have a bunch of advanced Claude code courses that teach you things like, you know, how to optimize skills for

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knowledge work, um, how to get up and running with like advanced sub agent or multi- aent orchestration and so on and so forth. Now, if you guys have got to

Chapter 36: Monetizing Your Skills

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the point where you know how to build stuff, the natural next question is, what do I do with all of my skills? And a route that many people take is monetizing those skills through

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providing it as a service. And if you guys want to learn how to do that, I also run a community called Maker School that is a 90-day money back guarantee where if you don't get your first client

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by the end of a 90-day period, I will actually give you all of your money back. Has probably the highest completion rates of any online community out there. We've won the school games

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many times in a row. I was actually the number one community on school for gosh, I don't know, probably three or four months. I went down to Los Angeles, met

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with Alex Ramoszi and Sam Evans and so on and so forth and actually like taught a bunch of people how to build great high uh engagement, highquality

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communities where people actually achieve their stated outcomes. So yeah, definitely check that out. Over 2,000 other people are currently learning how to sell these sorts of technologies

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through a variety of means. We do lead genen through like cold email. I show you guys how to construct, you know, DM sequences to send to people. I show you

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guys how to build your own social media uh brands and and and and also build like lead generation funnels that don't rely on referrals like I think a lot of people are unfortunately forced to.

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Anyway, I'll stop tooting my own horn, but uh you know, you make it to the end of a 4-hour course, probably going to be at least a couple minutes of advertising. Hopefully you guys appreciated it as mentioned. And if you

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guys have any ideas for future videos, just leave them down below. I'm more than happy to read the comments and then get inspired. I'll see all y'all in the next one. Take care. Oh, and please

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subscribe to the channel. Hate asking for people, but uh that significantly helps my reach.

Sync to video time

All

From Nick Saraev

Watched

LIVE

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